OSCAT

Network:LIBRARY

Documentation In English

Version 1.12

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1. Legal

Die OSCAT Network Bibliothek definiert neben den Standard Datentypen weitere Datentypen. Diese werden innerhalb der Bibliothek verwendet, können aber jederzeit von Anwender für eigene Deklarationen verwendet werden. Ein Löschen oder verändern von Datentypen kann dazu führen das Teile der Bibliothek sich nicht mehr kompilieren lassen.

1.1. Disclaimer

The Modules in the OSCAT library are intended as a design template and a guide for software development according to IEC61131-3. A guaranty for function in any way is not given by the developers and is explicitly excluded. The softwaremodules included in the Library are free of charge and has no approach for any waranty as far as defined legally. If not agreed explicitly in writing the copyright holders and / or third party software modules to available "as is" without any warranty, either express or implied, including - but not limited - to market or use for a particular purpose . The entire risk and full responsibility for quality, accuracy and performance of the software modules is at the user. Should the library, or parts of the library prove to be defective, the cost of all necessary servicing, repair or correction is by the user. Should parts or the entire library used for the creation of application software or are used in software projects, the user shall be liable for any errors, function and quality of the application. Indemnity by OSCAT is excluded. The user of the library OSCAT has to ensure, through appropriate testing and quality assurance measures and approvals that may arise from possible errors in the library of OSCAT no damage. The aforementioned licensing terms and disclaimers apply equally to the software library, and in this guide offered descriptions and explanations, even if not explicitly mentioned.

1.2. License

The use of the library OSCAT is free and can be used without a license agreement for private or commercial purposes. Diversion of the library is encouraged, but free and with reference to our Website WWW.OSCAT.DE to take place. If the library is available for download in electronic form or distributed on disks, it is necessary to ensure that a clearly identifiable reference to OSCAT and a web link to WWW.OSCAT.DE is included.

1.3. Intended Use

The software module contained in the OSCAT library and described in this document are exclusively for use by professionals with training in PLC programming. Users are responsible for compliance with all applicable standards and regulations come to bear in the application. OSCAT refers neither in the manual nor in the software to these standards and regulations.

1.4. Registered trademarks

All used brand names are used in this description, without reference to its registration or owner. The existence of such rights can not be excluded. The used brand names are property of their owners. A description of the use for commercial purposes is prohibited, therefore, not even in part.

1.5. Others

All arrangements legally binding are only in chapter 1 of this handbook. A derivation or reference to legal claims based on the content of this manual in addition to the provisions of Chapter 1 is excluded entirely.

2. Introduction

2.1. Objectives

OSCAT is for " Open Source Community for Automation Technology ".

OSCAT created a Open Source Library referenced to the IEC61131-3 standard, which can be dispensed with vendor-specific functions and therefore ported to all IEC61131-3-compatible programmable logic controllers. Although trends for PLC in the use of vendor-specific libraries are usually solved efficiently and these libraries are also provided in part free of charge, there are still major disadvantages of using it:

1. The libraries of almost all manufacturers are being protected and the Source Code is not freely accessible, which is in case of a error and correction of the error extremely difficult, often impossible.
2. The graphic development of programs with vendor-specific libraries can quickly become confusing, inefficient and error-prone, because existing functions can not be adjusted and expanded to the actual needs. The Source codes are not available.
3. A change of hardware, especially the move to another manufacturer, is prevented by the proprietary libraries and the benefits that a standard such as IEC61131 offer would be so restricted. A replacement of a proprietary library of a competitor is excluded, because the libraries of the manufacturers differ greatly in scope and content.
4. The understanding of complex modules without an insight into the source code is often very difficult. Therefore the programs are inefficient and error prone.

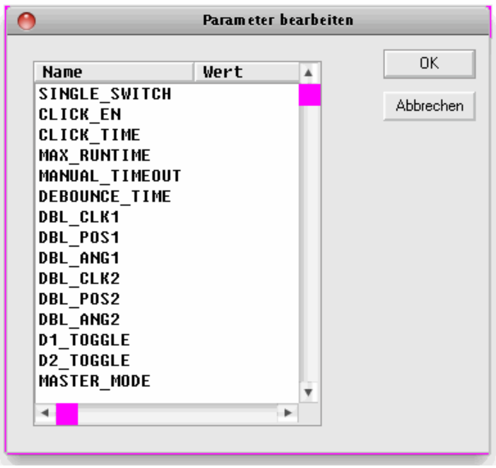
OSCAT will create with the open OSCAT Library a powerful and comprehensive standard for the programming of PLC, which is available in the Source Code and verified and tested by a variety of applications in detail. Extensive knowledge and suggestions will continue to flow through a variety of applications to the library. Thus, the library can be described as very practical. OSCAT understands his library as a development template and not as a mature product. The user is solely responsible for the tests in its application modules with the appropriate procedures and to verify the necessary accuracy, quality and functionality. At this point we reference to the license and the disclaimer mentioned in this documentation.

2.2. Conventions

1. Direct modification in memory:
   Functions, which modify input values with pointer like _Array_Sort, starts with an underscore "_". _Array_Sort sorts an array directly in memory, which has the significant advantage that a very large array may not be passed to the function and therefore memory of the size of the array and the time is saved for copying. However, it is only recommended for experienced users to use these functions, as a misuse may lead to serious errors and crashes! In the application of functions that begin with "_", special care is appropriate and in particular to ensure that the call parameters never accept undefined values.

2. Naming of functions:
   Function modules with timing manner, such as the function PT1 are described by naming FT_<modulname> (ie. FT_PT1). Functions without a time reference are indicated with F_<modulename>.

3. Logical equations:
   Within this guide, the logical links are used & for AND , + for OR, /A for negated A and # for a XOR (exclusive OR).

4. Setup values for modules:
   

   To achieve that the application and programming remains clear and that complex functions can be represented simply, many of the modules of the library OSCAT have adjustable parameters that can be edited in application by double-clicking on the graphic symbol of the module. Double-clicking on the icon opens a dialog box that allows you to edit the Setup values. If a function is used multiple times, so the setup values are set individually for each module. The processing by double-clicking works on CoDeSys exclusively in CFC. In ST, all parameters, including the setup parameters may passed in the function call. The setup parameters are simply added to the normal inputs. The parameters are in the graphical interface entered by double click and then processed as constants under IEC61131. It should be noted that time values has to be written with syntax "T#200ms" and TRUE and FALSE in capital letters.

5. Error and status Reporting (ESR):

   More complex components are largely contributed a Error or status output. A Error Output is 0 if no error occurs during the execution. If, however, in a module a error occurs, this output takes a value in the range 1 ..99 and reports a error with a error number. A status or Error Collection module may collect these messages and time-stamped, store them in a database or array, or by TCP/IP forward it to higher level systems. An output of the type Status is compatible with a Error starting with identical function. However, a status output reports not only errors but also leads on activities of the module log. Values between 1..99 are still error messages. Between 100..199 are located the reports of state changes. The range from 200..255 is reserved for Debug Messages. With this, within the library OSCAT standard functionality, a simple and comprehensive option is offered to integrate operational messages and error messages in a simple manner, without affecting the function of a system. Modules that support this procedure, as of revision 1.4 are marked "ESR-ready." For more information on ESR modules, see the section "Other functions".

2.3. Test environment and conditions

Available platforms and related dependencies

CoDeSys:

Needs the libraries " SysLibFile.lib " and " SysLibSockets.lib "

Runs on

WAGO 750-841

CoDeSys SP PLCWinNT V2.4

and compatible platforms

PCWORX:

No additional library needed

Runs on all PLC iwith file system and Ethernet controllers with firmware >= 3.5x

BECKHOFF:

Development Environment	Target Platform	PLC libraries to include

TwinCAT v2.8.0 or higher	PC or CX (x86)	TcSystem.Lib
TwinCAT v2.10.0 Build >= 1301 or higher	CX (ARM)	TcSystem.Lib

Requires the installation of "TwinCAT TCP/IP Connection Server"
Thus requires the Library "TcpIp.Lib"
(Standard.lib; TcBase.Lib; TcSystem.Lib then automatically be included)

Programming environment:
NT4, W2K, XP, Xpe;
TwinCAT system version 2.8 or higher;
TwinCAT Installation Level: TwinCAT PLC or higher;
Target platform:
TwinCAT PLC runtime system version 2.8 or higher.
PC or CX (x86)

TwinCAT TCP/IP Connection Server v1.0.0.0 or higher;
NT4, W2K, XP, XPe, CE (image v1.75 or higher);
CX (ARM)

TwinCAT TCP/IP Connection Server v1.0.0.44 or higher;
CE (image V2.13 or later);

2.4. Releases

This manual is updated by OSCAT continuously. It is recommended to download the latest version of the OSCAT manual under www.OSCAT.DE . Here the most current Manual is available for download. In addition to the Manual OSCAT prepared a detailed revision history. The OSCAT revisionhistory lists all revisions of individual modules, with amendments and at what release the library of this component is included.

2.5. Support

Support is given by the users in the forum WWW.OSCAT.DE. A claim for support does not exists, even if the library or parts of the library are faulty. The support in the forum under the OSCAT is provided for users voluntarily and with each other. Updates to the library and documentation are usually made available once a month on the home page of OSCAT under WWW.OSCAT.DE. A claim for maintenance, troubleshooting and software maintenance of any kind is generally not existing from OSCAT. Please do not send support requests by email to OSCAT. Requests can be processed faster and more effectively when the inquiries are made in our forum.

3. Demo-Programs

3.1. Demo programs

The OSCAT Network Library contains components and functions that deal with the issue of file-handling and ethernet communications, and sometimes require enhanced base knowledge . In order to allow the user easy access are possible, for many theme demo programs are prepared.

The demo programs are in network.lib in the folder "DEMO" included. If they are used, the needed programs should be copied to your project and then can be adjusted to your needs. Some modules require the disclosure of its own specific parameters so that they are fully functional.

BASE64_DEMO

CSV_PARSER_BUF_DEMO

CSV_PARSER_FILE_DEMO

DLOG_FILE_CSV_FTP_DEMO

DLOG_FILE_CSV_SMTP_DEMO

DLOG_RRD_DEMO

DNS_DYN_DEMO

DNS_REV_DEMO

DNS_SNTP_SYSLOG_DEMO

FILE_BLOCK_DEMO

FTP_CLIENT_DEMO

GET_WAN_IP_DEMO

HTTP_DEMO

INI_PARSER_BUF_DEMO

INI_PARSER_FILE_DEMO

IP2GEO_DEMO

IRTRANS_DEMO

MB_CLIENT_DEMO

MB_SERVER_DEMO

MD5_CRAM_AUTH_DEMO

RC4_CRYPT_DEMO

SHA_MD5_DEMO
SMTP_CLIENT_DEMO
SPIDER_DEMO
TELNET_LOG_DEMO
TELNET_PRINT_DEMO
TN_VISION_DEMO_1
TN_VISION_DEMO_2
WORLD_WEATHER_DEMO

4. Data Types of the NETWORK-Library

4.1. DLOG_DATA

The Structure DLOG_DATA is used for communication of the DLOG_ * modules.

DLOG_DATA:

Data field	Data Type	Description
ADD_HEADER	BOOL	Header data store
ADD_DATA	BOOL	Cyclic data store
CLOCK_TRIG	BOOL	DTI (Date-Time) New value
ID_MAX	USINT	Number of blocks DLOG_* modules
DTI	DT	current Date-Time Value
UCB	UNI_CIRCULAR_BUFFER_DATA	Data Storage
NEW_FILE_STRING	STRING	New file name
NEW_FILE_RTRIG	BOOL	Edge new file was created

4.2. us_LOG_VIEWPORT

us_LOG_VIEWPORT

Name	Type	Properties
LINE_ARRAY	ARRAY [1..40] OF INT	LOG-index references
COUNT	INT	Number of visible messages
UPDATE_COUNT	UINT	Update count
MOVE_TO_X	INT	Control of the message display
UPDATE	BOOL	Data has been changed -> redraw

4.3. URL

The Structure URL stores the individual parts of a URL.

URL:

Data field	Data Type	Description
PROTOCOL	STRING (10)	Protocol
USE R	STRING(32)	User Name
PASSWORD	STRING(32)	Passwort
DOMAIN	STRING(80)	Domain
PORT	WORD	Port Nummer
PATH	STRING(80)	Pfadangabe
QUERY	STRING(120)	Query
ANCHOR	STRING (40)	Anker
HEADER	STRING(160)	Header

4.4. us_TN_INPUT_CONTROL

A variable of type us_TN_INPUT_CONTROL can be used to parameterize and manage various INPUT_CONTROL elements, and as well as to represent the ToolTip information.

us_TN_INPUT_CONTROL:

Data field	Data Type	Description
bo_Enable	BOOL	Processing enable / disable
bo_Update_all	BOOL	All elements redraw
bo_Reset_Fokus	BOOL	set Focus on first Element
in_Fokus_at	INT	Element with an active focus
in_Count	INT	Number of INPUT_CONTROL elements
in_ToolTip_X	INT	ToolTip Text X Offset
in_ToolTip_Y	INT	ToolTip Text Y Offset
by_ToolTip_Attr	BYTE	ToolTip text attributes (color)

in_ToolTip_Size	INT	ToolTip text length
usa_TN_INPUT_CONTROL_DATA	ARRAY [1..20] OF us_TN_INPUT_CONTROL_DATA	

4.5. us_TN_INPUT_CONTROL_DATA

A variable of type us_TN_INPUT_CONTROL_DATA can use to parameterize a INPUT_CONTROL element and to process element related inputs / events.

us_TN_INPUT_CONTROL_DATA:

Data field	**Data Type**	**Description**
by_Input_Exten_Code	BYTE	Extended Key Code
by_Input_ASCII_Code	BYTE	Key Code ASCII
bo_Input_ASCII_IsNum	BOOL	Key code is a digit
in_Title_X_Offset	INT	Title Text X Offset
in_Title_Y_Offset	INT	Title Text Y Offset
by_Title_Attr	BYTE	Title text attributes
st_Title_String	STRING	st_Title_String
in_Cursor_X	INT	current cursor X position
in_Cursor_Y	INT	current cursor Y position
IN_TYPE	INT	Element Type
in_X	INT	Element X position
in_Y	INT	Element Y position
in_Cursor_Pos	INT	current cursor position
by_Attr_mF	BYTE	Attributes for element with focus
by_Attr_oF	BYTE	Attributes for element without focus
in_selected	INT	Text element is selected
st_Input_Mask	STRING	Input mask

st_Input_Data	STRING(STRING_LENGTH)	Text input current
st_Input_String	STRING	text copy after entering
st_Input_ToolTip	STRING	Text for ToolTip
in_Input_Option	INT	Text Options
bo_Input_Entered	BOOL	Text RETURN key pass
bo_Input_Hidden	BOOL	Text hidden input with '*'
bo_Input_Only_Num	BOOL	Text only allow number entry
bo_Focus	BOOL	Element has focus
bo_Update_Input	BOOL	Element due to input redraw
bo_Update_All	BOOL	Element draw from scratch

4.6. us_TN_MENU

A variable of type us_TN_MENU can be used to parameterize a MENU item, to display it and to process element related inputs.

us_TN_MENU:

Data field	**Data Type**	**Description**
st_Menu_Text	STRING(STRING_LENGTH)	Menu items
in_Menu_E_Count	INT	Number of menu items
in_Y	INT	Menu Y position
in_X	INT	Menu X position
by_Attr_mF	BYTE	Text attributes with focus
by_Attr_oF	BYTE	Text attributes without focus
in_X_SM_new	INT	Sub-menu, new X-position
in_Y_SM_new	INT	Sub-menu, new Y-position
in_X_SM_old	INT	Sub-menu old X-Position
in_Y_SM_old	INT	Sub-menu old Y position
in_Cur_Menu_Item	INT	current main menu item

in_Cur_Sub_Item	INT	current sub-menu item
in_State	INT	menu status
in_Menu_Selected	INT	selected menu item
Menu, number of lines	BOOL	action: create menu
bo_Destroy	BOOL	action: remove menu
bo_Update	BOOL	action: refresh menu

4.7. us_TN_MENU_POPUP

A variable of type us_TN_MENU_POPUP can be used to parameterize a PO-PUP item, to display it and to process element related inputs.

us_TN_MENU_POPUP:

Data field	Data Type	Description
st_Menu_Text	STRING(STRING_LENGTH)	Menu items
in_Menu_E_Count	INT	Number of menu items
in_X	INT	Menu X position
in_Y	INT	Menu Y position
in_Cols	INT	INT
INT	INT	Menu, number of lines
in_Cur_Item	INT	Current menu item
by_Attr_mF	BYTE	Text attributes with focus
by_Attr_oF	BYTE	Text attributes without focus
by_Input_Exten_Code	BYTE	keycode - special keys
Menu, number of lines	BOOL	action: create menu
bo_Destroy	BOOL	action: remove menu
bo_Update	BOOL	action: refresh menu
bo_Activ	BOOL	Menu is active

4.8. us_TN_SCREEN

A variable of type us_TN_SCREEN can be used to manage display the graphical user interface (GUI).

us_TN_SCREEN:

Data field	Data Type	Description
bya_CHAR	ARRAY [0..1919] OF BYTE	Screen character
bya_COLOR	ARRAY [0..1919] OF BYTE	screen color codes
bya_BACKUP	ARRAY [0..1919] OF BYTE	screen backup memory
bya_Line_Update	ARRAY [0..23] OF BYTE	screen lines update
by_Input_Exten_Code	BYTE	Key code special keys
by_Input_ASCII_Code	BYTE	Key Code ASCII
bo_Input_ASCII_IsNum	BOOL	Key code is a digit
in_Page_Number	INT	current page number
in_Cursor_X	INT	Cursor X Position
in_Cursor_Y	INT	Cursor Y Position
in_EOS_Offset	INT	End of String Offset
by_Clear_Screen_Attr	BYTE	screen delete color
bo_Clear_Screen_Attr	BOOL	delete screen
bo_Modul_Dialog	BOOL	modal dialog active
bo_Menu_Bar_Dialog	BOOL	Menu dialog active

4.9. FILE_PATH_DATA

The Structure FILE_PATH_DATA is used by the the module FILE_PATH_SPLIT to store each item.

FILE_PATH_DATA:

Data field	Data Type	Description
DRIVE	STRING (3)	Drive Name
DIRECTORY	STRING(STRING_LENGTH)	Directory Name

FILE NAME	STRING	File Name

4.10. FILE_SERVER_DATA

FILE_SERVER data structure:

Name	Type	Properties
File_open	BOOL	File is open
FILE NAME	STRING	File Name
MODE	BYTE	Mode - command
OFFSET	UDINT	File offset for reading and writing
FILE_SIZE	UDINT	Current size of the file in bytes
AUTO_CLOSE	TIME	Timing for the automatic closure
ERROR	BYTE	Error codes (system dependent)

4.11. IP2GEO

IP2GEO data structure:

Name	Type	Properties
STATE	BOOL	Data is valid
Data is valid	DWORD	IP address of the geographical data
COUNTRY_CODE	STRING(2)	Country code (ISO format) eg AT = Austria
COUNTY_NAME	STRING(20)	Name of the country
REGION_CODE	STRING(2)	Region Code (FIPS format) eg 09 = Vienna
REGION_NAME	STRING(20)	Name of region
CITY	STRING(20)	Name of the city
GEO_LATITUDE	REAL	Latitude of the place

GEO_LONGITUDE	REAL	Longitude of the place
TIME_ZONE_NAME	STRING(20)	Time zone name
GMT_OFFSET	INT	Offset from Universal Time in minutes
IS_DST	BOOL	DST active

4.12. IP_C

IP_C data structure:

Data field	**Data Type**	**Description**
C_MODE	BYTE	Type of connection
C_PORT	WORD	Port Number
C_IP	DWORD	coded IP v4 address
C_STATE	BYTE	Status of the connection
C_ENABLE	BOOL	Connection release
R_OBSERVE	BOOL	Receive data monitor
TIME_RESET	BOOL	Reset the monitoring times
ERROR	DWORD	Error Code
FIFO	IP_FIFO_DATA	Data structure of the access management (No user access required)
MAILBOX	ARRAY [1..16] OF BYTE	Mailbox: data area for module data exchange

4.13. IP_FIFO_DATA

IP_FIFO_DATA data structure:

Data field	Data Type	Description
X	ARRAY [1..128] OF BYTE	FIFO memory with registered ID's

Y	ARRAY [1..128] OF BYTE	Number of entries per ID's
ID	BYTE	Last assigned ID (highest ID)
MAX_ID	BYTE	Maximum number of applications per ID
INIT	BOOL	Initialization performed
EMPTY	BOOL	FIFO is empty
FULL	BOOL	FIFO is full (should not happen!)
TOP	INT	Maximum number of entries in FIFO
NW	INT	write-index FIFO
NR	INT	read-index FIFO

4.14. LOG_CONTROL

us_LOG_VIEWPORT data structure:

Name	Type	Properties
NEW_MSG	STRING(STRING_LENGTH)	New Message - Text
NEW_MSG_OPTION	DWORD	New message - Option BYTE 3: Reserve BYTE 2: Level BYTE 1: Backcolor BYTE 0: Frontcolor
LEVEL	BYTE	Given log level
SIZE	INT	Size of the array (maximum index)
RESET	BOOL	Reset / delete the entries
ENABLE	BOOL	Clearance to submit notifications
PRINTF	ARRAY[1.11] OF STRING(STRING_LENGTH)	Parameter data for PRINT_SF block
MSG	ARRAY[0.N] OF STRING(STRING_LENGTH)	Array for message - text
MSG_OPTION	ARRAY[0.N] OF DWORD	Array ofor messages - Option BYTE 3: Reserve BYTE 2: Level BYTE 1: Back Color BYTE 0: Color front

UPDATE_COUNT	UINT	Update-counter (increased with each new message)
IDX	INT	Current Issue Index
RING_MODE	BOOL	BUFFER enabled overflow / Ringmode enab-led

4.15. PRINTF_DATA

PRINTF_DATA data structure:

Data field	**Data Type**	Description
PRINTF	ARRAY [1..11] OF STRING(LOG_SIZE)	Array for passing parameters

4.16. UNI_CIRCULAR_BUFFER_DATA

The Structure UNI_CIRCULAR_BUFFER_DATA is used for data management for the module UNI_CIRCULAR_BUFFER

UNI_CIRCULAR_BUFFER_DATA:

Data field	**Data Type**	**Description**
D_MODE	INT	MODE (command number)
D_HEAD	WORD	Header information Read / Write
D_STRING	STRING(STRING_LENGTH)	STRING Read / Write
D_REAL	REAL	REAL Read / Write
D_DWORD	DWORD	DWORD Read / Write
BUF_SIZE	UINT	Number of bytes in the buffer
BUF_COUNT	UINT	Number of elements in the buffer
BUF_USED	USINT	Level (0-100%)
BUF	NW_BUF_LONG	Data BUFFER
_GetStart	UINT	Internal: read pointer

_GetEnd	UINT	Internal: read pointer
_Last	UINT	Intern: Data pointer
_First	UINT	Intern: Data pointer

4.17. VMAP_DATA

VMAP_DATA data structure:

Name	Type	Properties
FC	DWORD	function code: release bit mask
V_ADR	INT	Virtual Address Range: Start address
V_SIZE	INT	Virtual address space: number of WORD
P_ADR	INT	Real address space: Start address
TIME_OUT	TIME	Watchdog

4.18. XML_CONTROL

XML_CONTROL data structure:

Data field	Data Type	Description
COMMAND	WORD	Control commands (binary occupancy)
START_POS	UINT	(Buffer index of first character)
STOP_POS	UINT	(Buffer-index of the last characters)
COUNT	UINT	Element number
TYPE	INT	Type code of the current element
LEVEL	UINT	Current hierarchy / level
PATH	STRING(STRING_LENGTH)	Hierarchy as TEXT (PATH)
ELEMENT	STRING(STRING_LENGTH)	current item as TEXT

ATTRIBUTE	STRING(STRING_LENGTH)	Current attributes as TEXT
VALUE	STRING(STRING_LENGTH)	Current value as TEXT
BLOCK1_START	UINT	Start position of block 1
BLOCK1_STOP	UINT	Stop position of Unit 1
BLOCK2_START	UINT	Start position of block 2
BLOCK2_STOP	UINT	Stop position of Unit 2

4.19. WORLD_WEATHER_DATA

WORLD_WEATHER_DATA data structure:

Name	**Type**	**Properties**
CUR	WORLD_WEATHER_CUR	Current weather conditions
DAY	ARRAY [0..4] OF WORLD_WEATHER_DAY	Next 5 days of weather forecast

WORLD_WEATHER_CUR data structure:

Name	**Type**	**Properties**
OBSERVATION_TIME	STRING(8)	Observation time (UTC)
TEMP_C	INT	Temperature (°C)
WEATHER_CODE	INT	Unique Weather Code
WEATHER_DESC	STRING(30)	Weather description text
WIND_SPEED_MILES	INT	Wind speed in miles per hour
WIND_SPEED_KMPH	INT	Wind speed in kilometre per hour
WIND_DIR_DEGREE	INT	Wind direction in degree
WIND_DIR16POINT	STRING (3)	16-Point wind direction compass
PRECIPMM	REAL	Precipitation amount in millimetre

HUMIDITY	INT	Humidity (%)
VISIBILITY	INT	Visibility (km)
PRESSURE	INT	Atmospheric pressure in milibars
CLOUDOVER	INT	Cloud cover (%)

WORLD_WEATHER_DAY data structure:

Name	**Type**	**Properties**
DATE_OF_DAY	STRING (10)	Date for which the weather is forecasted
TEMP_MAX_C	INT	Day temperature in °C(Celcius)
TEMP_MAX_F	INT	Day temperature in °F(Fahrenheit)
TEMP_MIN_C	INT	Night temperature in °C(Celcius)
TEMP_MIN_F	INT	Night temperature in °F(Fahrenheit)
WIND_SPEED_MILES	INT	Wind Speed in mph (miles per hour)
WIND_SPEED_KMPH	INT	Wind Speed in kmph (Kilometer per hour)
WIND_DIR_DEGREE	INT	Wind direction in degree
WIND_DIR16POINT	STRING (3)	16-Point wind direction compass
WEATHER_CODE	INT	A unique weather condition code
WEATHER_DESC	STRING(30)	Weather description text
PRECIPMM	REAL	Precipitation Amount (millimetre)

4.20. YAHOO_WEATHER_DATA

YAHOO_WEATHER data structure:

Name	Type	Properties

TimeToLive	INT	Time to Live: how long in minutes this feed should be cached
location_city	STRING (40)	The location of this forecast: city: city name
location_region	STRING(20)	The location of this forecast: region: state, territory, or region, if given
location_country	STRING(20)	The location of this forecast: country:
unit_temperature	STRING (1)	temperature: degree units, for f c for Celsius Fahrenheit or
unit_distance	STRING(2)	distance: distance units for MI for miles or km for kilo-meters
unit_pressure	STRING(2)	pressure: barometric pressure units of, for in pounds per square inch or mb for milli bars
unit_speed	STRING (3)	speed: units of speed, mph for miles per hour or kilo-meters per hour for kph
wind_chill	INT	Forecast information about wind chill in degrees
wind_direction	INT	Forecast information about wind direction in degrees
wind_speed	REAL	Forecast information about wind speed, in the units (mph or kph)
atmosphere_humidity	INT	Forecast information about current atmospheric humid-ity: humidity, in percent
atmosphere_pressure	INT	Forecast information about current atmospheric pres-sure: barometric pressure, in the units (in or mb)
atmosphere_visibility	REAL	Forecast information about current atmospheric visibil-ity, in the units (mi or km)
atmosphere_rising	INT	Forecast Information about rising: state of the baromet-ric pressure: Steady (0), rising (1), or falling (2). (In-teger: 0, 1, 2)
astronomy_sunrise	STRING (10)	sunrise: today's sunrise time. The time is a string in a local time format of "h: mm am / pm"
astronomy_sunset	STRING (10)	sunset: today's sunset time. The time is a string in a loc al time format of "h: mm am / pm"
geo_latitude	REAL	The latitude of the location
geo_longitude	REAL	The longitude of the location
cur_conditions_temp	INT	cur_conditions_text

cur_conditions_text	STRING (40)	The current weather conditions: text: a textual description of conditions
cur_conditions_code	INT	The current weather conditions: code: the code for this condition forecast
forcast_today_low_temp	INT	The weather conditions today forcast: the forecasted low temperature for this day in the units (f or c)
forcast_today_high_temp	INT	The forcast today weather conditions: the forecasted high temperature for this in the day units (f or c)
forcast_today_text	STRING (40)	The forcast today weather conditions: text: a textual description of conditions
forcast_today_code	INT	The current weather conditions: code: the code for this condition forecast
forcast_tomorrow_low_temp	INT	The forcast tomorrow weather conditions: the forecasted low temperature for this day in the units (f or c)
forcast_tomorrow_high_temp	INT	The forcast tomorrow weather conditions: the forecasted high temperature for this day in the units (f or c)
forcast_tomorrow_text	STRING (40)	The forcast tomorrow weather conditions: text: a textual description of conditions
forcast_tomorrow_code	INT	The current weather conditions: code: the code for this condition forecast

Condition Codes:

The fields cur_conditions_code, forcast_today_code and forcast_tomorrow_code describe the weather in text form by " Condition Codes "

Value	Description
0	tornado
1	tropical storm
2	hurricane
3	severe thunderstorms
4	Temp
5	mixed rain and snow
6	mixed rain and sleet
7	mixed snow and sleet

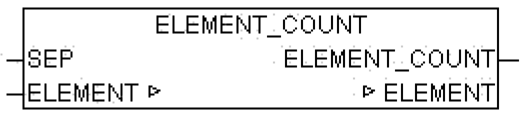
8	freezing drizzle
9	drizzle
10	freezing rain
11	showers
12	showers
13	Snow Flurries
14	Light snow showers
15	blowing snow
16	snow
17	hail
18	sleet
19	20
20	foggy
21	haze
22	smoky
23	blustery
24	windy
25	cold
26	cloudy
27	mostly cloudy (night)
28	mostly cloudy (day)
29	partly cloudy (night)
30	partly cloudy (day)
31	clear (night)
32	sunny
33	fair (night)
34	fair (day)
35	mixed rain and hail

36	hot
37	isolated thunderstorms
38	scattered thunderstorms
39	scattered thunderstorms
40	scattered showers
41	heavy snow
42	scattered snow showers
43	heavy snow
44	mostly cloudy
45	46
46	snow showers
47	isolated thundershowers
3200	not available

5. Other Functions

5.1. ELEMENT_COUNT

Type	Function: INT
Input	SEP: BYTE (separation character of the elements)
I / O	ELEMENT: STRING(ELEMENT_LENGTH) (input list)
Output	INT (number of items in the list)



ELEMENT_COUNT determines the number of items in a list.

If the parameter ELEMENT is an empty string 0 is passed as result. If at least one character is in ELEMENT it is evaluated as a single element and ELEMENT_COUNT = 1 is passed to output.

Examples:

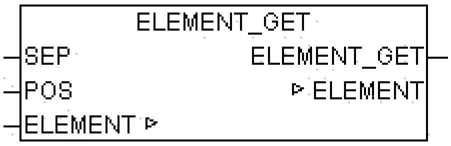
ELEMENT_COUNT('0,1,2,3',44) = 4

ELEMENT_COUNT('',44) = 0

ELEMENT_COUNT('x',44) = 1

5.2. ELEMENT_GET

Type	Function: STRING(ELEMENT_LENGTH)
Input	SEP: BYTE (separation character of the elements)
	POS: INT (of the item)
I / O	ELEMENT: STRING(ELEMENT_LENGTH) (input list)
Output	STRING (String output)



ELEMENT_GET passes the item at the position POS from a list. The list consists of strings which are separated by the separation character SEP. The first element of the list has the position 0

Examples:

ELEMENT_GET('ABC,23,,NEXT', 44, 0) = 'ABC'

ELEMENT_GET('ABC,23,,NEXT', 44, 1) = '23'

ELEMENT_GET('ABC,23,,NEXT', 44, 2) = ''

ELEMENT_GET('ABC,23,,NEXT', 44, 3) = 'NEXT'

ELEMENT_GET('ABC,23,,NEXT', 44, 4) = ''

ELEMENT_GET('', 44, 0) = ''

5.3. NETWORK_VERSION

Type	Function: DWORD
Input	IN : BOOL (if TRUE the module provides the release date)
Output	(Version of the library)

NETWORK_VERSION provides if IN = FALSE the current version number as DWORD. If IN is set to TRUE then the release date of the current version as a DWORD is returned.

Example: NETWORK_VERSION(FALSE) = 111 for version 1.11

DWORD_TO_DATE(NETWORK_VERSION(TRUE)) = 2011-2-3

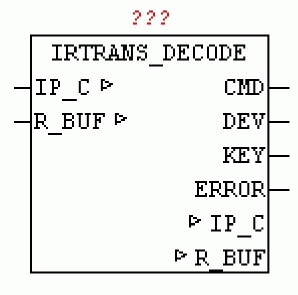
6. Device Driver

6.1. IRTRANS

The module IRTRANS_? provide an interface for infrared Transmitter Company IRTrans GmbH. IRTrans offers transmitter for RS232 and TCP/IP, all of which can be operated with the following driver components. The basic connection to RS232 or TCP/IP must be made with the appropriate manufacturer routines. The interface modules rely on a Buffer Interface to which provides in a Buffer (Array of Byte) data and in a Counter the length of the data packet in bytes. The IRTrans devices learn the IR key codes and translate them in ASCII Strings using a configurable database. With the Ethernet variant, this Strings then sent over UDP and can be received from a PLC and be evaluated. Thus, for example, the blinds are automatically shut down when someone turns on the TV without this additional action would be necessary. The PLC can listen in this manner any number of remote controls in different areas and derive appropriate actions from it. Conversely, of course, the release of key codes on the Transmitter modules is possible.

6.2. IRTRANS_DECODE

Type	Function module
I/O	IP_C: data structure 'IP_CONTROL ' (Parameterization)
	R_BUF: data structure NETWORK_BUFFER_SHORT '
	(Receive data)
Output	CMD: BOOL (TRUE if valid data are present at the output)
	DEV: STRING (name of the remote control)
	KEY: STRING (name of the key codes)
	ERROR: BOOL (TRUE if a invalid data packet is present)



IRTRANS_DECODE receives the data from the module IRTRANS_SERVER present in BUFFER, checks if a valid data package is available and decodes the name of the remote control and the name of the button form the data packet. If a valid data packet has been decoded, the name of the remote control is passed at the output DEV and the name of the button on the output KEY. The output CMD signals that the new output data are present. The ERROR output is then set when a data packet was received that is not in the correct format.

The format is defined as follows:

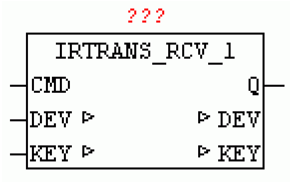
'Name of the remote control', 'Name of the key code' RN

A data packet consists of the name of the remote control, followed by a comma and then the name of the key codes. The data packet is a completed by Carriage Return and a Line Feed .

To ensure that IRTRANS_DECODE works in the IRTrans configuration the Check box BROADCAST IR RELAY must be checked and in the corresponding Device database under the DEFAULT ACTION the String '%r%c\r\n' must be registered. IRTRANS_DECODE evaluates just this String and decodes %r as the name and %c as pressed a button of the remote control.

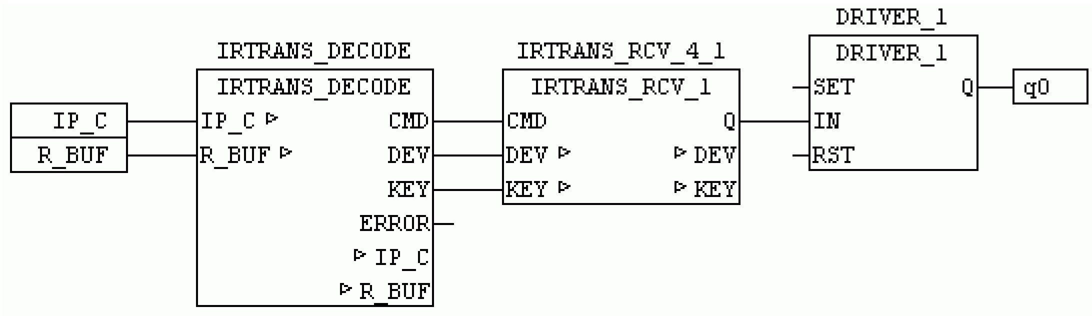
6.3. IRTRANS_RCV_1

Type	Function module
Input	CMD : BOOL (TRUE if data for evaluating are available)
I/O	DEV: STRING (name of the remote control)
	KEY: string (name of button)
Setup	DEV_CODE: STRING (to be decoded remote control name)
	KEY_CODE: STRING (key code to be decoded)
Output	Q: BOOL (output)



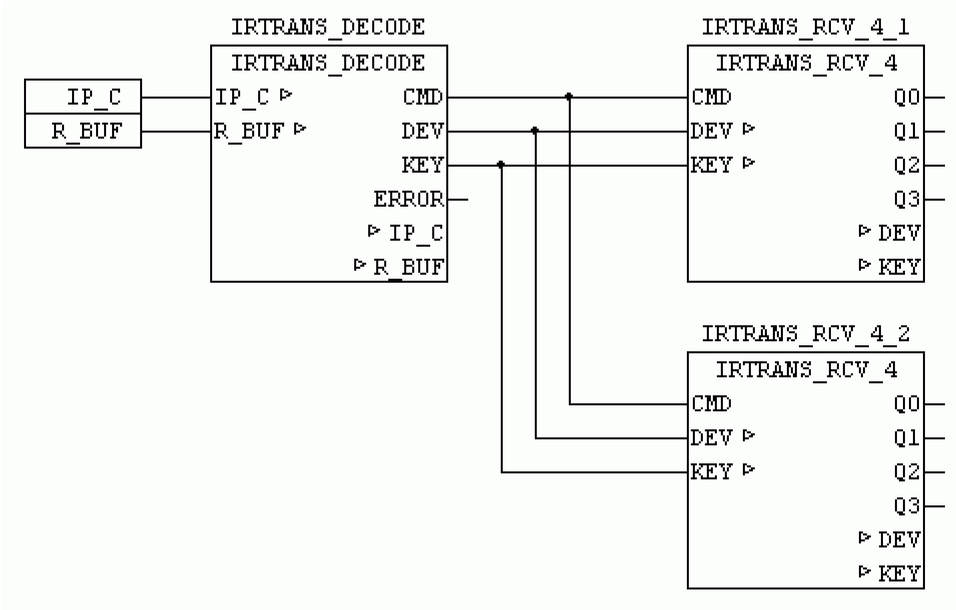
IRTRANS_RCV_1 checkes when CMD = TRUE if the string matches the input DEV corresponds to DEV_CODE (device code) and the string at the input KEY corresponds to the KEY_CODE. If the codes match and CMD = TRUE, then the output Q for a cycle is set to TRUE.

The following example shows the application of IRTRANS_RCV_1:



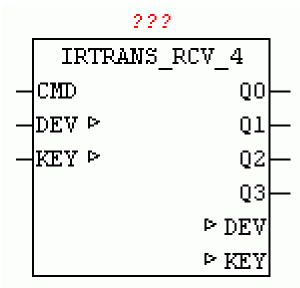
In this example, the receive data buffer to IRTRANS_DECODE is passed. The decoder determines from the valid data packets String DEV and KEY and passes them with CMD to IRTRANS_RCV_1 . IRTRANS_RCV_1 or alternatively IRTRANS_RCV_4 and IRTRANS_RCV_ checks whether DEV and KEY match and then switches the output Q for a cycle to TRUE. in the example a DRIVER_1 is controlled which enables the remote control to switch the output with each received log.

If multiple Key Codes are to be evaluated alternatively the modules IR-TRANS_RCV_4 or IRTRANS_RCV_8 can be used or more of these modules can be used in parallel mode.



6.4. IRTRANS_RCV_4

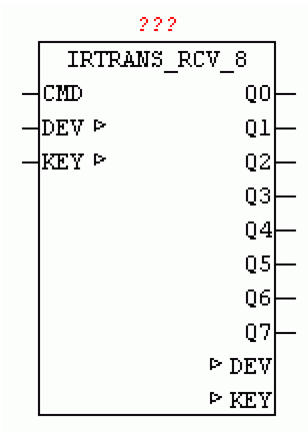
Type	Function module
Input	CMD: BOOL (TRUE if data for evaluating are available)
I/O	DEV: STRING (name of the remote control)
	KEY: string (name of button)
Setup	DEV_CODE: STRING (to be decoded remote control name)
	KEY_CODE_0..3: STRING (key code to be decoded)
Output	Q0..Q3: BOOL (output)



IRTRANS_RCV_4 checkes when CMD = TRUE if the string matches the input DEV corresponds to DEV_CODE (device code) and the string at the input KEY corresponds to the KEY_CODE. If the codes match and CMD = TRUE, then the output Q for a cycle is set to TRUE. For more information about the function of the device are under IRTRANS_RCV_1.

6.5. IRTRANS_RCV_8

Type	Function module
Input	CMD : BOOL (TRUE if data for evaluating are available)
I/O	DEV: STRING (name of the remote control)
	KEY: string (name of button)
Setup	DEV_CODE: STRING (to be decoded remote control name)
	KEY_CODE_0..7: STRING (key code to be decoded)
Output	Q0..Q7: BOOL (output)



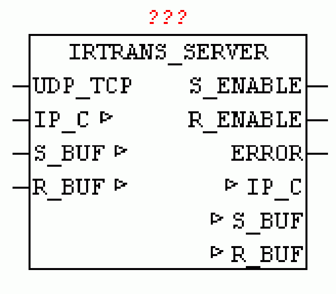
IRTRANS_RCV_8 checkes when CMD = TRUE if the string matches the input DEV corresponds to DEV_CODE (device code) and the string at the input KEY corresponds to the KEY_CODE. If the codes match and CMD = TRUE, then the output Q for a cycle is set to TRUE. For more information about the function of the device are under IRTRANS_RCV_1.

6.6. IRTRANS_SERVER

Type	Function module
Input	UDP_TCP : BOOL (FALSE = UDP / TRUE = TCP)
In_Out	IP_C: data structure 'IP_CONTROL ' (Parameterization)
	S_BUF: data structure 'NETWORK_BUFFER_SHORT'
	(Transmit data)
	R_BUF: data structure NETWORK_BUFFER_SHORT '
	(Receive data)
Output	S_ENABLE: BOOL (release IRTRANS data send)

R_ENABLE: BOOL (IRTRANS data receive enabled)

ERROR: DWORD (Error code: Check IP_CONTROL)

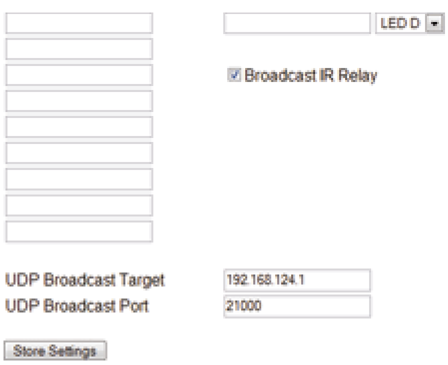


IRTRANS_SERVER can be used as both a receiver and a transmitter of IRTRANS commands. Is UDP_TCP = TRUE is a passive TCP connection, otherwise set up a passive UDP connection. The type of operation must also be configured with IRTRANS device. Once a data connection is available and sending commands is allowed, S_ENABLE = TRUE. In UDP mode, after the initial data received from IRTRANS, data can be sent, since in the passive mode, the UDP-IP parameter is initially not known. The receiving mode is indicated with R_ENABLE. If data are received they are available in R_BUF for further processing for other modules. Send data has to be entered by the modules in the S_BUF, so they are then sent automatically from IRTRANS_SERVER. If transmission errors occurs, they are issued with "ERROR" (see module IP_CONTROL2). Existing errors are acknowledged automatically every 5 seconds by the module.

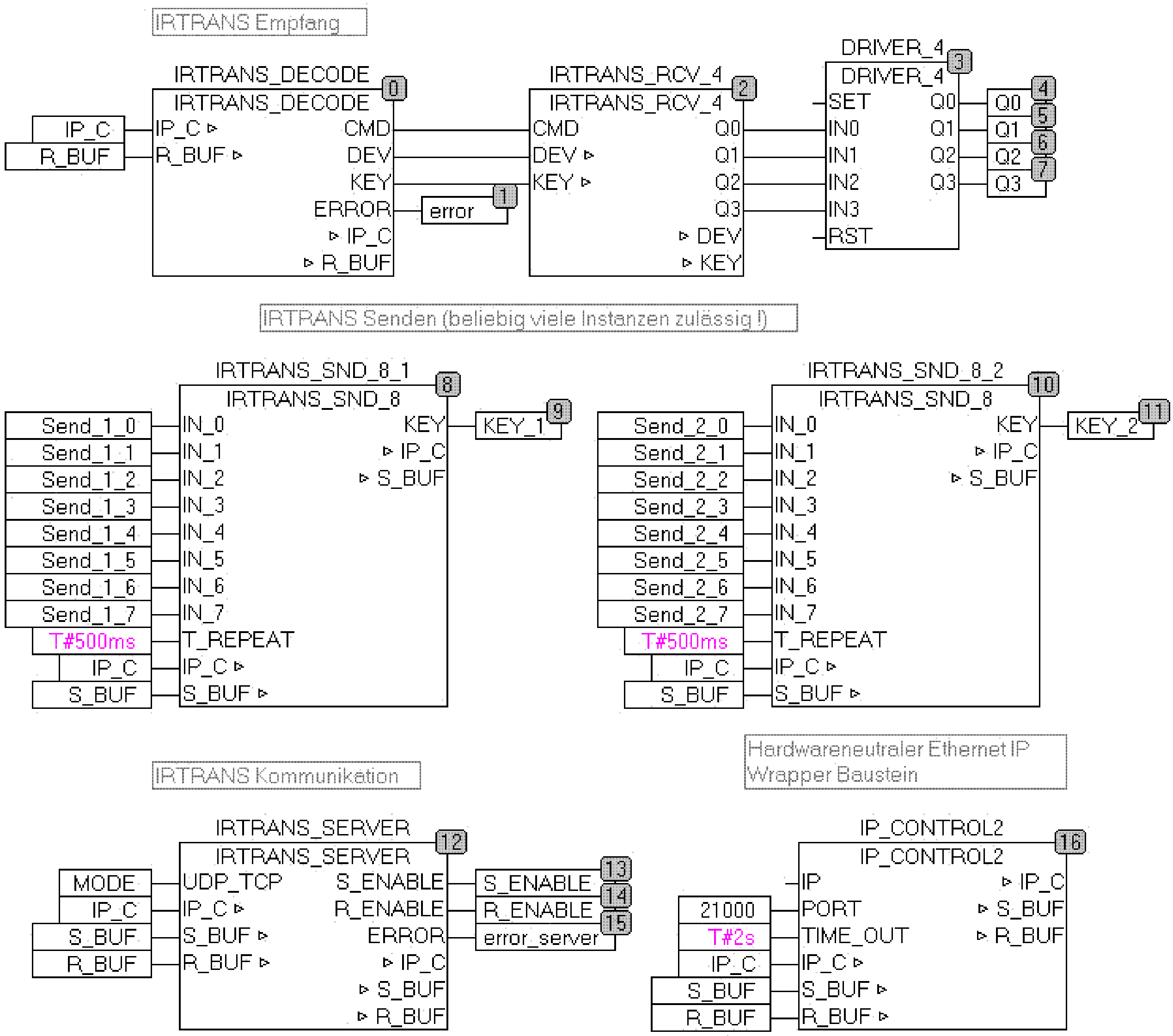
UDP server mode:

In the IRTRANS Web configuration, the IP address of the PLC is entered as a broadcast address.

IRTRANS Web Configuration:



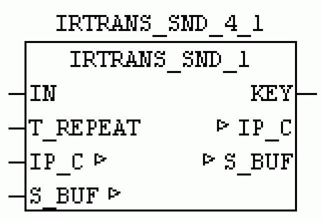
The following example shows the application of IRTRANS Devices



6.7. IRTRANS_SND_1

Type	Function module
Input	IN: BOOL (TRUE = Send key code)
	T_REPEAT: TIME (time to re-send the key code)
I/O	IP_C: data structure 'IP_CONTROL ' (Parameterization)

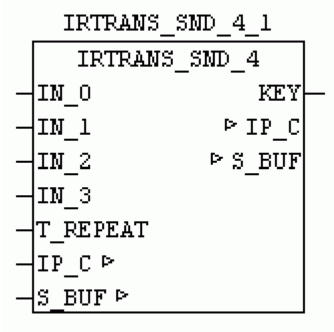
	S_BUF: data structure 'NETWORK_BUFFER_SHORT' (Transmit data)
Setup	DEV_CODE: STRING (to be decoded remote control name)
	KEY_CODE: STRING (key code to be decoded)
Output	KEY: BYTE (output of the currently active key codes)



IRTRANS_SND_1 allows you to send a remote command to the IRTrans. If IN TRUE the specified device and key code in setup is sent to the IRTrans which outputs in turn as a real remote control commands. With T_REPEAT the repeat time for sending can be specified . If IN remains constant to TRUE so always this key code sent repeated after the time T_REPEAT. At output KEY in active control "1" is passed. KEY = 0 means that the IN is not active.

6.8. IRTRANS_SND_4

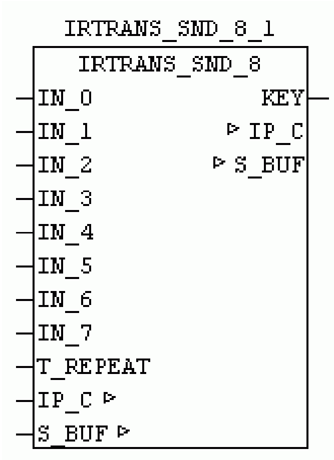
Type	Function module
Input	IN_0..3 : BOOL (TRUE = Send keycode x)
	T_REPEAT: TIME (time to re-send the key code)
I/O	IP_C: data structure 'IP_CONTROL ' (Parameterization)
	S_BUF: data structure 'NETWORK_BUFFER_SHORT' (Transmit data)
Setup	DEV_CODE: STRING (to be decoded remote control name)
	KEY_CODE_0..3: STRING (key code to be sent)
Output	KEY: BYTE (output of the currently active key codes)



IRTRANS_SND_4 allows users to send remote control commands to the IRTrans. If IN_x is TRUE the specified device and key code in setup is sent to the IRTrans which outputs in turn as a real remote control commands. With T_REPEAT the repeat time for sending can be specified . If IN_0 remains constant to TRUE so always this key code sent repeated after the time T_REPEAT. If a change to a different IN_x occures this code will send immediately and then again delayed with T_REPEAT, if it remains a long period of time. At output KEY the currently controlled KEY will be displayed. KEY = 0 means that no IN_x is active. The values 1-3 are the IN_0 - IN_3.

6.9. IRTRANS_SND_8

Type	Function module
Input	IN_0..7 : BOOL (TRUE = Send keycode x)
	T_REPEAT: TIME (time to re-send the key code)
I/O	IP_C: data structure 'IP_CONTROL ' (Parameterization)
	S_BUF: data structure 'NETWORK_BUFFER_SHORT' (Transmit data)
Setup	DEV_CODE: STRING (to be decoded remote control name)
	KEY_CODE_0..7: STRING (key code to be sent)
Output	KEY: BYTE (output of the currently active key codes)



IRTRANS_SND_8 allows users to send remote control commands to the IRTrans. If IN_x is TRUE the specified device and key code in setup is sent to the IRTrans which outputs in turn as a real remote control commands. With T_REPEAT the repeat time for sending can be specified . If IN_0 remains constant to TRUE so always this key code sent repeated after the time T_REPEAT. If a change to a different IN_x occures this code will send immediately and then again delayed with T_REPEAT, if it remains a long period of time. At output KEY the currently controlled KEY will be displayed. KEY = 0 means that no IN_x is active. The values 1-3 are the IN_0 - IN_7.

7. Data Logger

7.1. DATA-LOGGER

The data logger modules enable the collection and storage of process data in real time. After triggering the storage pulse all parameterized process values are stored in a data buffer, as various storage media are often not fast enough. Up to 255 process values are processed in one package. The calling order of the modules determines automatically the ranking of the process values (take care of data-flow order)

Fore storing the various data types, the following modules are provided.

DLOG_STRING

DLOG_REAL

DLOG_DINT

DLOG_DT

Other data types convert first manually, and transferred as STRING. The collected data can then be forwarded to a data target.

DLOG_STORE_FILE_CSV (data store in CSV file in control module)

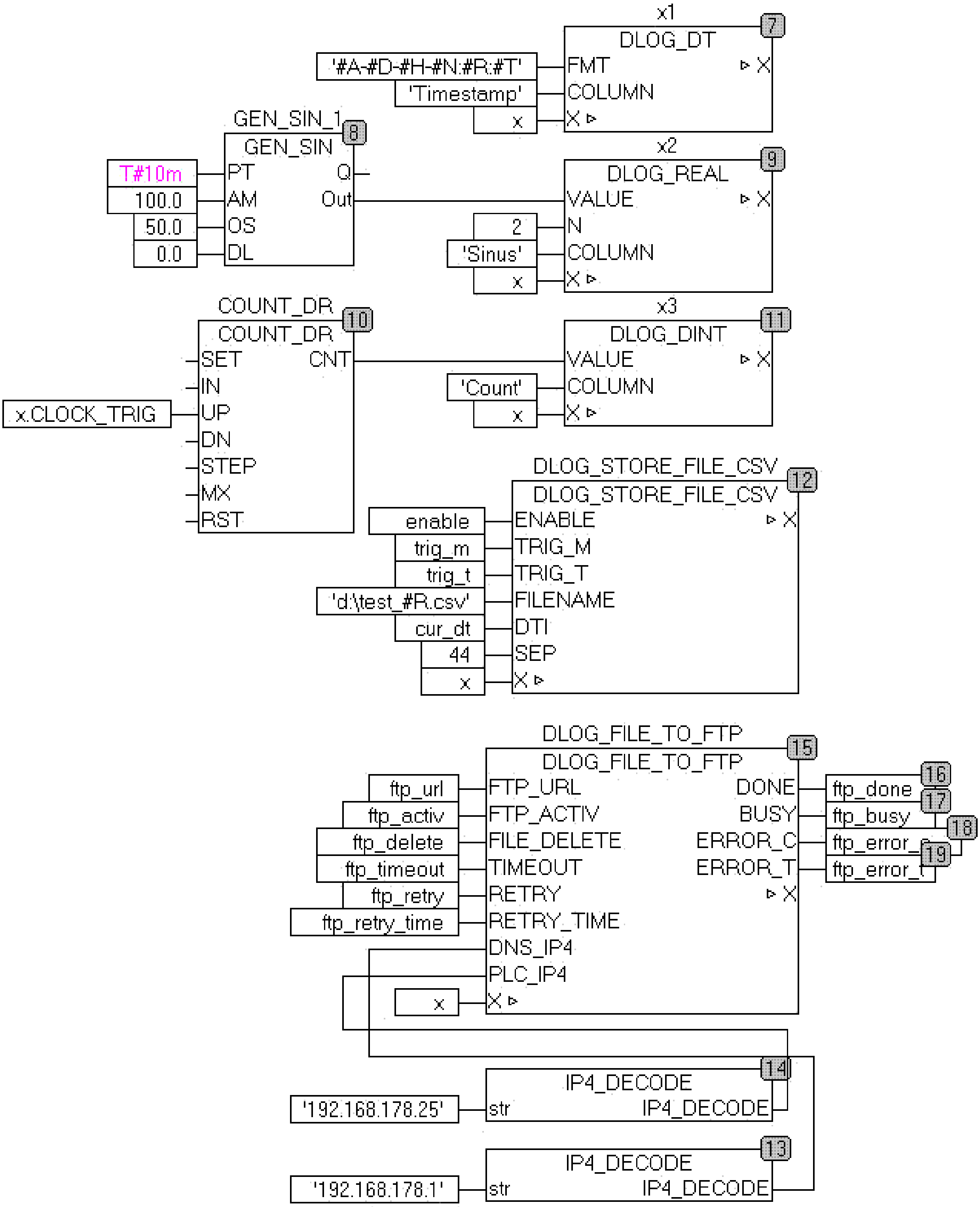
DLOG_STORE_RRD (data to store RRD server)

The files that are stored on the controller can then be forwarded to external data targets.

DLOG_FILE_TO_SMTP (File as Email)

DLOG_FILE_TO_FTP (copy file to an external FTP server)

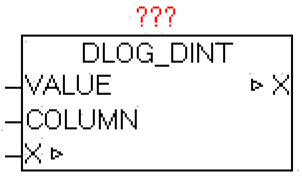
The following example shows the recording of a time stamp, a REAL and DINT counter. Here, the process data is stored after each minute in a new CSV formatted file. Once a file is ready, it will be moved automatically to an FTP server.



7.2. DLOG_DINT

Type	Function module:
IN_OUT	X: DLOG_DATA (DLOG data structure)
INPUT	VALUE: DINT (process value)

COLUMN: STRING (process value name)



The block DLOG_DINT is for logging (recording) of a process value of type DINT, and can only be used in combination with a DLOG_STORE_* module, as this coordinates of the data structure X to record the data. In recording formats that support a process value name, such as at DLOG_STORE_FILE_CSV at "COLUMN" a name can be provided.

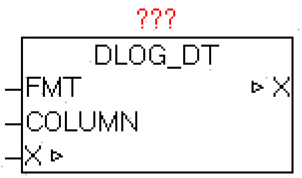
7.3. DLOG_DT

Type Function module:

IN_OUT X: DLOG_DATA (DLOG data structure)

INPUT FMT: STRING (formatting parameters)

COLUMN: STRING (process value name)



The module DLOG_DT is for logging (recording) of a date or time value of type STRING, and can only be used in combination with a DLOG_STORE_* module, as this coordinates the record the data by the data structure X. Using FMT parameter, the formatting will be set. In the FMT parameter can also be combined with normal text formatting parameters. See the documentation from the module DT_TO_STRF.
In recording formats that support a process value name, such as at DLOG_STORE_FILE_CSV at "COLUMN" a name can be provided.

Example:

FMT := '#A-#D-#H-#N:#R:#T'

resuls '2011-12-22-06:12:50'

7.4. DLOG_REAL

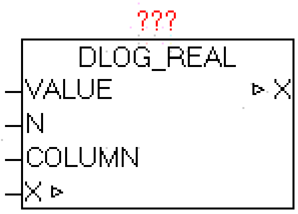
Type Function module:

IN_OUT X: DLOG_DATA (DLOG data structure)

INPUT VALUE: REAL (process value)

N: INT (number of decimal places)

COLUMN: STRING (process value name)



The block DLOG_REAL is for logging (recording) of a process value of type REAL, and can only be used in combination with a DLOG_STORE_* module, as this coordinates of the data structure X to record the data. Using parameter N defines the number of desired decimal places. See the documentation from the module REAL_TO_STRF.
In recording formats that support a process value name, such as at DLOG_STORE_FILE_CSV at "COLUMN" a name can be provided.

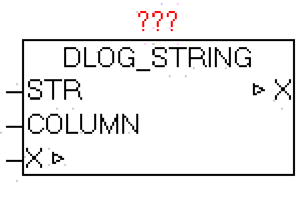
7.5. DLOG_STRING

Type Function module:

IN_OUT X: DLOG_DATA (DLOG data structure)

INPUT STR: STRING (process value)

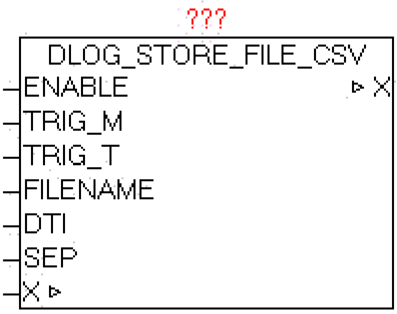
COLUMN: STRING (process value name)



The module DLOG_DINT is for logging (recording) of a process value of type DINT, and can only be used in combination with a DLOG_STORE_* module, as this coordinates of the data structure X to record the data. In recording formats that support a process value name, such as at DLOG_STORE_FILE_CSV at "COLUMN" a name can be provided.

7.6. DLOG_STORE_FILE_CSV

Type	Function module:
IN_OUT	X: DLOG_DATA (DLOG data structure)
INPUT	ENABLE: BOOL (release data recording)
	TRIG_M: BOOL (manual trigger)
	TRIG_T: UDINT (automatic trigger over time)
	FILE NAME: STRING (file name)
	DTI: DT (Current DATE-TIME value)
	SEP: BYTE (separator of the recorded elements)



The module DLOG_STORE_FILE_CSV is for logging (recording) of the process values in a CSV formatted file. The data can be passed with the modules DLOG_DINT, DLOG_REAL, DLOG_STRING, DLOG_DT. The parameter TRIG_M (positive pulse) is used to manually trigger (start) the storage of process data. With Parameters TRIG_T an automatic time-controlled release can be realized. If the current date / time value divided by the parameterized TRIG_T value with residual value is 0, then a Save is performed.
This also ensures that the store is always performed at the same time

Examples:

TRIG_T = 60

every 60 sec at each new minute in second 0 a store is performed.

TRIG_T = 10

In second 0,10,20,30,40,50 a store is performed.

TRIG_T = 3600

At after each new hour at minute 0 and second 0 a store is performed.

The triggers TRIG_T and TRIG_M can be used in parallel independent of each other.

With parameters FILENAME the file name (including path if necessary) is defined. If the filename is changed during the recprding, it will automatically on-the-fly changed to the new record file (with no data loss). This change can also be automated. The parameter FILE NAME supports the use of date / time parameter (see documentation from the module DT_TO_STRF)

Example: FILE NAME = 'Station_01_#R.csv'

At position of '#R' automatically the current minute number is entered. This means that automatically every minute the file name changes, and therefore the data is written into the file. Thus, within an entire hour 60 files are created and filled with data, and in the ring buffer manner overwritten again and again.

A recording can be done automatically and creates every day, week, month, etc. a new file as desired. If a new FILE NAME is detected, a possibly existing file is erased and rewritten.

On DTI parameters, the current date / time value has to be transferred. In SEP the ASCII code of the delimiter is given.

CSV file format:

See: http://de.wikipedia.org/wiki/CSV_(Dateiformat)

Example of a CSV delimited file ';' and column headings

Date / Time;Z1;Z2;seconds

2010-10-22-06:00:00;1;2;00

2010-10-22-06:00:06;1;2;06

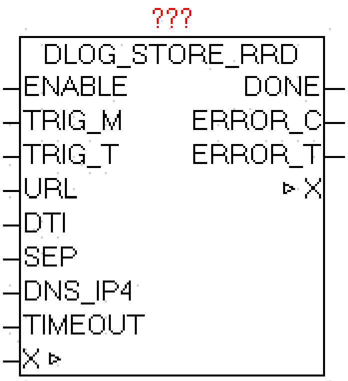
2010-10-22-06:00:12;1;2;12

2010-10-22-06:00:18;1;2;18

7.7. DLOG_STORE_RRD

Type	Function module:
IN_OUT	X: DLOG_DATA (DLOG data structure)

INPUT	ENABLE BOOL (Enable data recording)
	TRIG_M: BOOL (manual trigger)
	TRIG_T: UDINT (automatic trigger over time)
	URL: STRING(string_length) (URL address of the server)
	DTI: DT (Current DATE-TIME value)
	SEP: BYTE (separator of the recorded elements)
	Dns_ip4: DWORD (IP address of the DNS server)
	TIMEOUT: TIME (monitoring time)
OUTPUT	DONE: BOOL (Data transfer completed without error)
	ERROR_C: DWORD (Error code)
	ERROR_T: BYTE (Problem type)



The module DLOG_STORE_RRD serves for logging (recording) of the process values in an RRD database. The data can be passed with the modules DLOG_DINT, DLOG_REAL, DLOG_STRING, DLOG_DT. The parameter TRIG_M (positive pulse) is used to manually trigger (start) the storage of process data. With Parameters TRIG_T an automatic time-controlled release can be realized. If the current date / time value divided by the parameterized TRIG_T value with residual value is 0, then a Save is performed.
This also ensures that the store is always performed at the same time

Examples:

TRIG_T = 60

every 60 sec at each new minute in second 0 a store is performed.

TRIG_T = 10

In second 0,10,20,30,40,50 a store is performed.

TRIG_T = 3600

At after each new hour at minute 0 and second 0 a store is performed.

The triggers TRIG_T and TRIG_M can be used in parallel independent of each other.

On DTI parameters, the current date / time value has to be transferred. In SEP the ASCII code of the delimiter is given.

If an error occurs during the query it is reported in ERROR_C in combination with ERROR_T.

ERROR_T:

Value	Properties
1	The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	The exact meaning of ERROR_C can be read at module HTTP_GET
3	ERROR_C = 1: Data from the RRD-Server (PHP script) are not adopted.
4	ERROR_C = 1: The data could be passed in the URL string. Number of parameters or reduce the amount of data (URL + data <= 250 characters)

With the parameter URL, the access path and the php-script-call is passed.

An example URL:

http://my_servername/myhouse/rrd/test_rrd.php?rrd_db=test.rrd&value=

DNS server or IP address

Access path and name of the php-script

php-script parameter 1 = Database Name

php-script parameter 2 = Process values

The module automatically copies all process values behind "&value="

so that then the following data (example) used

http://my_servername/myhouse/rrd/test_rrd.php?rrd_db=test.rrd&value=10:20:30:40:50:60:70

The individual process data are, using the parameter SEP (separator), separated from each other.

It is important that the passed URL string and the process data are not longer than 250 characters.

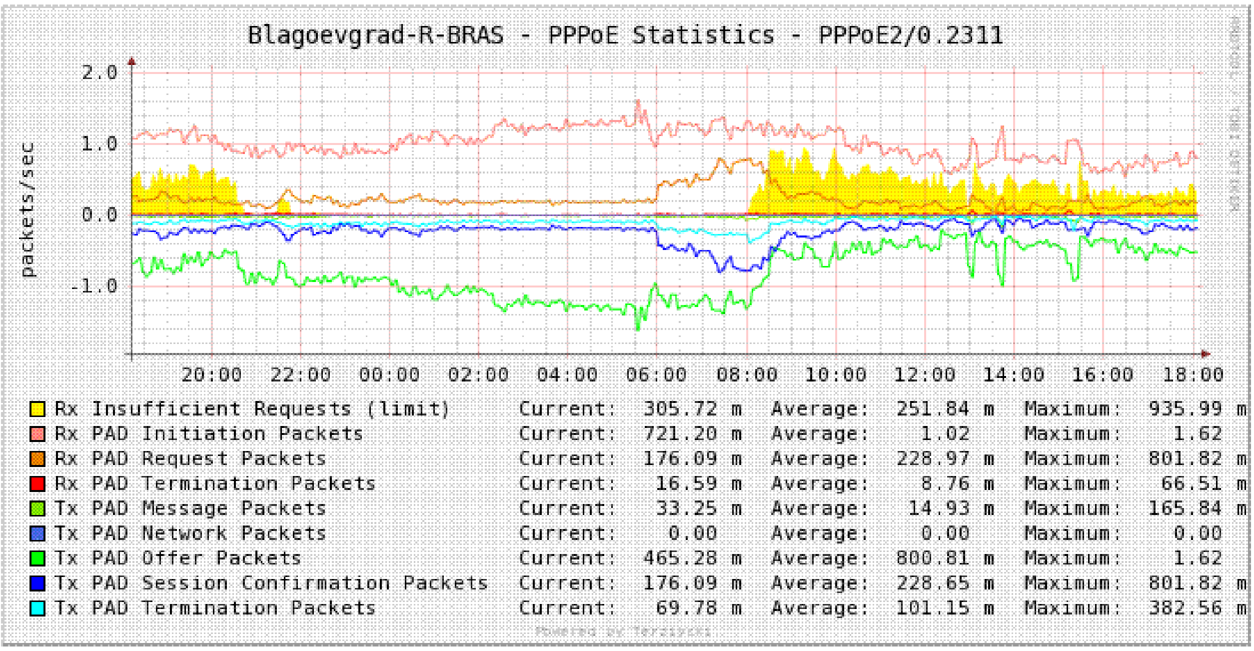
The structure of the URL is only an example, and can in principle be designed with own free server applications and scripts in conjunction with their.

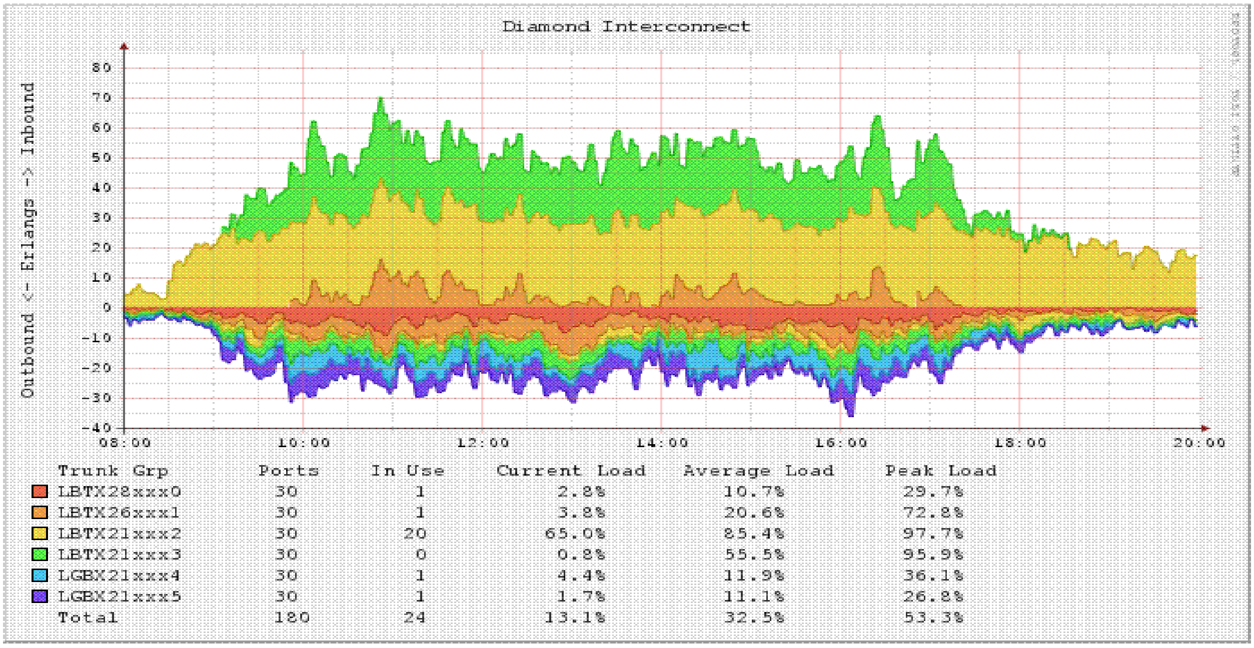
What are the possibilities for and benefits rrdtool

rrdtool is a program that saves the time-related measurement data, and summarizes and visualizes the data. The program was originally developed by Tobias Oetiker and under the GNU General Public License (GPL). By publishing a free software now many other authors new functionality and bug fixes have contributed. rrdtool is available as source code and an executable program for many operating systems.

Source: http://de.wikipedia.org/wiki/RRDtool

Sample graphs:





Source - http://www.mrtg.org/

What is required: hardware, software, tools, etc.

PLC with Network OSCAT-lib

A power-saving PC for the duration of operation (24/7).

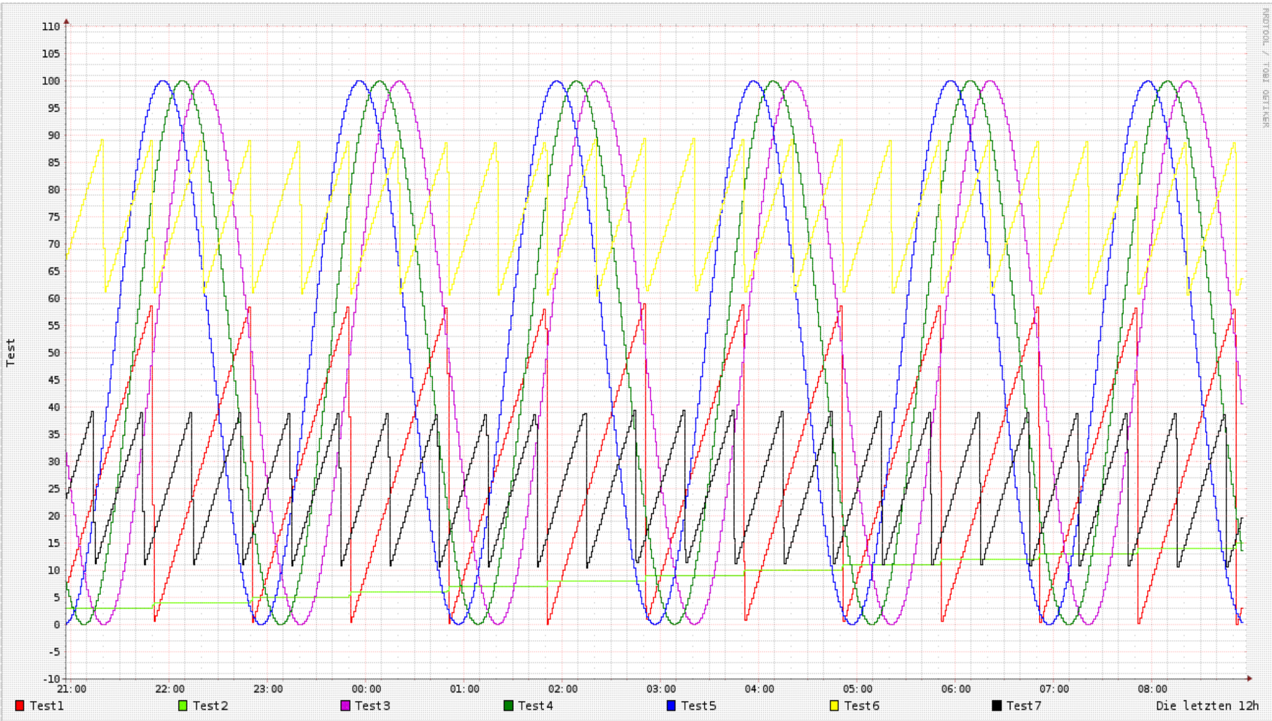
On the PC, rrdtol and the php scripts are installed.

The scripts have been developed on a Linux-Xubuntu-PC with PHP.

Quick Start:

- A sample program with some values may be recorded, is found the OSCAT network.lib under demo/DLOG_RRD_DEMO.
- The rrdtool installation on Xubuntu (DEBIAN) PC is processed either with the Synaptic package manager and select install rrdtool, or in the console with "apt-get install rrdtool".
- Script: create_test_rrd_db.php = Creates a new rrd database, and once must be adapted if necessary.
- Script: test_rrd.php = This script is called by the PLC with the OSCAT function module via HTTP-GET. Must usually are not adjusted, and outputs an error code. If error-free, then output is a "0".
- Script: chart_test.php = Script to create the charts from the rrd-DB and display it on a website. Must usually are not adjusted, and outputs an error code. If error-free is then output a "0".
- Download the three php scripts in the folder to a PC, ie./ var / www / rrd / and do not forget to adjust the appropriate access rights.

The demo program in conjunction with the demo php scripts create the following data or graphic.



Links

http://www.mrtg.org/rrdtool/

http://de.wikipedia.org/wiki/RRDtool

http://www.rrze.uni-erlangen.de/dienste/arbeiten-rechnen/linux/howtos/rrdtool.shtml

http://arbeitsplatzvernichtung-durch-outsourcing.de/marty44/rrdtool.html

php-script - Examples / Templates

create_test_rrddb.php

```
#!/etc/php5/cli -q
<?php
error_reporting(E_ALL);
# ==================================
# Creates a rrd database
# Is called once from the console
# 12/11/2010 by NetFritz
# ==================================
# Create wp.rrd creates the database test.rrd
# - Step 60 all 60 sec, a value is expected
# DS:t1:GAUGE:120:0:100 a data source named t1 is created
#    the type is gauge. It is waiting 120sec for new data, if not,
#    the data is written into the database as UNKNOWN.
#    the minimum and maximum reading
# RRA:AVERAGE:0.5:1:2160 this is the rrd-Archiv  AVERAGE=average 0.5= ave-
rage interval deviation
#    36h archive every minute, a value, 1:2160=1h =36h 3600sec*3600=129 600
1Minute =60seconds every minute a value, 129600/60 = 2160 Entries
# RRA:AVERAGE:0.5:5:2016 1week archive all 5minutes 1value, 3600*24*7 days
= 604800Sec / (5 minutes +60 sec = 2016 entries
# RRA: AVERAGE: 1Values 0.5:15:2880 30Days archive all 15minutes,
# RRA: AVERAGE: 1 year 0.5:60:8760 archive all 60Minuten a value
# It is now starting
$ Command = "rrdtool create test.rrd \
         - Step 60 \
         DS:t1:GAUGE:120:0:100 \
         DS:t2:GAUGE:120:0:100 \
         DS:t3:GAUGE:120:0:100 \
         DS:t4:GAUGE:120:0:100 \
         DS:t5:GAUGE:120:0:100 \
         DS:t6:GAUGE:120:0:100 \
         DS:t7:GAUGE:120:0:100 \
         RRA:AVERAGE:0.5:1:2160 \
         RRA:AVERAGE:0.5:5:2016 \
         RRA:AVERAGE:0.5:15:2880 \
         RRA:AVERAGE:0.5:60:8760";
```

```
system($command);
?>
```

test_rrd.php

```
<?php
# Called by control with
# Http://mein_server/test_rrd.php?rrd_db=test.rrd&value=10:20:30:40:50:60
   $ Rrd_db = urldecode($ _GET['rrd_db']);  # Name of the RRD database
   $ Value = urldecode($ _GET['value']);   # Submitted values
   # $ array_values = explode(":",$value);
   # echo "$rrd_db <br>";
   # print_r($array_value);
   # echo "<br>";
   $commando = "/usr/bin/rrdtool update " . $rrd_db . " N:" . $value;
   system($commando,$fehler);
   echo $fehler . $commando;
?>
```

chart_test_rrd.php

```
<?php
/ / Create chart for test scores, and is invoked by the browser
$command="/usr/bin/rrdtool graph test0.png \
          --vertical-label=Test \
          --start end-12h \
          --width 600 \
          --height 200 \
          --alt-autoscale \
DEF:t1=test.rrd:t1:AVERAGE \
DEF:t2=test.rrd:t2:AVERAGE \
DEF:t3=test.rrd:t3:AVERAGE \
DEF:t4=test.rrd:t4:AVERAGE \
DEF:t5=test.rrd:t5:AVERAGE \
DEF:t6=test.rrd:t6:AVERAGE \
DEF:t7=test.rrd:t7:AVERAGE \
LINE1:t1#FF0000:Test1 \
LINE1:t2#6EFF00:Test2 \
LINE1:t3#CD04DB:Test3 \
```

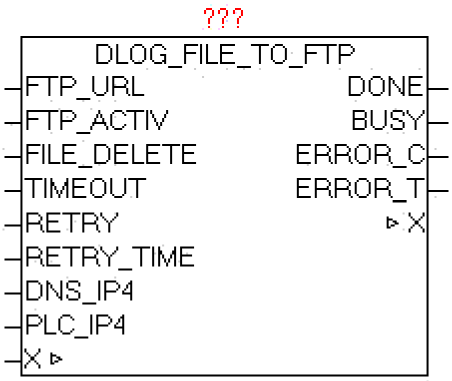
```
LINE1:t4#008000:Test4 \
LINE1:t5#0000FF:Test5 \
LINE1:t6#0000FF:Test6 \
LINE1:t7#0000FF:Test7 \
          COMMENT: 'The last 12 hours' ";
system($command);

echo "<!DOCTYPE HTML PUBLIC \"-//W3C//DTD XHTML 1.0 Transitional//EN\"
          \"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd\">\n";
echo "<html xmlns=\"http://www.w3.org/1999/xhtml\">\n";
echo "  <head>\n";
echo "    <title>Test</title>\n";
echo "  </head>\n";
echo "  <body>\n";
echo ("<center><img src='test0.png'></center>\n");
echo "    <center>Die letzten 12h</center>\n";
# Echo "Error =". $fehler;
echo "  </body>\n";
echo "</html>\n";
?>
```

7.8. DLOG_FILE_TO_FTP

Type	Function module:
IN_OUT	X: DLOG_DATA (DLOG data structure)
INPUT	FTP_URL: STRING(STRING_LENGTH) (FTP access path)
	FTP_ACTIV : BOOL (PASSIV = 0 / ACTIV = 1)
	FILE_DELETE: BOOL (delete files after transfer)
	TIMEOUT: TIME (time)
	RETRY: INT (number of repetitions)
	RETRY_TIME: TIME (waiting period before repetition)
	Dns_ip4: DWORD (IP4 address of the DNS server)

	Dns_ip4: DWORD (IP4 address of the DNS server)
OUTPUT	DONE: BOOL (Transfer completed without error)
	BUSY: BOOL (Transfer active)
	ERROR_C: DWORD (Error code)
	ERROR_T: BYTE (Problem type)



The module DLOG_FILE_TO_FTP is used to automatically transfer the by DLOG_STORE_FILE_CSV generated from files to an FTP-server. The FTP_URL parameter contains the name of the FTP server and optionally the user name and password, an access path and an additional port number for the data channel. If no Username or password is transferred, the device automatically tries to register as "Anonymous". The parameter FTP_ACTIV determined whether the FTP server is operated in active or passive mode. In the ACTIV mode, the FTP server tries to establish the data channel for control, however these may cause problems by security software, firewall, etc. because it could block the connection request. For this purpose, in the firewall a corresponding exception rule has to be defined. In the passive mode, this problem is alleviated since the controller establishes the connection, and can easily pass through the firewall. The control channel is always set up on port 20, and the data channel via standard PORT21, but this is in turn is depending whether active or passive mode is used, or optional PORT number in the FTP-URL is specified. With the parameter FILE_DELETE can be determined whether the source file should be deleted after successful transfer. This works on FTP and even on the control side. In specifying FTP directories the behavior depends on FTP server, whether they exist in this case or are created automatically. Normally, these should be already available. The size of files is no limit per se, but there are practical limits: Space on PLC, FTP storage and the transmission time. With dns_ip4 the IP address of the DNS server must be specified, if in the FTP URL a DNS name is given, alternatively, an IP address can be entered in the FTP URL. At parameters PLC_IP4 the own IP addresses has to be supplied. If errors occur during transmission these are passed to the output ERROR_C and ERROR_T. As long as the transfer is running, BUSY =

TRUE, and after an error-free completion of the operation, DONE = TRUE. Once a new transfer is started, DONE, ERROR_T and ERROR_C are reseted.

If parameter RETRY = 0, then the FTP transfer was repeated until it completes successfully. If RETRY state at a value > 0, the FTP transfer is just as often repeated in transmission failure. Then this job is simply discarded and the process continues with the next file. With RETRY-TIME the waiting time between the repetitions can be defined.

The module has integrated the IP_CONTROL and must not be externally linked to this, as it by default would be necessary.

Background: http://de.wikipedia.org/wiki/File_Transfer_Protocol

URL examples:

ftp://username:password@servername:portnummer/directory/

ftp://username:password@servername

ftp://username:password @ servername / directory /

ftp://servername

ftp://username:password@192.168.1.1/directory/

ftp://192.168.1.1

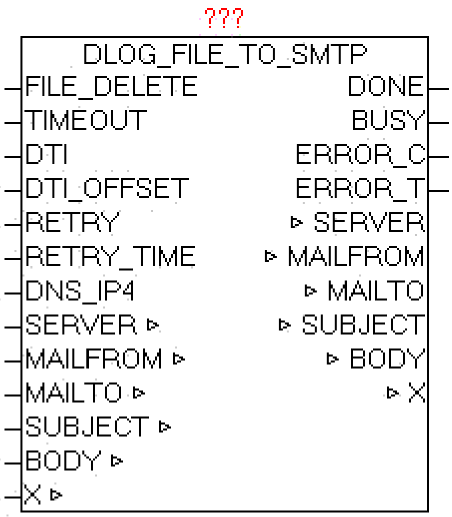
ERROR_T:

Value	Properties
1	Problem: DNS_CLIENT The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	Problem: FTP control channel The exact meaning of ERROR_C can be read at module IP_CONTROL
3	Problem: FTP data channel The exact meaning of ERROR_C can be read at module IP_CONTROL
4	Problem: FILE_SERVER The exact meaning of ERROR_C can be read at block FILE_SERVER
5	Problem: END - TIMEOUT ERROR_C contains the left WORD of the step number, and the right WORD has the response code received by the FTP server. The parameters must be considered first as a HEX value, divided into two WORDS, and then be considered as a decimal value. Example: ERROR_T = 5 ERROR_C = 0x0028_00DC End-step number 0x0028 = 40 Response-Code 0x00DC = 220

7.9. DLOG_FILE_TO_SMTP

Type	Function module:
IN_OUT	SERVER: STRING (URL of the SMTP server)
	MAIL FROM: STRING (return address)
	MAILTO: STRING (string_length) (recipient address)
	SUBJECT: STRING (subject text)
	SUBJECT: STRING (subject text)
	FILES: STRING (string_length) (files to be sent)
	X: DLOG_DATA (DLOG data structure)

INPUT	FILE_DELETE: BOOL (delete files after transfer)
	TIMEOUT: TIME (time)
	DTI: DT (current date-time)
	DTI_OFFSET: INT (time zone offset from UTC)
	RETRY: INT (number of repetitions)
	RETRY_TIME: TIME (waiting period before repetition)
	Dns_ip4: DWORD (IP4 address of the DNS server)
OUTPUT	DONE: BOOL (Transfer completed without error)
	BUSY: BOOL (Transfer active)
	ERROR_C: DWORD (Error code)
	ERROR_T: BYTE (Problem type)



The module DLOG_FILE_TO_SMTP is used to automatically transfer the of DLOG_STORE_FILE_CSV generated files as e-mail to an e-mail server.

The module uses internally the SMTP_CLIENT for sending.

The SERVER parameter contains the name of the SMTP server and optionally the user name and password and a port number. If you pass a user name and password, the procedure is according to standard SMTP.

SERVER: URL Examples:

username:password@smtp_server

username:password@smtp_server:portnumber

smtp_server

Special case:

If in the username is a '@' included this must be passed as '%' - character, and is then automatically corrected by the module again.

By specifying user and password the Extend-SMTP is used, and automatically the safest possible Authentication method is used. If parameter is to specify the MAIL FROM sender address:

i.e. oscat@gmx.net

Optionally, an additional "Display Name" be added This is displayed the email client automatically instead of the real return address. Therefore, always an easily recognizable name to be used.

i.e.. oscat@gmx.net;Station_01

The email client shows as the sender then "Station_01". Thus, more people will use the same email address but send a own "Alias".

At the MAILTO parameter can To, Cc, Bc be specified. The different groups of recipients are specified by '#' as the separator in a list. Multiple addresses within the same group are divided with the separator ";" . In each group can be defined unlimited count of recipients, the only limitation is the length of the mailto string.

To;To..#Cc;Cc...#Bc;Bc...

Examples.

o1@gmx.net;o2@gmx.net#o1@gmx.net#o2@gmx.net

defines two TO-addresses, one CC-address and a Bc-address

##o2@gmx.net

defines only one BC-address.

With subject, a subject text will be specified, as well as with BODY an email text content. The current Date / Time value must be definded at DTI, and at DTI_OFFSET the correction value as an offset in minutes from UTC (Universal Time). If the DTI in UTC time is passed, at DTI_OFFSET a 0 must be passed.

The monitoring time can be specified with parameter TIMEOUT. At dns_ip4 must be specified the IP address of the DNS server, if in SERVER a DNS name is specified. If errors occur during the transmission, they are passed at ERROR_C and ERROR_T. As long as the transfer is running, BUSY = TRUE, and after an error-free completion of the operation, DONE = TRUE. Once a new transfer is started, DONE, ERROR_T and ERROR_C are reseted.

If parameter RETRY = 0, then the SMTP transfer was repeated until it completes successfully. If RETRY state at a value > 0, then the SMTP transfer is just as often repeated at transmission failure. Then this job is simply discarded and the process continues with the next file. With RETRY-TIME the waiting time between the repetitions can be defined.

The parameter FILE_DELETE = TRUE a file is deleted on the controller after successful transfer via email.

The module has integrated the IP_CONTROL and so must not be externally linked to this, as it would be at default necessary.

Basics:

http://de.wikipedia.org/wiki/SMTP-Auth

http://de.wikipedia.org/wiki/Simple_Mail_Transfer_Protocol

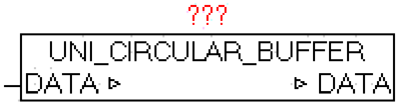
ERROR_T:

Value	Properties
1	Problem: DNS_CLIENT The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	Problem: SMTP Channel The exact meaning of ERROR_C can be read at module IP_CONTROL
4	Problem: FILE_SERVER The exact meaning of ERROR_C can be read at block FILE_SERVER
5	Problem: END - TIMEOUT ERROR_C contains the left WORD the end of the step number, and in the right WORD the last response code received by the SMTP server. The parameters must be considered first as a HEX value, divided into two WORDS, and then be considered as a decimal value. Example: ERROR_T = 5 ERROR_C = 0x0028_00FA End-step number 0x0028 = 40 Response-Code 0x00DC = 250

7.10. UNI_CIRCULAR_BUFFER

Type Function module:

IN_OUT DATA: UNI_CIRCULAR_BUFFER_DATA (data storage)



The module UNI_CIRCULAR_BUFFER is a ring buffer in the FIFO (first in - first out) principle, and can process any data as a byte stream.

For this purpose, in the data structure UNI_CIRCULAR_BUFFER_DATA all can be processed.

The following commands are supported on DATA.D_MODE.

01 Element to write to buffer

10 Element of Buffer read but not to remove

11 The above command 10 read with item is removed.

99 Buffer is reset. All data is deleted

With DATA.D_HEAD (WORD) in the right byte can be provided the element type, and in the left byte optional an additional user ID.

D_HEAD = LEFT-BYTE (ADD-Ino), RIGHT-BYTE (Type ID)

Type codes:

1 = STRING (For DATA.D_STRING, the string must be provided)

2 = REAL (For DATA.D_REAL, the REAL value is passed)

3 = DWORD (In the DWORD the DATA.D_DWORD must be passed)

X = header information without data

In DATA.BUF_SIZE the number of bytes output, to show the dropped items in total. With DATA.BUF_COUNT the number of in Buffer contained elements is provided. And on BUF_USED will issue the occupancy of the buffer as a percentage value.

When an item is written in the buffer, and the required free space (memory) does not exist, after calling the module, the DATA.D_MODE remains unchanged. The command was successfu only if D_MODE contains 0 after module call.

When reading elements, the same operation is essential.

Only if D_MODE subsequently is 0, in D_HEAD the data type can be found, and if necessary, the data from D_STRING, D_REAL, D_DWORD can be read. After successful reading step, the deletion of the element to be performed with command 11.

Example: Writing element:

```
DATA.D_MODE: = 1; (* command to write data *)
DATA.D_HEAD: =  1; (* element-type = STRING *)
DATA.D_STRING:= 'This is the text';
module-call()
if DATA.D_MODE = 0, then the element was successfully saved
```

Example: Reading element:

```
DATA.D_MODE:= 10; (* read command * element)
module-call()

result
DATA.D_HEAD =  1; (* Element-Type = STRING *)
DATA.D_STRING = 'This is the text';
DATA.D_MODE = 0
```

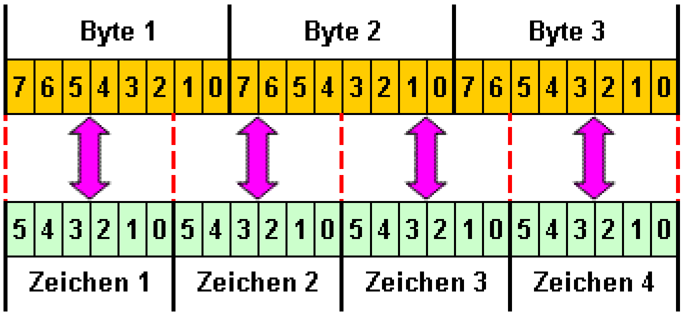
Example: Delete element:

```
DATA.D_MODE:= 11; (* command * delete item)
module-call()
DATA.D_MODE = 0; (* item was deleted *)
```

8. Converter

8.1. BASE64

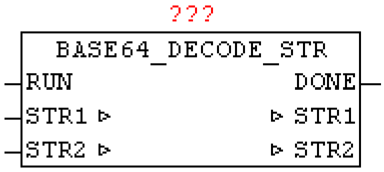
The BASE64 encoding is a process for Encoding of 8-bit binary data into a string consisting of only 64 globally available ASCII characters. Application is HTTP Basic Authentication, PGP signatures and keys, and MIME encoding for e-mail. To enable the SMTP protocol as the easy transport of binary data, a conversion is necessary, as foreseen in the original version, only 7-bit ASCII characters.



When encoding always three bytes of byte stream (24 bit) are devided in 6-bit blocks. Each of these 6-bit blocks results in a number between 0 and 63. This results in the following 64 printable ASCII characters [A-Z] [a-z] [0-9], [+/]. The encoding increases the space requirements of the data stream by 33%, from 3 characters each to 4 characters each. If the length of the coding is not divisible by 4, filler characters will be appended at the end. In this case the sign "=" is used .

8.2. BASE64_DECODE_STR

Type	Function module
Input	RUN: BOOL (positive edge starts conversion)
Output	DONE: BOOL (TRUE if conversion is completed)
I/O	STR1: STRING(192) (text in BASE64 format)
	STR2: STRING(144) (converted normal text)



With a BASE64_DECODE_STR encoded in BASE64 text can be converted back to plain text. With a positive edge of RUN the process starts. Here DONE is immediately reseted, if it has been set by a previous conversion. The BASE64 encoded text is passed on STR1, and after the conversion the plain text is available in STR2, and DONE is set to TRUE.

Example:

Text in STR1

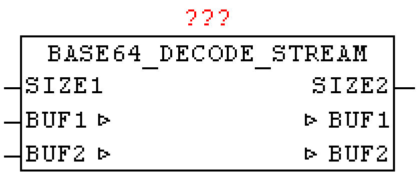
'T3BlbiBTb3VyY2UgQ29tbXVuaXR5IGZvciBBdXRvbWF0aW9uIFRlY2hub2xv-Z3k='

Result in STR2

Text in STR2 = 'Open Source Community for Automation Technology'

8.3. BASE64_DECODE_STREAM

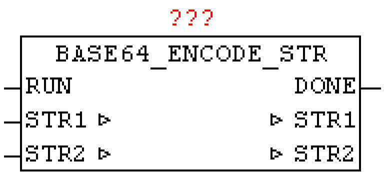
Type	Function module
Input	SIZE1: INT (number of bytes in the BUF1 for decode)
Output	SIZE2: INT (number of bytes in BUF2 of the decoded results)
I/O	BUF1: ARRAY [0..63] OF BYTE (BASE64 data for conversion)
	BUF2: ARRAY [0..47] OF BYTE (converted data)



With BASE64_DECODE_STREAM arbitrarily long BASE64 byte streams are decoded. In one pass, up to 64 bytes are decoded, which in turn emerged from a maximum of 48 bytes each. Here, the source data is passed to the decoder over BUF1 in the data-stream manner as individual blocks of data, and in decoded form re-issued in BUF2. The user has to provide the further processing of the BUF2 data before the next block of data is converted. The number of bytes in BUF2 is issued by SIZE2 from the module.

8.4. BASE64_ENCODE_STR

Type	Function module
Input	RUN: BOOL (positive edge starts conversion)
Output	DONE: BOOL (TRUE if conversion is completed)
I/O	STR1: STRING (144) (Text to convert)
	STR2: STRING (192) (converted text in BASE64 format)



With BASE64_ENCODE_STR a standard text can be converted to a BASE64 encoded text. With a positive edge of RUN the process starts. Here DONE is immediately reseted, if it has been set by a previous conversion. The BASE64 encoded text is passed on STR1, and after the conversion the BASE64 text is available in STR2, and DONE is set to TRUE.

Example:

Text in STR1 = 'Open Source Community for Automation Technology'

Result in STR2

'T3BlbiBTb3VyY2UgQ29tbXVuaXR5IGZvciBBdXRvbWF0aW9uIFRlY2hub2xv-
Z3k='

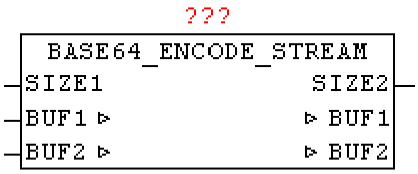
8.5. BASE64_ENCODE_STREAM

Type	Function module
Input	SIZE1: INT (number of bytes in the BUF1 to encode)

Output SIZE2: INT (number of bytes in the encoded BUF2 results)

I/O BUF1: ARRAY [0..47] OF BYTE (data to convert)

BUF2: ARRAY [0..63] OF BYTE (BASE64 converted data)



With BASE64_ENCODE_STREAM arbitrarily long byte data stream according to BASE64 can be encoded. In one pass, up to 48 bytes are converted, in turn, result more than 64 bytes. Here, the source data is passed to the encoder over BUF1 in the data-stream manner as individual blocks of data, and in coded form re-issued in BUF2. The user has to provide the further processing of the BUF2 data before the next block of data is converted. The number of bytes in BUF2 is issued by SIZE2 from the module.

8.6. HTML_DECODE

Type Function : STRING(string_length)

Input IN: STRING(String)

Output STRING(string_length) (string)

HTML_DECODE converts reserved characters which are in the form &name; stored HTML code, in the original character. In addition, all coded characters are converted into the corresponding ASCII code. Special characters can be represented by the following string in HTML:

- &#NN, where NN represents the position of the character within the character map in decimal notation.

- &#xNN, or &#XNN where NN represents the position of the character within the character table in hexadecimal notation.

&name; Special characters have names like € for €.

The reserved characters in HTML are:

& Is encoded as &

> Is encoded as >

< Is encoded as <

" is coded as "

Examples:

HTML_DECODE('1 ist >als 0') = '1 is > als 0';

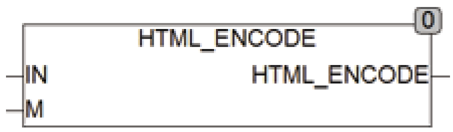
HTML_DECODE('&#D79;&#D83;&#D67;&#D65;&#D84;') = 'OSCAT';

HTML_DECODE('&#xH4F;&#xH53;&#xH43;&#xH41;&#xH54;') = 'OSCAT';

HTML_DECODE('&#XH4F;&#XH53;&#XH43;&#XH41;&#XH54;') = 'OSCAT';

8.7. HTML_ENCODE

Type	F unction : STRING(string_length)
Input	IN: STRING(String)
	M: BOOL (mode)
Output	STRING(string_length) (string)



Html_encode converts in HTML reserved characters to form &Name;. If the input M is set to TRUE also all the characters with the code 160-255 and 128 are implemented in the &Name convention.

Caution should be exercised in the use of character sets because they are not the same on all systems and deviations are common in special characters. Thus, for example, not all systems the € character at position 128 in the character map.

The reserved characters in HTML are:

& Is encoded as &

> Is encoded as >

< Is encoded as <

" is coded as "

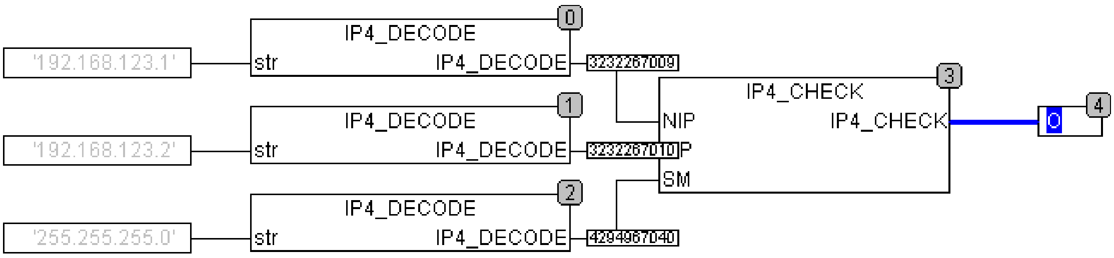
Html_encode converts the string '1 > than 0 'into '1 is > than 0'.

8.8. IP4_CHECK

Type	Function : BOOL
Input	NIP: DWORD (network IP address)
	LIP: DWORD (Local IP address)
	SM: DWORD (Subnet Mask)
Output	BOOL (TRUE if NIP and LIP are in the same Subnet)

IP4_CHECK checks if a network address of the NIP and the local address LIP are in the same Subnet lie. Both addresses will be first masked with the Subnet mask and then tested for equality. Only the bits which are in the Subnet Mask TRUE are examined for equality. The network addresses must correspond to the IPv4 format and presented as a DWORD. If IP addresses must be tested that are String they are to be converted to DWORD before.

The following example shows 2 IP addresses and a Subnet Mask as String are tested after appropriate conversion to DWORD there. The output is TRUE because both addresses are in the same Subnet .



8.9. IP4_DECODE

Type	Function : DWORD
Input	STR: STRING(15) (string that contains the IP address)
Output	DWORD (decoded IP v4 address)

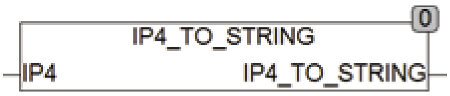
IP4_DECODE decodes the in STR stored string as a IP v4 address and returns it as a DWORD. A return of 0 means an invalid address or an address of '0.0.0.0 ' was evaluated. IP4 may be used for evaluating a Subnet Mask of the IP v4 format.

8.10. IP4_TO_STRING

Type Function : STRING(15)

Input IP4: BOOL (string that contains the IP address)

Output DWORD (decoded IP v4 address)



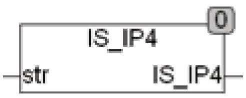
IP4_TO_STRING converts the IP4 address stored as DWORD in a string. The format is 'NNN.NNN.NNN.NNN'.

8.11. IS_IP4

Type Function : BOOL

Input STR: STRING (string to be tested)

Output BOOL (TRUE if STR contains a valid IP v4 address)



IS_IP4 checks if the string str contains a valid IP v4 address, if not FALSE is returned. A valid IP v4 address consists of 4 numbers from 0 - 255 and they are separated each with one point. The address 0.0.0.0 is there classified as wrong.

IS_IP4(0.0.0.0) = FALSE

IS_IP4(255.255.255.255) = TRUE

IS_IP4(256.255.255.255) = FALSE

IS_IP4(0.1.2.) = FALSE

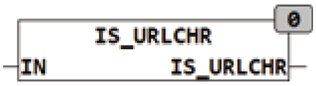
IS_IP4(0.1.2.3.) = FALSE

8.12. IS_URLCHR

Type Function : BOOL

Input IN: STRING (string to be tested)

Output BOOL (TRUE if STR contains a valid IP v4 address)



IS_URLCHR checks if the string contains only valid characters for a URL encoding. If the string contains reserved characters it returns FALSE, otherwise TRUE.

For a URL following characters are valid:

[A..Z]
[a..z]
[0..9]
[-._~]

all other characters are reserved or not allowed.

8.13. MD5_AUX

Type Function: DWORD

Input N: INT (internal use)

A: DWORD (Internal use)

B: DWORD (Internal use)

C: DWORD (internal use)

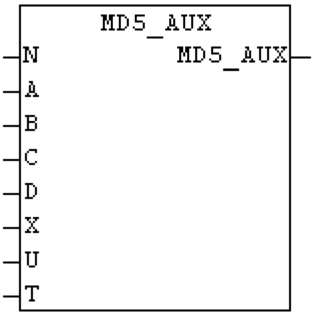
D: DWORD (internal use)

X: DWORD (Internal use)

U: INT (internal use)

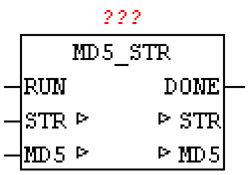
T: DWORD (Internal use)

At the MD5 hash generation several cycles through the complex mathematical calculations which are processed. Thus, the amount of redundant code in the module MD5_STREAM remains small, periodic calculations have been displaced as a macro in the MD5_AUX. This module has only in conjunction with the block MD5_STREAM a useful application.



8.14. MD5_STR

Type	Function module
Input	RUN: BOOL (Positive edge starts the calculation)
Output	DONE: BOOL (TRUE if calculations are complete)
	MD5: ARRAY[0..15] OF BYTE (current MD5 hash)
I/O	STR: STRING(string_length) (Text for HASH creation)



With MD5_STR a string of the MD5 hash can be calculated by. In the STR a string is passed to the module, and a positive edge at input "RUN", the calculation starts. DONE is immediately reset at startup, and after the process is DONE is set to TRUE. Then, at the parameter HASH the actual calculated HASH value is available. (See module MD5-STREAM).

Example:

the MD5 hash of 'OSCAT' is 30f33ddb9f17df7219e1acdea3386743

8.15. MD5_STREAM

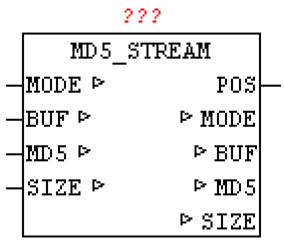
Type	Function module
I/O	MODE: INT(mode: 1 = init / 2 = Data Block / 3 = Complete)

BUF: ARRAY[0..63] OF BYTES (source data)

MD5: ARRAY [0..15] OF BYTE (current MD5-HASH)

SIZE: UDINT(number of data)

Output POS: UDINT(start address of the requested data block)



The module MD5_STREAM allows the calculation of the MD5 (*Message-Digest Algorithm 5*) of a cryptographic hash function.

This can be created from any data stream a unique check value. It is virtually impossible to find two different messages with the same test value, this is referred to as collisions free. This can be used to check a configuration file for change or manipulation.

With the hash algorithm (MD5) a hash value is generated from 128 bits in length for any data. The maximum length of the stream is on this module is limited to 2^32 (4 gigabyte). The result is a 16 bytes hash value at parameters MD5.

Example:

There are 2000 bytes in a buffer and are read using the file system in blocks.

User set MODE to 1 and SIZE to 2000. Calling the MD5_STREAM

MD5 STREAM performs a initialization and set MODE to 2 and passes at POS the index (base 0) of the desired data. At SIZE the number of data is set, which are copied into the data memory BUF.

User copies the requested data in the BUF and calls the module MD5_STREAM repeatedly. This step is repeated until MODE remains at 2.

If the MD5_STREAM has processed the last data block, this set MODE to 3. It can also happen that the last block, that at the SIZE length zero is set, therefore, so no data are copied into BUF .

The current hash value can then be processed as a result.

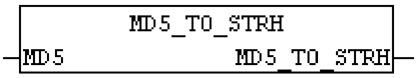
Example:

the MD5 hash of 'OSCAT' is 30f33ddb9f17df7219e1acdea3386743

8.16. MD5_TO_STRH

Type Function: STRING(32)

Input MD5 : ARRAY[0..15] OF BYTE (MD5-HASH)



The module MD5 MD5_TO_STRH converts the MD5 byte array to a hex string.

8.17. RC4_CRYPT_STREAM

Type Function module

I / O MODE: INT(mode: 1 = init / 2 = Data Block / 3 = Complete)

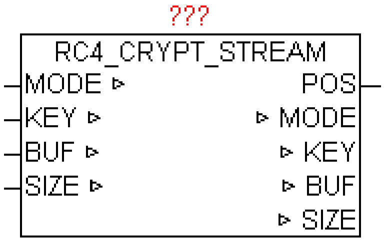
KEY: STRING(40) (320-bit long secret key)

BUF: ARRAY[0..63] OF BYTES (data block to process)

SIZE: UDINT (number of data)

Output POS: UDINT(start address of the requested data block)

The module RC4_CRYPT_STREAM uses the RC4 data encryption to process an almost arbitrarily long data stream. This standard is used for example in an SSH, HTTPS, and WEP or WPA, and is thus widely used. The algorithm can in principle operate at up to 2048 bit key, but this is limited to the mo-



dule on a 40-character key (but it can always be adjusted to up to 250 characters). Thus, it presents a key length of 320 bits, which are designed for applications on a PLC more than adequate. The maximum length of the stream is on this module is limited to 2^32 (4 gigabyte). The module can be used for encryption as well as to decrypt RC4 data. 64 bytes per cycle can still be processed, they will be processed in serial block mode. The data been encrypted or decrypted, remains in the module BUF for further processing, and must, of course, processed previously by the user before each new block of data.

Example:

There are 2000 bytes in a buffer and are read using the file system in blocks.

User sets mode is to 1 and SIZE to 2000. Calling the RC4_CRYPT_STREAM

RC4_CRYPT_STREAM performs initialization and set MODE to 2 and passes at the POS the index (base 0) to the desired data. At SIZE the number of data, to be copied into the data memory BUF, is set .

User copies the requested data in the BUF and calls the module RC4_CRYPT_STREAM repeatedly. This step is repeated until MODE remains at 2. If the RC4_CRYPT_STREAM has processed the last data block, this set MODE to 3.

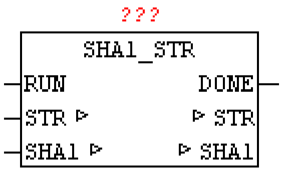
8.18. SHA1_STR

Type	Function module
Input	RUN: BOOL (Positive edge starts the calculation)

Output DONE: BOOL (TRUE if calculations are complete)

HASH : ARRAY[0..19] OF BYTE (actual SHA1-HASH)

I / O STR: STRING(string_length) (Text for HASH creation)



With SHA1_STR the SHA1 hash can be calculated in a string. In the STR a string is passed to the module, and a positive edge at input "RUN", the calculation starts. DONE is immediately reset at startup, and after the process is DONE is set to TRUE. Then, at the parameter HASH the actual calculated HASH value is available. (See module SHA1-STREAM).

Example:

Hash value of 'OSCAT' is 4fe4fa862f2e7b91bf95abe9f22831247a3afd35

8.19. SHA1_STREAM

Type Function module

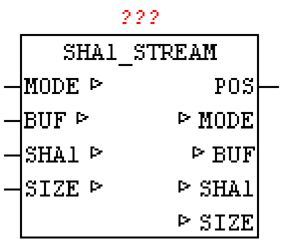
I / O MODE: INT(mode: 1 = init / 2 = Data Block / 3 = Complete)

BUF: ARRAY[0..63] OF BYTES (source data)

SHA1: ARRAY [0..19] OF BYTE (current SHA1-HASH)

SIZE: UDINT (number of data)

Output POS: UDINT(start address of the requested data block)



The module SHA1_STREAM allows the calculation of standard cryptographic hash function SHA-1 (Secure Hash Algorithm).

This can be created from any data stream a unique check value. It is virtually impossible to find two different messages with the same test value, this is referred to as collisions free. This can be used to check a configuration file for change or manipulation.

With the secure hash algorithm (SHA1) a hash value is generated from 160 bits in length for any data. The maximum length of the stream is on this module is limited to 2^32 (4 gigabyte). The result is a 20-byte hash value, issued as ARRAY [0..19] OF BYTE.

Example:

There are 2000 bytes in a buffer and are read using the file system in blocks.

User sets MODE to 1 and SIZE to 2000. Calling the SHA1_STREAM

SHA1_STREAM performs initialization and set MODE to 2 and passes at the POS the index (base 0) of the desired data. At SIZE the number of data, to be copied into the data memory BUF, is set .

User copies the requested data in the BUF and calls the module SHA1_STREAM repeatedly. This step is repeated until MODE remains at 2.

[fzy] If the SHA1_STREAM has processed the last data block, this set MODE to 3. It can also happen that the last block, that at the SIZE length zero is set, therefore, so no data are copied into BUF .

The current hash value can then be processed as a result.

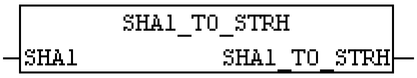
Example:

Hash value of 'OSCAT' is 4fe4fa862f2e7b91bf95abe9f22831247a3afd35

8.20. SHA1_TO_STRH

Type Function: STRING (40)

Input MD5 : ARRAY[0..19] OF BYTE (SHA1 hash)



The module converts the SHA1_TO_STRH SHA1 byte array to a hex string.

8.21. STRING_TO_URL

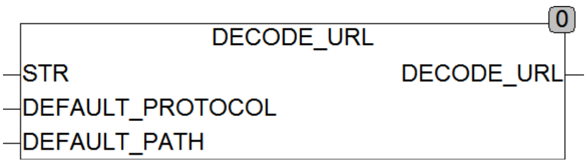
Type Function : URL

Input STR: STRING (string_length) (Unified Resource Locator)

DEFAULT_PROTOCOL: STRING(replacement protocol)

DEFAULT_PATH: STRING (alternate path)

Output URL (URL)



STRING_TO_URL split a URL (Uniform Resource Locator) into its components and stores it in the data type URL. If in STR no path or protocoll is specified, so the function sets the missing values automatically with the specified replacement values.

A URL is as follows:

Protocol : / / user : Password @ domain : port / path ? query # anchor

Example: ftp://hugo:nono@oscat.de:1234/download/manual.html

some parts of the URL are optional, such as user name, password, Anchor, Query ...

8.22. URL_DECODE

Type Fu FUNCTIONS: STRING(string_length)

Input IN: STRING (String)

Output STRING(string_length) (string)

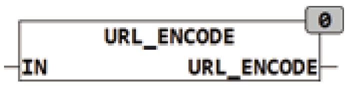
URL_DECODE converts the in %HH encoded special characters in the string IN in the appropriate ASCII code. In a URL encoding only the characters [A.. Z, a.. Z found, 0 .9, -._~] can occur. Other characters with a % character, followed by 2 characters long Hexadecimal of the character are shown. The reserved character '#' is encoded as a '%23'.

8.23. URL_ENCODE

Type Function : STRING(STRING_LENGTH)

Input IN: STRING (String)

Output STRING (STRING_LENGTH) (string)



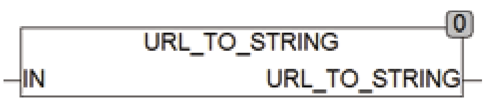
URL_ENCODE converts reserved characters in the string IN in the string '%HH'. In a URL encoding only the characters [A.. Z, a.. Z found, 0 .9, -._~] can occur. Other characters with a % sign followed by the two-character hexadecimal code of the character are shown. The reserved character '#' is encoded as a '%23'.

8.24. URL_TO_STRING

Type Function: STRING (string_length)

Input IN: STRING(Unified Resource Locator)

Output URL (URL)



URL_TO_STRING generates from stored data in IN with type URL a Unified Resource Locator as String .

A URL is as follows:

protokoll://user:passwort@domain:port/path?query#anchor

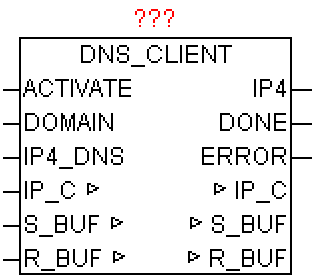
Example: ftp://hugo:nono@oscat.de:1234/download/manual.html

some parts of the URL are optional, such as user name, password, Anchor, Query ...

9. Network and Communication

9.1. DNS_CLIENT

Type	Function module
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
INPUT	ACTIVATE: BOOL (Query start by positive edge)
	DOMAIN: STRING (Domain name or IP as String)
	IP4_DNS: DWORD (IPv4 address of the DNS server)
OUTPUT	IP4: DWORD (IPv4 address of the requested domain)
	DONE: BOOL (IP of the domain has been queried successfully)
	ERROR: DWORD (error code)



DNS_CLIENT determine from the given qualified DOMAIN name the associated IPv4 address eg "www.oscat.de" . For this purpose, a DNS query to a DNS server for configured DOMAIN name with is made. With positive edge of ACTIVATE the specified DOMAIN is stored so that they no longer must be present. If the query provide more IP addresses, so always he highest value of the TTL (Time To Live) is used. As IP4_DNS can be used any public DNS servers. If the PLC is sitting behind a DSL router, this router can be used through its gateway address as a DNS server. Which ultimately leads to faster even with repeated requests response times because they are managed in the router cache. With positive results DONE = TRUE the IP4 contains the requested IP address until the start of the next query by positive edge of ACTIVATE. If in the DOMAIN name a valid IPv4 address is detected, no more DNS query is made and it is passed in converted type to IPv4 and DONE is set to TRUE. ERROR gives, if an error occurs, the exact cause.

Error Codes:

Value				Source	Description
B3	B2	B1	B0		
nn	**nn**	**nn**	xx	IP_CONTROL	Error from module IP_CONTROL
xx	xx	xx	**00**	DNS_CLIENT	No error: The request completed successfully
xx	xx	xx	**01**	DNS_CLIENT	Format error: The name server was unable to interpret the query.
xx	xx	xx	**02**	DNS_CLIENT	Server failure: The name server was unable to process this query due to a problem with the nameserver.
xx	xx	xx	**03**	DNS_CLIENT	Name Error: Meaningful only for responses from an authoritative name server, this code signifies that the domain name referenced in the query does not exist
xx	xx	xx	**04**	DNS_CLIENT	Not Implemented: The name server does not support the requested kind of query
xx	xx	xx	**05**	DNS_CLIENT	Refused: The name server refuses to perform the specified operation for policy reasons
xx	xx	xx	**06**	DNS_CLIENT	YXDomain: Name Exists when it should not
xx	xx	xx	**07**	DNS_CLIENT	YXRRSet. RR: Set Exists when it should not
xx	xx	xx	**08**	DNS_CLIENT	Nxrrset. RR: Set that should exist does not
xx	xx	xx	**09**	DNS_CLIENT	Server Not Authoritative for zone
xx	xx	xx	**0A**	DNS_CLIENT	Name not contained in zone

9.2. DNS_REV_CLIENT

Type Function module

IN_OUT IP_C: IP_C (parameterization)

S_BUF: NETWORK_BUFFER (transmit data)

R_BUF: NETWORK_BUFFER (receive data)

INPUT ACTIVATE: BOOL (query start by positive edge)

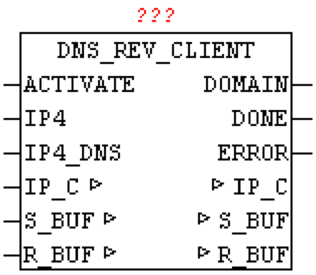
DOMAIN: STRING (Domain name or IP as String)

IP4_DNS: DWORD (IPv4 address of the DNS server)

OUTPUT IP4: DWORD (IPv4 address of the requested domain)

DONE: BOOL (IP of the domain has been queried successfully)

ERROR: DWORD (error code)



DNS_REV_CLIENT determine from the given IP address the officially registered domain name. For this purpose a reverse DNS query on the configured IP address with a DNS server is made. With positive edge of ACTIVATE the specified IP is stored so that they no longer must be present. If the query result in more matches, it will always use the last record. As IP4_DNS can be used any public DNS servers. If the PLC is sitting behind a DSL router, this router can be used as a DNS server through its gateway address. Which ultimately leads to faster even with repeated requests response times because they are managed in the router cache. With positive results DONE = TRUE the DOMAIN contains the officially registered domain name until the start of the next query by positive edge of ACTIVATE. ERROR gives ao error, the error code. (See error codes).

Error Codes:

Value				**Source**	Description
B3	B2	B1	B0		
nn	**nn**	**nn**	xx	IP_CONTROL	Error from module IP_CONTROL
xx	xx	xx	**00**	DNS_CLIENT	No error: The request completed successfully
xx	xx	xx	**01**	DNS_CLIENT	Format error: The name server was unable to interpret the query.
xx	xx	xx	**02**	DNS_CLIENT	Server failure: The name server was unable to process this query due to a problem with the nameserver.
xx	xx	xx	**03**	DNS_CLIENT	Name Error: Meaningful only for responses from an authoritative name server, this code signifies that the domain name referenced in

					the query does not exist
xx	xx	xx	**04**	DNS_CLIENT	Not Implemented: The name server does not support the requested kind of query
xx	xx	xx	**05**	DNS_CLIENT	Refused: The name server refuses to perform the specified operation for policy reasons
xx	xx	xx	**06**	DNS_CLIENT	YXDomain: Name Exists when it should not
xx	xx	xx	**07**	DNS_CLIENT	YXRRSet. RR: Set Exists when it should not
xx	xx	xx	**08**	DNS_CLIENT	Nxrrset. RR: Set that should exist does not
xx	xx	xx	**09**	DNS_CLIENT	Server Not Authoritative for zone
xx	xx	xx	**0A**	DNS_CLIENT	Name not contained in zone

9.3. DNS_DYN

Type Function module

INPUT ENABLE: BOOL (release of the module)

UPDATE: BOOL (Launches new DNS registration immediately)

T_UPDATE: TIME (waiting time for new DNS registration)

MODE: BYTE (selection of DynDNS provider)

HOST NAME: STRING (30) (own domain name)

USER NAME: STRING(20) (name for registration)

PASSWORD: STRING(20) (password for registration)

IP4: DWORD (Optional specify the IP address)

OUTPUT BUSY: BOOL (module is active, query is performed)

DONE: BOOL (performed DNS registration successful)

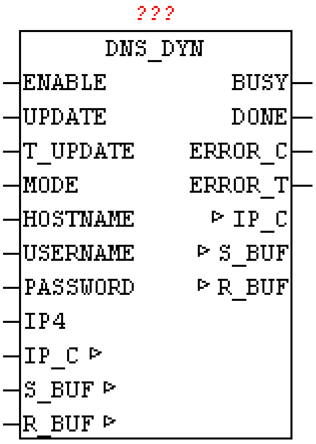
ERROR_C: DWORD (error code)

ERROR_T: BYTE (error type)

IN_OUT IP_C: IP_C (parameterization)

S_BUF: NETWORK_BUFFER (transmit data)

R_BUF: NETWORK_BUFFER (receive data)



WIth DNS_DYN dynamic IP addresses are registered as domain names. Many Internet providers assign a dynamic IP address when dialing into the Internet. To be visible and accessible for Internet Participants, one of the ways is to upgrade its current IP address via Dyn-DNS. The process is not standardized, unfortunately, so for every Dyn-DNS provider has to be created a individual solution. The module can be used in conjunction with DynDNS.org and Selfhost.de. These providers offer in addition to paid also free DynDNS services.

If ENABLE is set to TRUE, then the module is active. Using a positive edge to UPDATE any time an update can be started. If at T_UPDATE a time is specified, always an update is done after that time.

Caution, most DynDNS providers rates a frequent or unnecessary update as an attack, and block the account for a certain time.

The time T_UPDATE should not be set below an hour. If the parameter T_UPDATE is not connect it is assumed as an update time of 1 hour. If no update is needed on time, then T#0ms should be passed.

The MODE parameter allows the selection of DynDNS Provider

(0 = DynDNS.org, 1 = SELFHOST.DE)

The own domain name must be passed by the hostname. For security reasons, USERNAME and PASSWORD as authorization data must be specified to the DynDNS provider. If the parameter IP4 is not used, so DynDNS provider automatically adopts the current registration-IP as WAN IP with which the update is performed. By specifying an IP address also an individual IP address may be assigned.

With flawless execution the parameter DONE = TRUE, else ERROR_C and ERROR_T passes the error code and error type. (See error codes).

ERROR_T:

Value	Properties

1	The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	The exact meaning of ERROR_C can be read at module HTTP_GET
3	The DynDNS provider has refused registration

9.4. FTP_CLIENT

Type Function module:

INPUT ACTIVATE: BOOL (positive edge starts the query)

FILE NAME: STRING (file path/ filename)

FTP_URL: STRING(STRING_LENGTH) (FTP access path)

FTP_DOWNLOAD : BOOL (UPLOAD = 0 / DOWNLOAD = 1)

FTP_ACTIV : BOOL (PASSIV = 0 / ACTIV = 1)

FILE_DELETE: BOOL (delete files after transfer)

TIMEOUT: TIME (time)

Dns_ip4: DWORD (IP4 address of the DNS server)

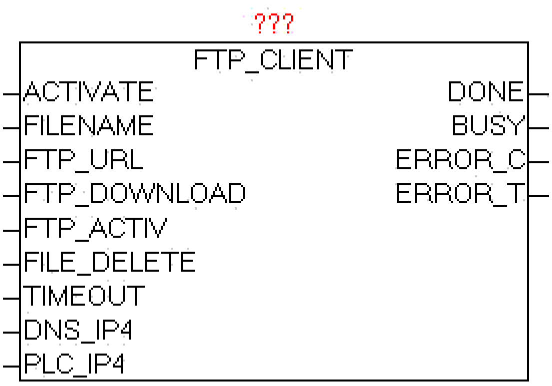
Dns_ip4: DWORD (IP4 address of the DNS server)

OUTPUT DONE: BOOL (Transfer completed without error)

BUSY: BOOL (Transfer active)

ERROR_C: DWORD (Error code)

ERROR_T: BYTE (Problem type)



The module FTP_CLIENT is used to transfer files from the PLC to an FTP server and to transmitted from the FTP server to the PLC. A positive edge at ACTIVATE starts the transfer process. In FTP_DOWNLOAD the transmission direction can be specified. The parameter FTP_URL contains the name of the FTP server and pass the optional user name and password, an access path and an additional port number for the data channel. If no username or password is passed, the module trys automatically to register as "Anonymous" . The parameter FTP_ACTIV determines whether the FTP server operates in active or passive mode. In the ACTIV mode, the FTP server tries to establish the data channel for control, however these may cause problems by security software, firewall, etc. because these could block the connection request. For this purpose, in the firewall a corresponding exception rule has to be defined. In the passive mode, this problem is alleviated since the controller establishes the connection, and can easily pass through the firewall. The control channel is always set up on port 20, and the data channel via standard PORT21, but this is in turn is depending whether active or passive mode is used, or optional PORT number in the FTP-URL is specified. With the parameter FILE_DELETE can be determined whether the source file should be deleted after successful transfer. This works on FTP and even on the control side. In specifying FTP directories the behavior depends on FTP server, whether they exist in this case or are created automatically. Normally, these should be already available. The size of files is no limit per se, but there are practical limits: Space on PLC, FTP storage and the transmission time. With dns_ip4 the IP address of the DNS server must be specified, if in the FTP URL a DNS name is given, alternatively, an IP address can be entered in the FTP URL. At parameters PLC_IP4 the own IP addresses has to be supplied. If errors occur during transmission these are passed to the output ERROR_C and ERROR_T. As long as the transfer is running, BUSY = TRUE, and after an error-free completion of the operation, DONE = TRUE. Once a new transfer is started, DONE, ERROR_T and ERROR_C are reseted.

The module has integrated the IP_CONTROL and so must not be externally linked to this, as it would be at default necessary.

Background: http://de.wikipedia.org/wiki/File_Transfer_Protocol

URL examples:

ftp://username:password@servername:portnummer/directory/

ftp://username:password@servername

ftp://username:password @ servername / directory /

ftp://servername

ftp://username:password@192.168.1.1/directory/

ftp://192.168.1.1

ERROR_T:

Value	Properties
1	Problem: DNS_CLIENT The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	Problem: FTP control channel The exact meaning of ERROR_C can be read at module IP_CONTROL
3	Problem: FTP data channel The exact meaning of ERROR_C can be read at module IP_CONTROL
4	Problem: FILE_SERVER The exact meaning of ERROR_C can be read at block FILE_SERVER
5	Problem: END - TIMEOUT ERROR_C contains the left WORD of the step number, and the right WORD has the response code received by the FTP server. The parameters must be considered first as a HEX value, divided into two WORDS, and then be considered as a decimal value. Example: ERROR_T = 5 ERROR_C = 0x0028_00DC End-step number 0x0028 = 40 Response-Code 0x00DC = 220

9.5. GET_WAN_IP

Type Function module:

IN_OUT IP_C: data structure 'IP_CONTROL ' (Parameterization)

S_BUF: data structure NETWORK_BUFFER '(transmit data)

R_BUF: data structure 'NETWORK_BUFFER '(receive data)

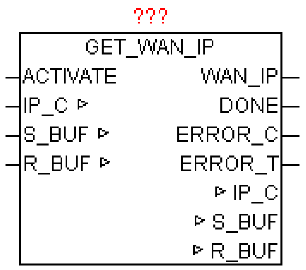
INPUT ACTIVATE: BOOL (release for query)

OUTPUT: WAN_IP: DWORD (Wide Area Network address)

DONE: BOOL (Query completed without errors)

ERROR_C: DWORD (Error code)

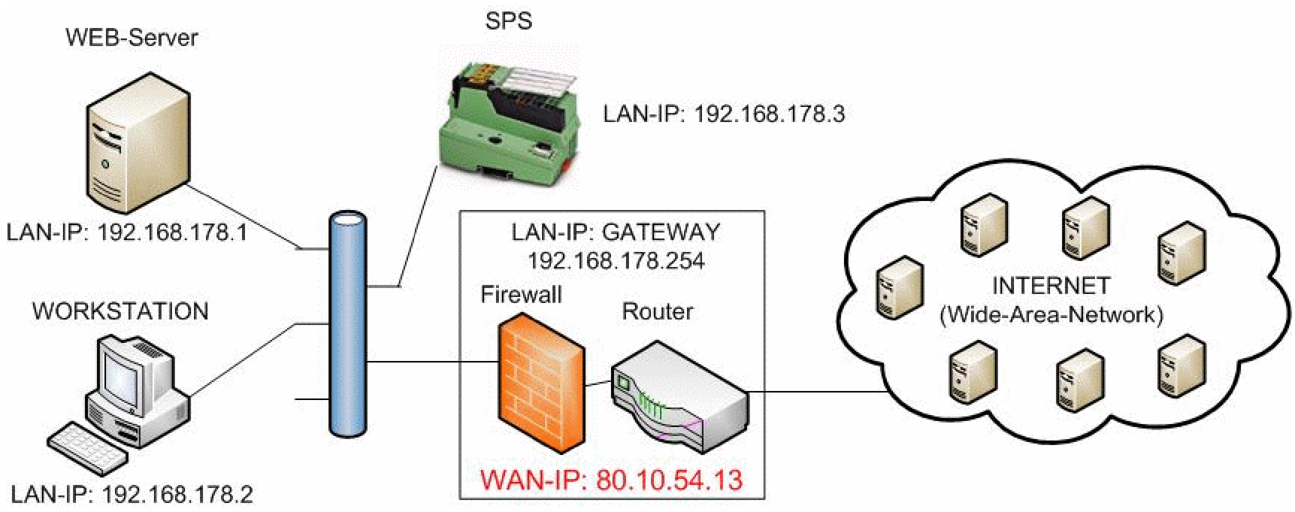
ERROR_T: BYTE (error type)



The module determines the IP address that the Internet router on the Wide Area Network (Internet) uses. This IP address is necessary for example to to make DynDNS declarations. With a positive edge of the ACTIVATE the request is started. After successful completion of the query DONE = TRUE, and the parameters WAN_IP the current WAN IP address displayed. If an error occurs during the query it is reported in ERROR_C in combination with ERROR_T.

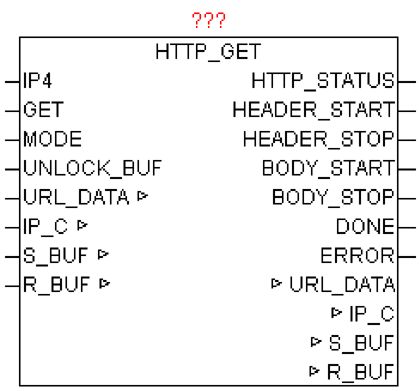
ERROR_T:

Value	Properties
1	The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	The exact meaning of ERROR_C can be read at module HTTP_GET



9.6. HTTP_GET

Type	Function module:
IN_OUT	URL_DATA: URL (data STRING_TO_URL)
	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
INPUT	IP4: DWORD (IP address of the HTTP server)
	GET: BOOL (Starts the HTTP query)
	MODE: BYTE (version of the HTTP GET query)
	UNLOCK_BUF: BOOL (release of the receive data buffer)
OUTPUT	HTTP_STATUS: STRING (HTTP status code)
	HTTP_START: UINT (start position of the message header)
	HTTP_STOP: UINT (stop position of the message header)
	BODY_START: UINT (start position of the message body)
	BODY_STOP: UINT (stop position of the message body)
	DONE: BOOL (task performed without error)
	ERROR: DWORD (error code)



HTTP_GET does at positive edge of Get a GET-command on an HTTP server. IWith MODE the HTTP protocol version can be specified. The requested URL (web link) must be submitted completely processed in the URL_DATA structure. The full URL should therefore be processed before the module call by "STRING_TO_URL. After a successful query DONE = TRUE, and the parameters HTTP_START and HTTP_STOP point to the data area in which the message header data for further processing and evaluation are to be found. Normally, a message body is present, which in turn is transmitted via BODY_START and BODY_STOP. Also, on HTTP_STATUS is reported the HTTP status code as a string. One of the difficulties in receiving the

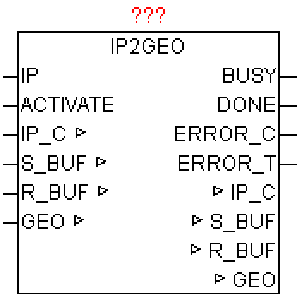
HTTP data is the end of the data stream. This module pursued multiple strategies. In the process of the HTTP/1.0 the end of receiving data is detected by disconnection of the host. Furthermore, always the information in the header "Content-Length" is checked, and with this can be clearly recognized, that all data is received. If none of the previous versions is true, so a simple Receive Timeout Error detectes the end of data transmission. The only downside is, that this takes time. Sometimes, depending on the timeout value longer than desired. Therefore it is not bad if a reasonable timeout value is set at IP_CONTROL. ERROR gives at errors, the exact cause (See module IP_CONTROL)

The following MODE can be used:

Mode	Protocol Version	Properties
0	HTTP/1.0	The host terminates automatically the TCP connection, after the transfer of data.
1	HTTP/1.0	By applying "Connection: Keep-Alive", the host is instructed to use a persistent connection. The client should end of the connection after stopping activities.
2	HTTP/1.1	The host uses a persistent connection and must be stopped by client.
3	HTTP/1.1	By use of "Connection: Close" the host is instructed to stop transmission of data, the TCP connection.

9.7. IP2GEO

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
	GEO: IP2GEO (Geographic Data)
INPUT	ACTIVATE: BOOL (release for query)
OUTPUT	BUSY: BOOL (Query is active)
	DONE: BOOL (Query completed without errors)
	ERROR_C: DWORD (Error code)
	ERROR_T: BYTE (error type)



The device supplies because of the wide-area network IP address, the geographic information of the Internet access. As the WAN IP addresses are registered worldwide, therefore can be determined the approximate geographical position of the PLC. Should access runs through a proxy server, so its geographic position is determined and not the PLC. Usually, but normally it is in the nearness of the PLC, and thus the deviation is not relevant. This results in calculated positions differ only a few miles from the true position, and is relatively accurate.

If the parameter "IP" specifies no IP address, automatically the current WAN IP is used, otherwise the information of the configured IP delivered. With a positive edge of the ACTIVATE the request is started. As long as the query is not complete, BUSY = TRUE is passed. After successful completion of the query DONE = TRUE, and the parameters WAN_IP the current WAN IP address displayed. If an error occurs during the query it is reported in ERROR_C in combination with ERROR_T.

ERROR_T:

Value	Properties
1	The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	The exact meaning of ERROR_C can be read at module HTTP_GET

The "country_code is coded according to ISO 3166 country code ALPHA-2".

http://www.iso.org/iso/english_country_names_and_code_elements
http://de.wikipedia.org/wiki/ISO-3166-1-Kodierliste

The "REGION_CODE" is coded to "FIPS region code".
http://en.wikipedia.org/wiki/List_of_FIPS_region_codes

9.8. IP_CONTROL

Type Function module

IN_OUT IP_C : IP_C (parameterization)

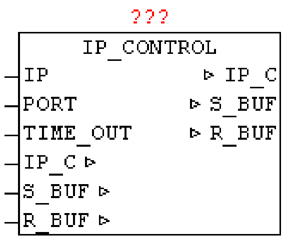
S_BUF: NETWORK_BUFFER (transmit data)

R_BUF: NETWORK_BUFFER (receive data)

INPUT IP: DWORD (encoded IP address as the default)

PORT: WORD (port number of the IP address)

TIME_OUT: TIME (monitoring time)



Available platforms and related dependencies

CoDeSys:

requires the library "SysLibSockets.lib"

Runs on

WAGO 750-841
CoDeSys SP PLCWinNT V2.4

and compatible platforms

PCWORX:

No library required

Runs on all controllers with file system from firmware >= 3.5x

BECKHOFF:

Requires the installation of "TwinCAT TCP/IP Connection Server"
Thus requires the Library "TcpIp.Lib"
(Standard.lib; TcBase.Lib; TcSystem.Lib are then automatically included)

Programming environment:
NT4, W2K, XP, Xpe;
TwinCAT system version 2.8 or higher;

TwinCAT Installation Level: TwinCAT PLC or higher;

Target platform:

TwinCAT PLC runtime system version 2.8 or higher.

PC or CX (x86)

TwinCAT TCP/IP Connection Server v1.0.0.0 or higher;

NT4, W2K, XP, XPe, CE (image v1.75 or higher);

CX (ARM)

TwinCAT TCP/IP Connection Server v1.0.0.44 or higher;

CE (image V2.13 or later);

The IP_CONTROL enables manufacturers and platform-neutral use of Ethernet communications. In order to unite the many different interfaces of the PLC-companies that IP_CONTROL is used as an adapter "wrapper" . This module UDP and TCP as well as active and passive connections can be handled. As in some small controls the number of simultaneous open sockets is very limited, so this module also supports the sharing of sockets. An integrated automatic coordination of requests allows to divide a socket to a number of client devices. Here is automatically recognized whether a client uses a different connection parameters than its predecessor. An existing connection is automatically terminated, and established with the new connection parameters . The type of connection can be set with C_MODE (see table). With C_PORT the desired port number is given, and by the C_IP the IP v4 address. With C_STATE can be determined whether the connection is established - closed, resp. the negative and positive edge on change of state. C_ENABLE agent will release the connection (establish) or close (removed). The send and receive data works independently of each another, which means it is also possible to send and receive asynchronous such as Telnet. In applications which only send data and no data receive is expected R_OBSERVE must be FALSE, so that no Timeout at receive occurs. At the start of transmit and receive activities TIME_RESET is set by the user a to TRUE once, so that all timeout start over with a defined start value. This is required due to the Sharing functionality, because established connections remains connected and the access rights are passed here only. The parameter IP serves as a possible default IP address. To avoid repeating the same IP address parameters, a Default - IP can be used. One possible use would be to specify the DNS server address. When the module recognizes as C_IP the IP 0.0.0.0, it automatically uses the default IP address. The same behavior is at the Port parameter. If at the port C_PORT a 0 is detected so the parameterized block port number of the module is used. The error code ERROR consists of several parts (see table ERROR). With TIMEOUT the overall monitoring time can be specified. This time value is independently used used for connection, send data and receive data. The transferred TIMEOUT value is automatically limited to 200 ms minimum. Thus, this parameter can remain free.

The data block is automatically sent if in a shared connection in the send buffer the transmit data and data length are entered. For data reveice, the data is appended to the already existing data in the buffer. By setting SIZE = 0, the receive data pointer is reset and the next received data is then stored at position 0.

The module supports the blocking of data messages, that means the S_BUF resp. R_BUF can be arbitrarily large. Individual received data frames are collected in R_BUF in stream form. Just the same when process data are sent. The data in S_BUF is sent individually as Stream allowed block size.

Application example:

```
     CASE state OF
00: (* On Wait for release *)
  IF RELEASE THEN
    state := 10;
    IP_STATE: = 1; (* Sign on *)
  END_IF;
10: (* Wait for clearance to access for connection and sending content *)
  IF IP_STATE = 3 THEN (* access permitted? *)
    (* IP set up data traffic *)
    IP_C.C_PORT: = 123; (* enter port number *)
    IP_C.C_IP = IP4; (* Enter IP *)
    IP_C.C_MODE:= 1; (* Mode: UDP+ACTIVE+Port+IP *)
    IP_C.C_ENABLE:= TRUE; (* Release connection *)
    IP_C.TIME_RESET: = TRUE; (Reset time monitoring * *)
    IP_C.R_OBSERVE: = TRUE; (* Monitor data receive *)
    R_BUF.SIZE:= 0; (* Reset Home length *)

    (* Send data register *)
    S_BUF.BUFFER[0] := BYTE#16#1B;
    (* Etc. ... *)
    S_BUF.SIZE: = xx; (* enter send length *)
    state := 30;
30:
  IF IP_C.ERROR <> 0 THEN
    (* Perform error analysis *)
    ELSIF S_BUF.SIZE = 0 AND R_BUF.SIZE >= xxx THEN
     (* evaluate received data *)
```

```
      (* Logout - release access for other *)
      IP_STATE := BYTE#4;
      state: = 0 0; (* process end *)
   END_IF;
END_CASE;

(* IP_FIFO call cyclic *)
IP_FIFO(FIFO:=IP_C.FIFO,STATE:=IP_STATE,ID:=IP_ID);
IP_C.FIFO:=IP_FIFO.FIFO;
IP_STATE := IP_FIFO.STATE;
IP_ID:=IP_FIFO.ID;
```

following C_MODE may be used:

TYPE	TCP / UDP	Aktiv / Passiv	Port number required	IP address required
0	TCP	Active	Yes	Yes
1	UDP	Active	Yes	Yes
2	TCP	Passive	Yes	**Yes** (Address of the active partner)
3	UDP	Passive	Yes	**Yes** (Address of the active partner)
4	TCP	Passive	Yes	**No** (Any active partner will be accepted)
5	UDP	Passive	Yes	**No** (Any active partner will be accepted)

C_STATE:

Value	State Message
0	connection is down
1	Connection has been broken down (negative edge) value exists only for one cycle, then returns 0.
254	Connection is established (positive edge) value exists for one cycle, then returns 255.
255	Connection is established
<127	query if connections is established

>127	query if connection is established

ERROR:

DWORD				Message Type	Description
B3	B2	B1	B0		
00	xx	xx	xx	Connection establish	Value 00 - No errors found
nn	xx	xx	xx	Connection establish	Value 01-252 system-specific error
FD	xx	xx	xx	Connection establish	Value 253 - Connection closed by remote
FF	xx	xx	xx	Connection establish	value 255 - Timeout Error
xx	**00**	xx	xx	Send data	Value 00 - No errors found
xx	**nn**	xx	xx	Send data	Value 01-252 system-specific error
xx	**FF**	xx	xx	Send data	value 255 - Timeout Error
xx	xx	**00**	xx	Receive data	Value 00 - No errors found
xx	xx	**nn**	xx	Receive data	Value 01-252 system-specific error
xx	xx	**FF**	xx	Receive data	value 255 - Timeout Error
xx	xx	**FE**	xx	Receive data	Value 254 - Receive buffer is full (overflow) (Buffer size is automatically set to 0)
xx	xx	xx	**nn**	Application- Error	In IP_CONTROL always 00 ERROR is transferred originally from the client application and optionally, at this point an application error is reported. This error code is entered, but only by the client devices.

System-specific error: (PCWorx / MULTIPROG)

Value	**State Message**
0x00	No error occurred.
0x01	Creation of at least one task has failed.
0x02	Initialization of the socket interface failed (only WinNT).
0x03	Dynamic memory could not be reserved.
0x04	FB can not be initialized because at start the asynchronous communication tasks, an error has occurred.

0x10	Socket initialization failed.
0x11	Error at sending a message.
0x12	Error when receiving a message.
0x13	Unknown service code in the message header.
0x21	Invalid state transition upon connection.
0x30	No more free channels available.
0x31	The connection was canceled.
0x33	General timeout, receiver or transmitter does not answer or sender has not completed transmission.
0x34	Connection request has been negatively acknowledged.
0x35	Recipient did not confirm transfer, possibly overloaded receivers (repeat transfer).
0x40	Partner-string is too long (255 characters max).
0x41	The specified IP address is not valid or could not be interpreted correctly.
0x42	not valid port number.
0x45	Unknown parameters to input "PARTNER".
0x50	Transmission attempt on invalid connection (sender or receiver).
0x53	All available connections are occupied.
0x61	Neg. confirmation of the recipient. It was used an incorrect sequence number.
0x62	Data type of transmitter and receiver are not equal.
0x63	Receiver is at the moment not ready to receive (poss. Cause: The recipient is disabled or is currently in the data transfer (NDR = TRUE).
0x64	Can not find a receiver module with the corresponding R_ID.
0x65	Another module instance is already working on this connection.
0x70	Partner control was not configured as a time server.

System-specific error: (CoDeSys)

0x00	No error occurred.
0x01	SysSockCreate unsuccessful
0x02	SysSockBind unsuccessful

0x03	SysSockListen unsuccessful

System-specific error: (Beckhoff)

0x00	No error occurred.
0x01	SocketUdpCreate unsuccessful
0x03	socket listen unsuccessful
0x04	SocketAccept unsuccessful

9.9. IP_CONTROL2

Type Function module

IN_OUT IP_C : IP_C (parameterization)

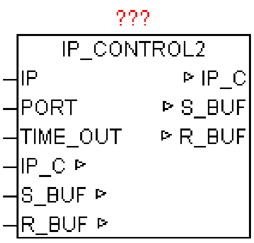
S_BUF: NETWORK_BUFFER_SHORT (transmit data)

R_BUF: NETWORK_BUFFER_SHORT (receive data)

INPUT IP: DWORD (encoded IP address as the default)

PORT: WORD (port number of the IP address)

TIME_OUT: TIME (monitoring time)



Available platforms and related dependencies

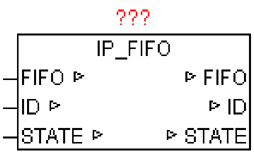
(See module IP_CONTROL)

The block has basically the same functionality as IP_CONTROL. However S_BUF and R_BUF are of type 'NETWORK_BUFFER_SHORT'
(See general description IP_CONTROL).

It is no blocking of the data supported by IP_CONTROL2 . The maximum data size for transmission and reception depends on the hardware platform and is in the range of < 1500 bytes. This module can always be used when no data stream mode is needed. The primary advantage is that less buffer memory is required, and data will not be copied between internal and external data buffer. Thus, the module is more economical with respect to memory consumption and system load.

9.10. IP_FIFO

Type	Function module:
IN_OUT	FIFO: IP_FIFO_DATA (IP-FIFO management data)
	ID: BYTE (current ID assigned by IP_FIFO module)
	STATE: BYTE (control commands and status messages)



This module is used in combination with IP_CONTROL for resource management. This makes it possible that client applications request exclusive access permissions and can also give back. By the FIFO is ensured that each participant equally often gets the resource access assigned.

In the first call of the module automatically a new unique application ID is assigned, which one the administration in FIFO is managed. The STATE parameter is changed by the application as well as from IP_FIFO module. Each application may register by default only once in the FIFO.

STATE:

Value	State Message
0	no action
1	Privilege request
2	Privilege request has been accepted in FIFO
3	Privilege obtained (allowing resource access)
4	Privilege remove

5	Privilege was again removed from FIFO

Procedure:

1. application set the STATE to 1
2. Access permission are required as is the STATE = 2
3. if resource is free, and access rights are present, then STATE=3
4. If the application has the resource resp. the access needs not anymore the application sets STATE to 4. Thereafter IP_FIFO releases the resource again and set STATE to 0.
5. Process is repeated (same or other application)

Example is found in the module IP_CONTROL!

9.11. LOG_MSG

Type	Function module:
IN_OUT	LOG_CL: LOG_CONTROL (log-data)

With LOG_MSG messages (STRINGS) are stored in a ring buffer. The messages can be provided with additional parameters such as the front color and back color for the output to TELNET and a filter by specifying an entry-level news. Is the level of the message larger than the default log level, the message is discarded. Furthermore,with Enable the logging will be disabled in general. Thus, it is not a problem to archive many messages per PLC cycle. The message buffer can be passed to a telnet client with the module TELNET_LOG. Details on the interface are shown in the table below.

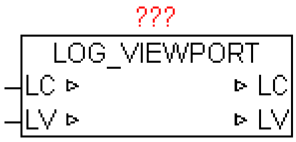
If messages are applied from various module instances to the same LOG_BUFFER, then the "LOG_CL" data structure has to be created Global.

9.12. LOG_VIEWPORT

Type Function module

IN_OUT LC: LOG_CONTROL

LV: us_LOG_VIEWPORT



The module LOG_VIEWPORT is used to index a list of LOG_CONTROL messages, which are currently in the virtual view. To move around within the message list (scroll), a scroll offset can be specified by LV.MOVE_TO_X. A positive value scroll in direction of newer reports and a negative value in the direction of the earlier messages. The number of lines in the message list of the virtual view is given by LV.COUNT. The current messages in the virtual view are stored in LV.LINE_ARRAY [x], and are available for further processing. A change in the message list is always announced with LV.UPDATE:= TRUE, and the user has to reset.

The following LV.MOVE_TO_X values produce a special behavior.

+30000 = display previous Messages (beginning of the ring buffer)

+30001 = display latest messages (end of the ring buffer)

+30002 = one full page in direction of recent messages.

+30003 = One full page in direction of older messages

9.13. MB_CLIENT (OPEN MODBUS)

Type Function module:

IN_OUT IP_C: IP_C (parameterization)

S_BUF: NETWORK_BUFFER_SHORT (transmit data)

R_BUF: NETWORK_BUFFER_SHORT (receive data)

DATA: ARRAY [0..255] OF WORD (MODBUS Register)

INPUT DATA_SIZE: INT (number of data words in structure MB_DATA)

ENABLE: BOOL (release)

UDP: BOOL (Prefix TCP / UDP, UDP = TRUE)

FC: INT (function number)

UNIT_ID: BYTE (Device ID)

R_ADDR: INT (Read command: MODBUS data point address)

R_POINTS: INT (Read command: MODBUS number of data points)

R_DATA_ADR: INT (Read command: DATA data point address)

R_DATA_BITPOS: INT (Read command: DATA bit position data point)

W_ADDR: INT (Read command: MODBUS data point address)

W_POINTS: INT (Read command: MODBUS number of data points)

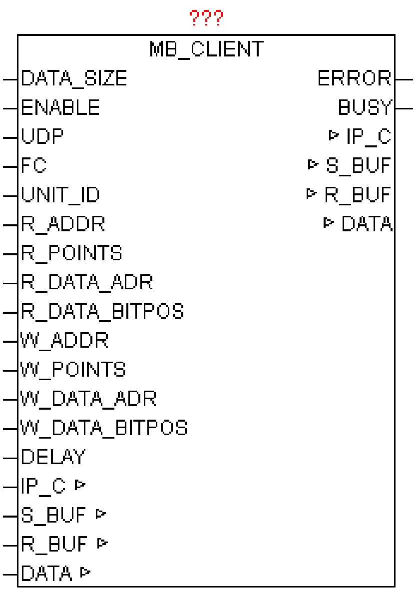
W_DATA_ADR: INT (Read command: DATA data point address)

W_DATA_BITPOS: INT (Read command: DATA bit position data point)

DELAY: TIME (repetition time)

OUTPUT ERROR: DWORD (error code)

BUSY: BOOL (module is active)



The module provides access to Ethernet devices, the MODBUS TCP or MODBUS UDP supported, or MODBUS RS232/485 devices are connected via Ethernet Modbus gateway. There commands from Classes 0,1,2 are supported. The parameters IP_C, S_BUF, R_BUF this form the interface to the module IP_CONTROL and used here for processing and coordination. The desired IP address and port number (for MODBUS default is 502) must be specified on IP_CONTROL centrally. The DATA structure is designed as a WORD array and contains the MODBUS data for reading and writing. The size of the WORD_ARRAY is given by DATA_SIZE. By ENABLE, the module is released, and by remove of the release a possibly still active query is ended. For devices that support MODBUS with UDP = TRUE this mode can be activated. The parameter UNIT_ID must only at use of Ethernet Modbus provided. The desired function is specified by FC (see function code table). Depending on the function, the R_xxx and W_xxx parameters has to be supplied with data. By specifying the DELAY the repetition time can be specified. If not specify the time an attempt is made as often as possible to execute the command. The integrated access management automatically guarantees to get the other module instances also to the series. A negative command execution is reported by ERROR (see ERROR-table). If the module actively performs a query, then BUSY = TRUE will be passed during this time.

Supported function codes and parameters used:

Function Code	Bit Access	16 Bit Access (Register)	Group	Function Description	R_ADDR	R_POINTS	R_DATA_ADR	R_DATA_BITPOS	W_ADDR	W_POINTS	W_DATA_ADR	W_DATA_BITPOS
1	x		Coils	Read Coils	x	x	x	x				
2	x		Input Discrete	Read Discrete Inputs	x	x	x	x				
3		x	Holding Register	Read Holding Registers	x	x	x					
4		x	Input Register	Read Input Register	x		x					
5	x		Coils	Write Single Coil					x		x	x
6		x	Holding Register	Write Single Register					x		x	
15	x		Coils	Write Multiple Coils					x	x	x	x
16		x	Holding Register	Write Multiple Register					x	x	x	

22		x	Holding Register	Mask Write Register					x		x	
23		x	Holding Register	Read/Write Multiple Register	x	x	x		x	x	x	

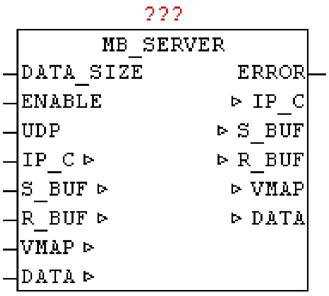
ERROR:

Value				**Source**	Description
B3	B2	B1	B0		
nn	**nn**	**nn**	xx	IP_CONTROL	Error from module IP_CONTROL
xx	xx	xx	**00**	MB_CLIENT	No Error
xx	xx	xx	**01**	MB_CLIENT	ILLEGAL FUNCTION: The function code received in the query is not an allowable action for the server (or slave). This may be because the function code is only applicable to newer devices, and was not implemented in the unit selected. It could also indicate that the server (or slave) is in the wrong state to process a request of this type, for example because it is unconfigured and is being asked to return register values.
xx	xx	xx	**02**	MB_CLIENT	ILLEGAL DATA ADDRESS: The data address received in the query is not an allowable address for the server (or slave). More specifically, the combination of reference number and transfer length is invalid. For a controller with 100 registers, the PDU addresses the first register as 0, and the last one as 99. If a request is submitted with a starting register address of 96 and a quantity of registers of 4, then this request will successfully operate (address-wise at least) on registers 96, 97, 98, 99. If a request is submitted with a starting register address of 96 and a quantity of registers of 5, then this request will fail with Exception Code 0x02 “Illegal Data Address” since it attempts to operate on registers 96, 97, 98, 99 and 100, and there is no register with address 100.
xx	xx	xx	**03**	MB_CLIENT	ILLEGAL DATA VALUE: A value contained in the query data field is not an allowable value for server (or slave). This indicates a fault in the structure of the remainder of a complex request, such as that the implied length is incorrect. It specifically does NOT mean that a data item submitted for storage in a register has a value outside the expectation of the application program, since the MODBUS protocol is unaware of the significance of any particular value of any particular register.
xx	xx	xx	**04**	MB_CLIENT	SLAVE DEVICE FAILURE: An unrecoverable error occurred while the server (or slave) was attempting to perform the requested action.
xx	xx	xx	**05**	MB_CLIENT	ACKNOWLEDGE: Specialized use in conjunction with programming commands. The

					server (or slave) has accepted the request and is processing it, but a long duration of time will be required to do so. This response is returned to prevent a timeout error from occurring in the client (or master). The client (or master) can next issue a Poll Program Complete message to determine if processing is completed.
xx	xx	xx	**06**	MB_CLIENT	SLAVE DEVICE BUSY: Specialized use in conjunction with programming commands. The server (or slave) is engaged in processing a long–duration program command. The client (or master) should retransmit the message later when the server (or slave) is free.
xx	xx	xx	**8**	MB_CLIENT	MEMORY PARITY ERROR: Specialized use in conjunction with function codes 20 and 21 and reference type 6, to indicate that the extended file area failed to pass a consistency check.The server (or slave) attempted to read record file, but detected a parity error in the memory.The client (or master) can retry the request, but service may be required on the server (or slave) device.
xx	xx	xx	**0A**	MB_CLIENT	GATEWAY PATH UNAVAILABLE: Specialized use in conjunction with gateways, indicates that the gateway was unable to allocate an internal communication path from the input port to the output port for processing the request. Usually means that the gateway is misconfigured or overloaded.
xx	xx	xx	**0B**	MB_CLIENT	GATEWAY TARGET DEVICE FAILED TO RESPOND: Specialized use in conjunction with gateways, indicates that no response was obtained from the target device. Usually means that the device is not present on the network.

9.14. MB_SERVER (OPEN-MODBUS)

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER_SHORT (transmit data)
	R_BUF: NETWORK_BUFFER_SHORT (receive data)
	VMAP: ARRAY [1..10] OF VMAP_DATA (virtual address table)
	DATA: ARRAY [0..255] OF WORD (MODBUS Register)
INPUT	DATA_SIZE: INT (number of data words in DATA)
	ENABLE: BOOL (release)
	UDP: BOOL (Prefix TCP / UDP, UDP = TRUE)
OUTPUT	ERROR: DWORD (error code)



The module provides access from external to local MODBUS data tables via Ethernet. It supports commands in categories 0,1,2. The parameters IP_C, S_BUF, R_BUF this form the interface to the module IP_CONTROL and used here for processing and coordination. The desired port number (for MODBUS default is 502) must be specified on IP_CONTROL centrally. The IP address is not required on IP_CONTROL, as this one operates in the PASSIVE mode. The DATA structure is designed as a WORD array and contains the MODBUS data. DATA_SIZE specifies the size of DATA . By ENABLE, the module is released, and by remove of the release a possibly still active query is ended. For devices that support MODBUS with UDP = TRUE this mode can be activated. A negative command execution is reported by ERROR (see ERROR table).

WIth entries in the data structure VMAP, virtual data areas are created, and the access to certain function codes and data regions is parameterized. Thus, it is very easy to map virtual address spaces into a coherent Data block (DATA), or write data areas. Or provide areas, that are connected to output peripherals, with a watchdog.

The handling of the VMAP data is described in more detail in the module MB_VMAP.

ERROR:

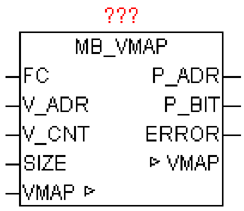
Value				**Source**	Description
B3	B2	B1	B0		
nn	**nn**	**nn**	xx	IP_CONTROL	Error from module IP_CONTROL
xx	xx	xx	**00**	MB_SERVER	NO ERROR:
xx	xx	xx	**01**	MB_SERVER	ILLEGAL FUNCTION:
xx	xx	xx	**02**	MB_SERVER	ILLEGAL DATA ADDRESS:
xx	xx	xx	**03**	MB_SERVER	ILLEGAL DATA VALUE:

Supported function codes and parameters used:

Function Code	Bit Access	16 Bit Access (Register)	Group	Function Description
1	x		Coils	Read Coils
2	x		Input Discrete	Read Discrete Inputs
3		x	Holding Register	Read Holding Registers
4		x	Input Register	Read Input Register
5	x		Coils	Write Single Coil
6		x	Holding Register	Write Single Register
15	x		Coils	Write Multiple Coils
16		x	Holding Register	Write Multiple Register
22		x	Holding Register	Mask Write Register
23		x	Holding Register	Read/Write Multiple Register

9.15. MB_VMAP

Type	Function module:
IN_OUT	VMAP : ARRAY [1..10] OF VMAP_DATA (VIRTUAL_MAP Data)
INPUT	FC: INT (function number)
	V_ADR: INT (virtual address range start address)
	V_CNT: INT (Virtual address space: number of data points)
	SIZE: INT (number of MODBUS registers in structure DATA)
OUTPUT:	P_ADR: INT (Real address space: Start address)
	P_BIT: INT (real address range: bit position)
	ERROR: DWORD (error code)



The module allows the conversion of virtual addresses at a real address space in the MODBUS DATA Structure. Virtual address ranges are defined in the VMAP data array. If the module is called and found that nothing in the VMAP data is entered, automatically a block is created, allowing full access to all the MODBUS data. In each address block also a watchdog timer is maintained that sets each time you access this block on the timer to zero. Thus, simply by comparing the TIME_OUT value to a cutoff value, at communication error (no update) can be responded.

By the parameter FC is detected the functional code and whether the register (16 bit) or individual bits must be processed. The bit number corresponds to the function code. This means that Bit5 = 1 in FC the function code 5 (Write Single Coil) enables. By V_ADR by the virtual start address is specified (At 16bit commands this is a register address and at bit commands an absolute bit number within a defined block.) The parameter V_CNT defines the number of data points (unit 16-bit or bits depending on the function code). The overall size is given by MODBUS_ARRAY SIZE (number WORDS). By using these parameters, the module searched the VMAP data table for a matching block of data, and passes from the correct data block P_ADR as a result. The value corresponds to the real index for MODBUS_DATA array. At a function code with bit access in addition the bit position within P_ADR is passed as well. A potential error occurring in the analysis is reported for the parameter "error" (see error table). The watchdog timer is reseted at each access to a function code from the group of write commands.

If no special treatment required, so in VMAP are not settings required, and then MODBUS_ARRAY is mapped 1:1 with the access.

ERROR:

Value	Description
0	No error
1	Invalid function code
2	Invalid Data Address

! Note the special treatment of function code 23!

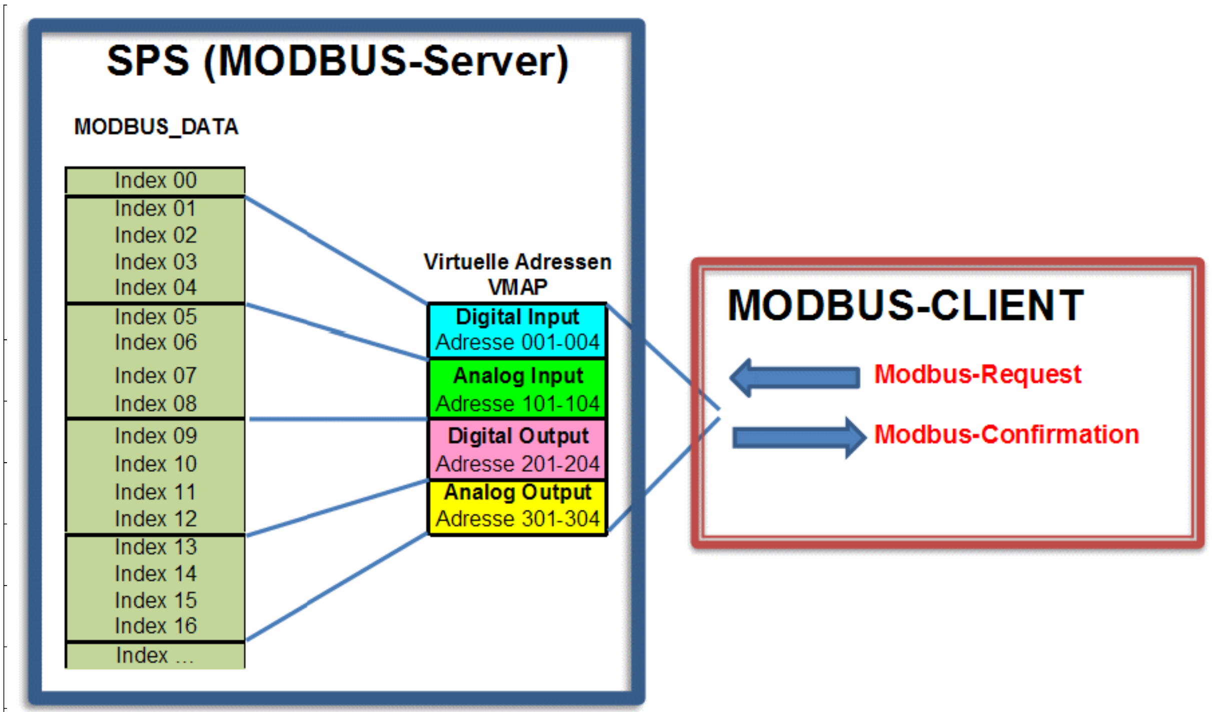
The Modbus Function Code 23 is a combined command, because it consists of two actions. First register are written and then the register are read. Found that the write or read parameter is not allowed, so neither of these actions is performed.

To distinguish between reading and writing by VMAP, the read command is checked in VMAP at FC 23 as BIT23 (Read/Write Multiple registers), and the write command on the other hand, is tested in Bit16 (Write multiple registers).

Example Configuration

```
(* Virtual block 1 *)
VMAP[1].FC := DWORD#2#00000000_10000000_00000000_00011100); (FC 2,3,4,23)
VMAP[1].V_ADR := 1;   (* Virtual Address Range: Start address *)
VMAP[1].V_SIZE := 4;  (* Virtual address space: number of WORD *)
VMAP[1].P_ADR := 1;   (* Real address space: Start address *)
(* Virtual Block 2 *)
VMAP[2].FC := DWORD#2#00000000_10000000_00000000_00011000); (FC 3,4,23)
VMAP [2] V_ADR. = 101; (* Virtual Address Range: Start address *)
VMAP[2].V_SIZE := 4;  (* Virtual address space: number of WORD *)
VMAP[2].P_ADR := 5;   (* Real address space: Start address *)
(* Virtual Block 3 *)
VMAP[3].FC := DWORD#2#00000000_11000001_10000000_01111010);(FC1,3-6,15-16,23)
VMAP [3] V_ADR. = 201; (* Virtual Address Range: Start address *)
VMAP[3].V_SIZE := 4;  (* Virtual address space: number of WORD *)
VMAP[3].P_ADR := 9;   (* Real address space: Start address *)
(* Virtual Block 4 *)
VMAP[4].FC := DWORD#2#00000000_11000001_00000000_01011000); (FC 3,4,6,16,23)
VMAP [4] V_ADR. = 301; (* Virtual Address Range: Start address *)
VMAP[4].V_SIZE := 4;  (* Virtual address space: number of WORD *)
VMAP[4].P_ADR := 12;      (* Real address space: home address *)
```

The configuration is following access matrix:



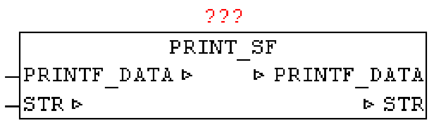
Write Multiple Coils	15	x		Schreiben			x	
Write Multiple Register	**16**		x	Schreiben			x	x
Mask Write Register	**22**		x	Schreiben				
Read/Write Multiple Register	**23**		x	Read / Write	x	x	x	x

9.16. PRINT_SF

Type Function module:

IN_OUT PRINTF_DATA: ARRAY[1..11] OF STRING(string_length) (Parameter data)

STR: STRING (String_length) (String result)



With PRINT_SF a STRING can be added dynamically with a part of a string. The position of the substring to be inserted is indicated by '~' tilde character and the subsequent number defines the parameter number. '~ 1' to '~ 9' are thus processed automatically. If the insert of the substring reached the maximum number of characters, so instead of the substring '..' is inserted.

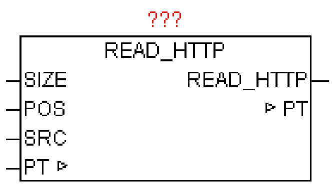
```
VAR
  LITER : REAL := 545.4;
  FUELLZEIT : INT := 25;
  NAME: STRING: = 'tank content';
  PARA: ARRAY[1..11] OF STRING(string_length);
  PS: PRINT_SF;
END_VAR
PARA[1]: = REAL_TO_STRING(liters); (* Parameter 1: string to convert *)
PARA[2]: = INT_TO_STRING(filling time); (* Parameter 2: string to convert
*)
PARA[3]: = NAME; (* Parameter 3: *)
PS.STR: = ' ~3: ~1 Liter, filling time: ~2 Min.' ; (* Text output-mask *)
PS.PRINTF_DATA := PARA; (* Pass parameter data structure *)
PS(); (* Module version *)
```

The string PS. STR then has the following content

'Tank Capacity: 545.4 liters, filling time: 25 min'

9.17. READ_HTTP

Type	Function module:	
INPUT	SIZE: UINT	(Buffer size)
	POS: INT	(position as of that the search is started)
	SRC: STRING	(Search string)
IN_OUT	PT: POINTER	(Address of the buffer)
OUTPUT	VALUE	(Parameters of the header information)



After a successful HTTP-GET Request always a HTTP header (message header) and a message body (message body) is available in the buffer. In the HTTP header various information about the requested HTTP page is stored. The following message body contains the actual requested data. With READ_HTTP the HTTP header information can be analyzed. The module searches any array of bytes on the contents of a string and then evaluates the following parameters, and returns that string as its result. The data in the buffer are automatically converted to upper case, so all search string at SRC has to be too, given in capital letters. With POS it can begin its search at any position. The first element in the array is at position number 1

Example of an HTTP response (header information):

HTTP/1.0 200 OK<CR><LF>

Content-Length: 2165<CR><LF>

Content-Type: text/html<CR><LF>

Date: Mon, 15 Sep 2008 16:59:08 GMT<CR><LF>

Last-Modified: Wed, 18 Jun 2008 12:35:52 GMT<CR><LF>

Mime-Version: 1.0<CR><LF><CR><LF>

If SRC does not include a search term, automatically the HTTP version and the HTTP status code in the buffer is searched and evaluated. As a result, according to the above example "1.0 200 OK" is returned. If SRC is in a search term, this header information is searched in the buffer and the value as a string eg 'Content-Length' = "2165" is returned.

9.18. SMTP_CLIENT

Type	Function module:
IN_OUT	SERVER: STRING (URL of the SMTP server)
	MAIL FROM: STRING (return address)
	MAILTO: STRING (string_length) (recipient address)
	SUBJECT: STRING (subject text)
	SUBJECT: STRING (subject text)
	FILES: STRING (string_length) (attached files)
INPUT	ACTIVATE: BOOL (positive edge starts the query)
	TIMEOUT: TIME (time)
	DTI: DT (current date-time)

DTI_OFFSET: INT (time zone offset from UTC)

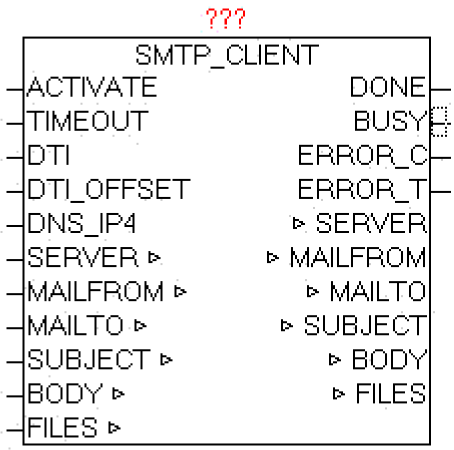
Dns_ip4: DWORD (IP4 address of the DNS server)

OUTPUT DONE: BOOL (Transfer completed without error)

BUSY: BOOL (Transfer active)

ERROR_C: DWORD (Error code)

ERROR_T: BYTE (Problem type)



The module SMTP_CLIENT is used to send of classic emails.

Following features are supported:

SMTP protocol

Extended SMTP protocol

Sending the subject line, text and content

Indication of email sender address (From:), including "Display Name"

Indication of the recipient (s) (To:)

Indication of carbon copy recipient (s) (Cc:)

Indication of blind copy recipient (s) (bc:)

Sending file (s) as an attachment

Authentication method: NO, PLAIN, LOGIN, CRAM-MD5

Specifying the port number

When positive edge at ACTIVATE the transfer process is started. The SERVER parameter contains the name of the SMTP server and optionally the user name and password and a port number. If you pass a user name and password, the procedure is according to standard SMTP.

SERVER: URL Examples:

username:password@smtp_server

username:password@smtp_server:portnumber

smtp_server

Special case:

If in the username is a '@' included this must be passed as '%' - character, and is then automatically corrected by the module again.

By specifying user and password the Extend-SMTP is used, and automatically the safest possible Authentication method is used. If parameter is to specify the MAIL FROM sender address:

i.e. oscat@gmx.net

Optionally, an additional "Display Name" be added This is displayed the email client automatically instead of the real return address. Therefore, always an easily recognizable name to be used.

i.e.. oscat@gmx.net;Station_01

The email client shows as the sender then "Station_01". Thus, more people will use the same email address but send a own "Alias".

At the MAILTO parameter can To, Cc, Bc be specified. The different groups of recipients are specified by '#' as the separator in a list. Multiple addresses within the same group are divided with the separator ";" . In each group can be defined unlimited count of recipients, the only limitation is the length of the mailto string.

To;To..#Cc;Cc...#Bc;Bc...

Examples.

o1@gmx.net;o2@gmx.net#o1@gmx.net#o2@gmx.net

defines two TO-addresses, one CC-address and a Bc-address

##o2@gmx.net

defines only one BC-address.

With subject, a subject text will be specified, as well as with BODY an email text content. The current Date / Time value must be definded at DTI, and at DTI_OFFSET the correction value as an offset in minutes from UTC (Universal Time). If the DTI in UTC time is passed, at DTI_OFFSET a 0 must be passed.

It can be sent files as attachment. The files must be passed in list form for parameter FILES. Any number of files are given, only limitation is the length of the file-strings, and the space of the e-mailbox (in practice 50-30 megabytes).

By an additional optional information of '#DEL# deleting the files can be triggered on the controller after the successful transfer of files via email.

eg

FILES: 'log1.csv **;** log2.csv **;** #DEL# '

The two files are deleted after successful transfer.

The monitoring time can be specified with parameter TIMEOUT. At dns_ip4 must be specified the IP address of the DNS server, if in SERVER a DNS name is specified. If errors occur during the transmission, they are passed at ERROR_C and ERROR_T. As long as the transfer is running, BUSY = TRUE, and after an error-free completion of the operation, DONE = TRUE. Once a new transfer is started, DONE, ERROR_T and ERROR_C are reseted.

The module has integrated the IP_CONTROL and so must not be externally linked to this, as it would be at default necessary.

Basics:

http://de.wikipedia.org/wiki/SMTP-Auth

http://de.wikipedia.org/wiki/Simple_Mail_Transfer_Protocol

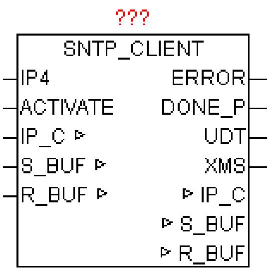
ERROR_T:

Value	Properties
1	Problem: DNS_CLIENT The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	Problem: SMTP Channel The exact meaning of ERROR_C can be read at module IP_CONTROL
4	Problem: FILE_SERVER The exact meaning of ERROR_C can be read at block FILE_SERVER
5	Problem: END - TIMEOUT ERROR_C contains the left WORD the end of the step number, and in the right WORD the last response code received by the SMTP server. The parameters must be considered first as a HEX value, divided into two WORDS, and then be considered as a decimal value. Example: ERROR_T = 5

	ERROR_C = 0x0028_00FA End-step number 0x0028 = 40 Response-Code 0x00DC = 250

9.19. SNTP_CLIENT

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
INPUT	IP4: DWORD (IP address of the SNTP server)
	ACTIVATE: BOOL (Starts the query)
OUTPUT	ERROR: DWORD (error code)
	DONE_P: BOOL (positive edge finish without error)
	UDT: DT (Date and time output as Universal Time)
	XMS: INT (millisecond of Universal Time UDT)

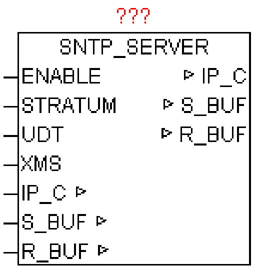


The SNTP_CLIENT is used to synchronize local time with an SNTP server. For this, the Simple Network Time Protocol is used which is designed to provide a reliable time information over networks with variable packet delay. The SNTP is technically completely identical with NTP, which here means no differences. Therefore, all known SNTP and NTP server can be used, whether it be on the local network or via the Internet. For IP4 a IP-address of a SNTP / NTP server is specified. A positive edge at ACTIVATE starts the query. The elapsed time between sending and receiving of the time is measured and a time correction is calculated. Then, the received time will be corrected by this value. Upon successful completion DONE_P is one positive edge, and the current time is passed at UDT. On XMS the associated fractional seconds as milliseconds are passed. The values of

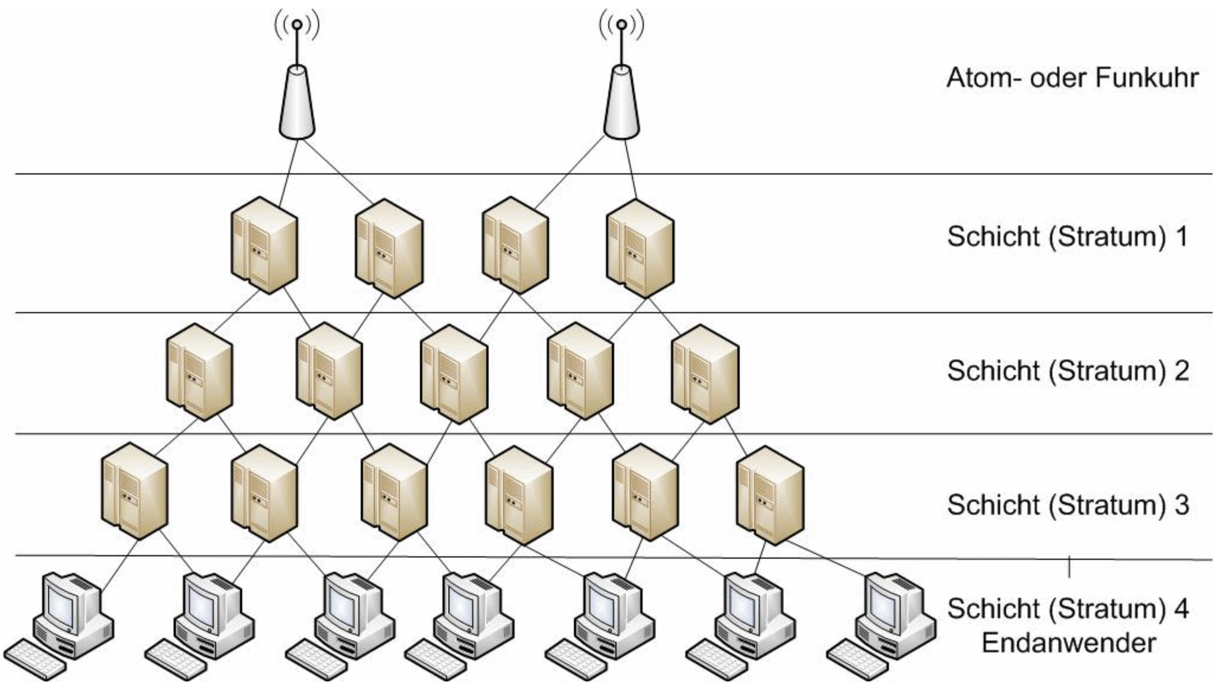
UDT and XMS are only valid when DONE_P = TRUE, since this is a static time value, and is only used for setting of pulse-controlled time. ERROR gives at error the exact cause (See block IP_CONTROL).

9.20. SNTP_SERVER

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
INPUT	ENABLE: BOOL (Starts SNTP server)
	STRATUM: BYTE (specify the hierarchical level or accuracy)
	UDT: DT (Date and time input as Universal Time)
	XMS: INT (millisecond of Universal Time UDT)



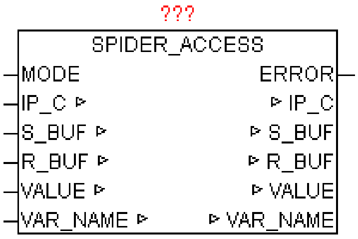
The module provides the functionality of an SNTP (NTP) server. With ENABLE = TRUE the module logs in at IP_CONTROL and waits for the release of the resource, if it occupied by other subscribers for now. Then the module is waiting for requests from other SNTP clients and answers it with the current time of UDT and XMS. As long as ENABLE = TRUE, the Ethernet access of this resource is permanently locked for other users (Exclusive Access - due to passive UDP mode). SNTP uses a hierarchical system of different strata. As stratum 0 is defined as the exact time standard. The directly coupled systems, such as NTP, GPS or DCF77 time signals are called Stratum 1.Each additional dependent unit causes an additional time lag of 10-100ms and is designate with a higher number (Stratum 2, Stratum 3 to 255). If no STRATUM is specified at the module, STRATUM = 1 is used as a standard.



If an SNTP client itself has a time with a higher stratum than an SNTP server, the time of this is sometimes rejected because it is less accurate than their own reference. It is therefore important to specify a logically correct STRATUM. The module SNTP_CLIENT ignores deliberately the STRATUM and synchronizes in each case with the SNTP server, because pretty much everyone SNTP server as a more precise time than a PLC.

9.21. SPIDER_ACCESS

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
	VALUE: STRING (Value of the variable)
	NAME: STRING(40) (Variable Name)
INPUT	MODE: BYTE (operating mode: 1 = read / write = 2)
ERROR	ERROR: DWORD (Error code)



ERROR:

Value	Properties
1	At writing variable values an error occured.
> 1	The exact meaning of ERROR is read at module HTTP_GET

With SPIDER_ACCESSvariables can be read wnd written from the PLC, which are provided by visualizations of web servers based on "spider control" from the company iniNet integrated Solution GmbH,

For the following PLC is this web server integration available:

Simatic S7 200/300/400
SAIA-Burgess PCD
Wago (750-841)
Beckhoff (CX series)
Phoenix Contact (ILC Reihe)
Selectron
Berthel
Tbox
Beck IPC

In the PLC program of target PLC, the desired variables must be released for web access. Since the communication is performed via HTTP (port 80), the data exchange is no problem, even across firewalls. Global and instance variables can be processed.

Format of variables:

At global variables, only the regular variable names has to be given. An instance variable must be specified below.
"instance name. variable name"

Mode: Read

If the MODE parameter is set to "1" and the variable name is quoted in "NAME", so cyclically a request to the HTTP to Web Server (PLC) is performed and the result is displayed the "VLAUE" as a string.

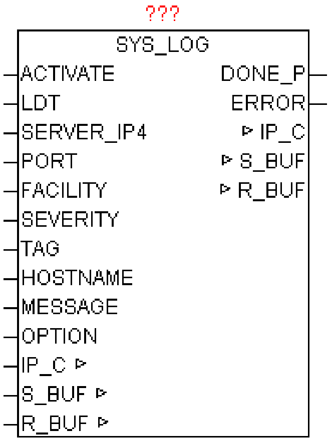
Mode: Write

If the parameter MODE is set to "2" and at "VALUE" the variable value and in "NAME" the variable name as string, then cyclically an HTTP request to the Web Server (PLC) is performed

The mode resp. the variable name can be changed in the cyclic mode at any time. If several variables have to be processed, thus only a many module instances as needed must be called.

9.22. SYS_LOG

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	R_BUF: NETWORK_BUFFER (receive data)
INPUT	ACTIVATE: BOOL (positive edge starts the query)
	LDT: DT (local time)
	SERVER_IP4: DWORD (IP address of the syslog server)
	PORT: WORD (Port number of the syslog server)
	FACILITY: BYTE (specifies the service or component)
	SEVERITY: BYTE (Classification of severity)
	TAG: STRING(32) (Process name, ID, etc.)
	HOST NAME: STRING (Name or IP address of the sender)
	MESSAGE: STRING(string_length) (Message)
	OPTION BYTE (Various)
OUTPUT	DONE: BOOL (Query completed without errors)
	ERROR: DWORD (Error code)



SYSLOG is a standard for transmitting messages in an IP computer network. The protocol is very simple - the client sends a short text message to the syslog receiver. The receiver is also called "syslog daemon" or "syslog server". The messages are sent using UDP port 514 or TCP port 1468 and includes the message in plain text. SYSLOG is typical used for computer systems management and security surveillance. This enables the easy integration of various log sources to a central syslog server. The server software is available for all platforms, sometimes known as free / shareware. Unix or Linux systems have a syslog server already integrated. Through a positive edge at ACTIVATE from the parameters of LDT, FACILITY, SEVERITY, TAG, HOST NAME, MESSAGE a syslog message is generated and sent to the SERVER_IP4 mail address. With OPTION various properties can still be controlled (See Table OPTION). After successfully sending DONE gets TRUE, otherwise ERROR is issued when the actual error message (See ERROR of module IP_CONTROL).

A syslog message has the following structure

FACILITY,SEVERITY,TIMESTAMP,HOSTNAME,TAG,MESSAGE

Example:

MAIL.ERR: Sep 10 08:31:10 149.100.100.02 PLANT2_PLC1 This is a test message generated by OSCAT SYSLOG

The following options can be used

BIT	Function
0	FALSE = with Facility, Severity code TRUE = No Facility, Severity code
1	FALSE = with RFC header TRUE = without RFC Header (only the MESSAGE alone sent)

2	FALSE = with CR,LF at end TRUE = without CR,LF end
3	FALSE = UDP Modus TRUE = TCP Modus

Severity is defined as the following standard:

Severity	Description
0	Emergency
1	Alert
2	Critical
3	Error
4	Warning
5	Notice
6	Informational
7	Debug

The following facility is defined as standard:

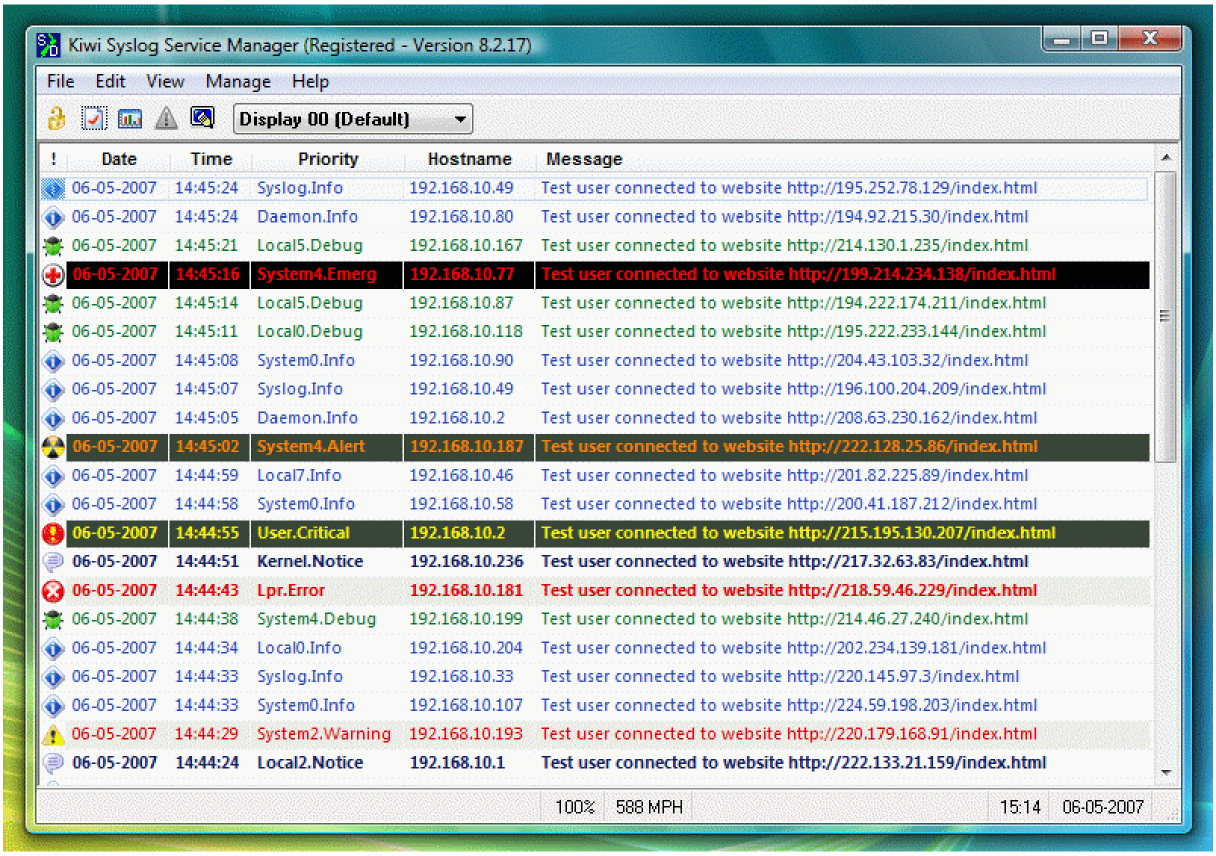
Facility	Description
00	Kernel message
01	user-level messages
02	mail system
03	system daemons
04	security/authorization messages
05	messages generated internally by syslogd
06	line printer subsystem
07	network news subsystem
08	UUCP subsystem
09	clock daemon
10	security/authorization messages
11	FTP daemon

12	NTP subsystem
13	log audit
14	log alert
15	clock daemon
16	local10
17	local11
18	local12
19	local13
20	local14
21	local15
22	local16
23	local17

For general syslog messages, the facility values 16-23 are provided (local0 to local7). But it is quite permissible to use the predefined values from 0 to 15 for own purposes.

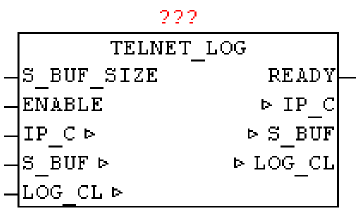
With Facility and Severity can be filtered on the SYSLOG server (database) according to certain reports, such as: "Record all error messages from the mail server with severity level.

Example (screenshot) of a syslog server for Windows



9.23. TELNET_LOG

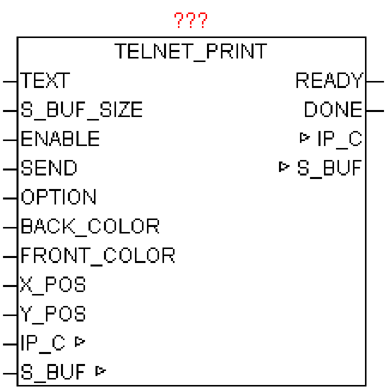
Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
	LOG_CL: LOG_CONTROL (log-data)
INPUT	S_BUF_SIZE: UINT (Size of S_BUF)
	ENABLE: BOOL (TELNET server released)
OUTPUT	READY: BOOL (TELNET client has established connection))



TELNET_LOG is used to pass all the messages in the ring LOG_CONTROL-buffer over TELNET. By "ENABLE", the module can be activated. As soon as a Telnet client connects on port 23 this is indicated by parameter "READY". Then be automatically all messages are passed to TELNET. Once occurred new reports in the course in LOG_CONTROL they are always passed automatically. When a new connection from/to rebuilds, all messages will be passed again. Most TELNET clients offer the opportunity to redirect the data stream to a file, just to make a long-term data archiving.

9.24. TELNET_PRINT

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (transmit data)
INPUT	TEXT: STRING(string_length) (output text)
	S_BUF_SIZE: UINT ((Size of the buffer S_BUF)
	ENABLE: BOOL (enable communication)
	SEND: BOOL (positive edge - Send offense)
	OPTION: BYTE (Send Options)
	BACK_COLOR: BYTE (background color)
	FRONT_COLOR: BYTE (foreground color)
	X_pos: BYTE (X-coordinate of the cursor position)
	Y_pos: BYTE (Y-coordinate of the cursor position)
OUTPUT:	READY: BOOL (module ready)
	DONE: BOOL (positive edge - Transmission completed)



The module enables easy output of text to a TELNET console. At the parameter TEXT is passed the desired string. To unlock the device for communication ENABLE = 1 must be set , so that the registration takes place at IP_CONTROL. With BACK_COLOR and FRONT_COLOR can be defined the colors you want, if the function parameter OPTION is activated. The parameters X_pos and Y_pos pass the desired coordinates of the text. If indicated in X_pos and Y_pos the value "0", the text position is inactive, and the text are always appended at the current cursor position. The standard Telnet console allows X_pos (horizontal) from 1 to 80 and a Y_pos (Vertical) 1 to 25 The behavior here can in turn be influenced by OPTION (Autowrap, carriage return, line feed, Buf_Flush etc..). If a large quantity of text will be issued, there may be a buffering enabled, so the data are written if either the buffer is full (this is from the module induced independently) or this is signaled by the amended OPTION parameter. By SEND = 1, the data is written into the buffer. The parameters may only be changed again if READY is 1, and with DONE the data acquisition was displayed as a positive edge.

OPTION:

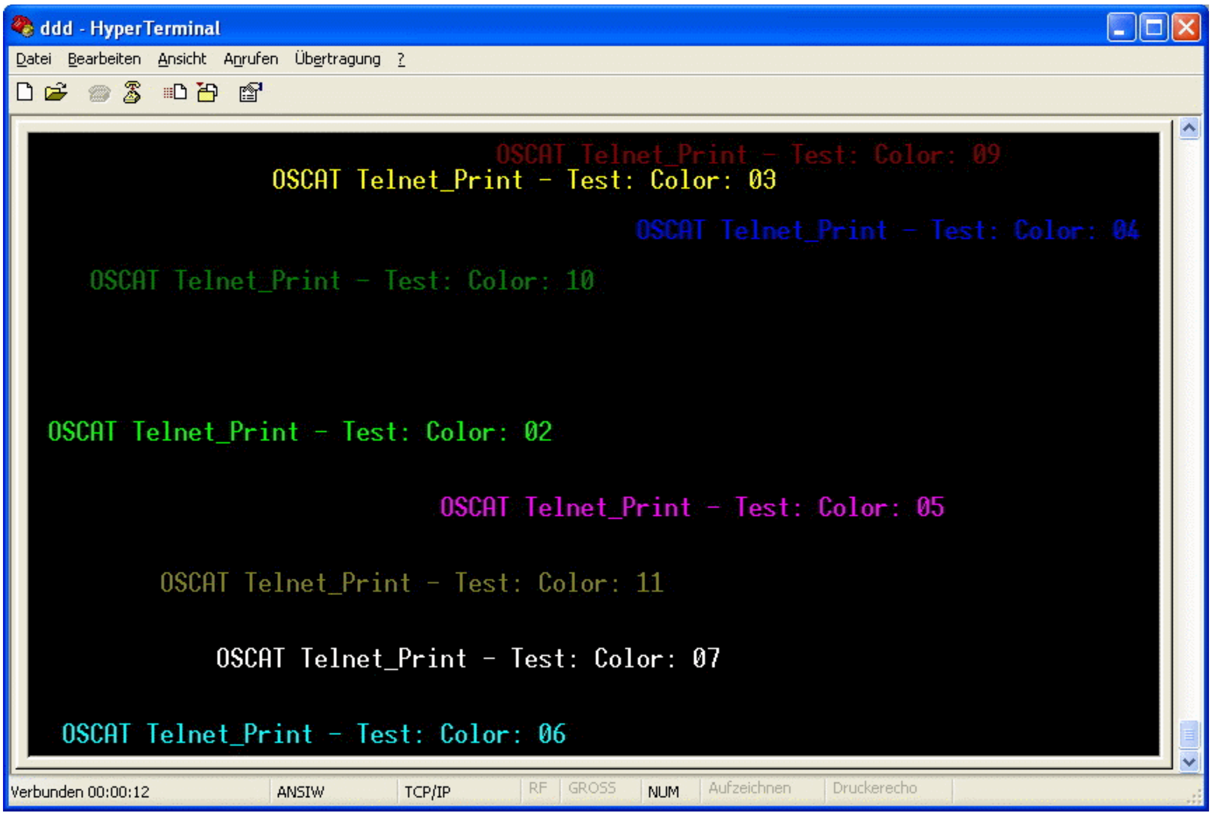
BIT	**Function**	Description
0	SCREEN_INIT	After connecting to the TELNET console the entire screen is cleared. If the COLOR OPTION is selected, the screen BACK_COLOR will be deleted.
1	AUTOWRAP	In AUTOWRAP = 1, the write cursor is on reaching the end of line is automatically set to a next line. If the text output the X,Y positions are always specified with, it is better when AUTOWRAP = 0.
2	COLOR	Enables the color mode, it will apply BACK_COLOR and FRONT_COLOR to the output.
3	NEW_LINE	In NEW_LINE = 1 is automatically a carriage return and line feed added to the end of the text. So the next text output starts a new line. But this is only useful if no X_pos and Y_pos be specified.
4	RESERVE	
5	RESERVE	
6	RESERVE	
7	NO_BUF_FLUSH	Prevents the data in the buffer to be sent immediately. Only if the buffer is completely full, or this option is disabled, the data is sent. Allows fast sending many texts in the same cycle

FRONT_COLOR:

Byte	Color	Byte	Color
0	Black	16	Flashing Black
1	Light Red	17	Flashing Light Red
2	Light Green	18	Flashing Light Green
3	Yellow	19	Flashing Yellow
4	Light Blue	20	Flashing Light Blue
5	Pink / Light Magenta	21	Flashing Pink / Light Magenta
6	Light Cyan	22	Flashing Light Cyan
7	White	23	Flashing White
8	Black	24	Flashing Black
9	Red	25	Flashing Red
10	Green	26	Flashing Green
11	Brown	27	Flashing Brown
12	Blue	28	Flashing Blue
13	Purple / Magenta	29	Purple / Magenta
14	Cyan	30	Flashing Cyan
15	Gray	31	Flashing Gray

BACK_COLOR:

Byte	Color
0	Black
1	Red
2	Green
3	Brown
4	Blue
5	Purple / Magenta
6	Cyan
7	Gray



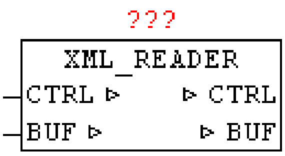
9.25. XML_READER

Type Function module:

IN_OUT CTRL: XML_CONTROL

(Control and status data)

BUF: NETWORK_BUFFER (Receive data)



XML_READER means it is possible to parse so-called 'well-formed' XML documents. Here, not as usual at high-level languages, the whole XML data is read as a data structure and stored in memory, but a very resource-friendly version is used. The XML_READER reads XML data as a sequential data stream from the buffer and signals the in COMMAND defined element types automatically back.

With XML is a strict distinction between upper and lower case. An XML document consists of just elements, attributes, their assignments, and the contents of the elements that can be text or child elements, which in turn can have attributes with assigned values and content. There are elements with and without attributes, elements can consist of many other elements, and those that may occur within the text only, and even empty elements that may have no content. The structure that emerges from these elements and their principles, can be understood as a tree structure. Elements always consist of tags and end tags. Attributes are additional information about items. There are also comment elements allowed, however, these may not between the start and end tags are of elements in XML_READER. The possible DTD - Document Type Definition only be reported as DTD, but not further evaluated and applied by XML_READER. With a CDATA section a parser is told that no markup follows, but normal text which is reported by start-end block.

Before the first call of the XML_READER a few parameters in the CTRL data structure needs to be initialized. CTRL.START_POS and CTRL.STOP_POS defines the beginning and the end of the XML data in the buffer. CTRL.-COMMAND with one hand, can be an initialization (Bit15 = TRUE) and with bit 0-14 can be defined which element / data types are reported. Here the type codes in the following table corresponds just the bit number, which has to be set to True in CTRL.COMMAND.

It is tried to pass the text of element, attribute, Value and Path in total length to the accompanying STRINGS. In STRINGS greater than 255 characters this will be cut off flush left, but with block-start and block-end parameters is reported back, so that they can subsequently be evaluated yet complete. The BLOCK-START/STOP index are is always passed parallel to the STRINGS. If the PATH STRING is greater than 255 characters so the PATH tracking is disabled and only "OVERFLOW" is entered as text.

Since for very large and complex XML data is not clear, how long it takes until the module find data to report back, an WATCHDOG function is integrated. A maximum processing time can be parameterized. When reaching the time limit the module call is automatically canceled, and the next cycle resumes at the same point. The type code 98 is returned.

The following type codes are defined.

Type (Code)	Data Type	Description
00	Unknown	Undefined item found
01	TAG (element)	Start - of Element Data pointer for the element of BLOCK1

02	END-TAG (Element)	End - of the element
03	TEXT	Content of an element Data pointer for value on BLOCK1
04	ATTRIBUTE	Attributes of an element Data pointer for attributes of BLOCK1 Data pointer for value on BLOCK2
05	TAG (Processing Instruction)	Instructions for processing Data pointer for the element of BLOCK1
12	CDATA	not analyzed content TEXT Data pointer for value on BLOCK1
13	COMMENT	COMMENT Data pointer for value on BLOCK1
14	DTD	Document Type Declaration Data pointer for value on BLOCK1
98	WATCHDOG	Maximum processing time reached - cancel
99	END	No more items available

Sample XML

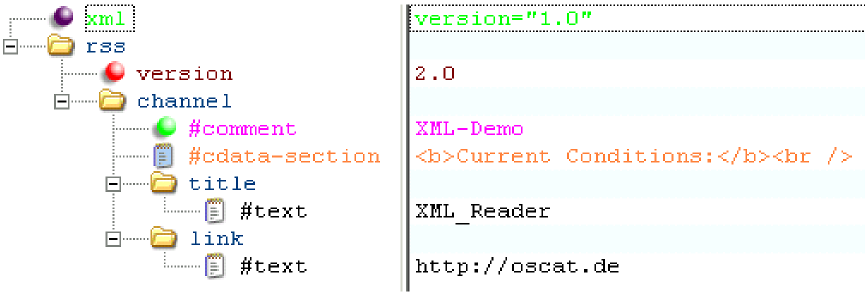
flat display

<?xml version="1.0" ?><rss version="2.0"><channel><!-- XML-Demo --><![CDATA[CurrentConditions:
]]><title>XML_Reader</title>
<link>http://oscat.de</link></channel></rss>

Representation of the levels (without processing Instruction)

```
- <rss version="2.0">
  - <channel>
      <!-- XML-Demo -->
      <b>Current Conditions:</b><br />
      <title>XML_Reader</title>
      <link>http://oscat.de</link>
    </channel>
  </rss>
```

View as tree of item types



Legend:

Element		Comment	
Attribute		Processing Instruction	
Text		CDATA Section	

Application example:

```
  CASE STATE OF
00:
  STATE := 10;
  CTRL.START_POS := HTTP_GET.BODY_START; (Index des ersten Zeichen *)
  CTRL.STOP_POS  := HTTP_GET.BODY_STOP;  (Index of last character *)
  CTRL.COMMAND   := WORD#2#11111111_11111111; (* Init + report all elements *)
10:
  (* XML * data read serial)
  XML_READER.CTRL := CTRL;
  XML_READER.BUF := BUFFER;
  XML_READER();
  CTRL := XML_READER.CTRL;
  BUFFER := XML_READER.BUF;

  IF CTRL.TYP = 99 THEN
    STATE := 20; (* Exit - no further elements available *)
  ELSIF CTRL.TYP < 98 THEN (* do nothing at timeout(Code 98) *)
    (* Evaluation of the XML elements by pressing CTRL-data structure *)
  END_IF;
20:
   (* sonstiges...... *)
END_CASE;
```

The following information is passed via the CTRL-data structure

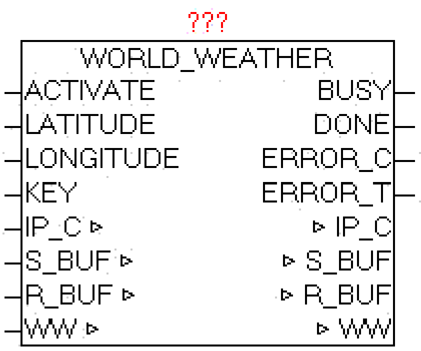
```
--------First pass --------
COUNT:          1
TYPE:           5        (OPEN ELEMENT - PROCESSING
INSTRUCTION)
LEVEL:          1
ELEMENT:        'xml'
PATH:           '/xml'
--------Next cycle--------
COUNT:          2
TYPE:           4        (ATTRIBUTE)
LEVEL:          1
ELEMENT:        'xml'
ATTRIBUTE:               'version'
VALUE:          '1.0'
PATH:           '/xml'
--------Next cycle--------
COUNT:          3
TYPE:           2        (CLOSE ELEMENT)
LEVEL:          0
ELEMENT:        'xml'
PATH:           ''
--------Next cycle--------
COUNT:          4
TYPE:           1        (OPEN ELEMENT - Standard)
LEVEL:          1
ELEMENT:        'rss'
PATH:           '/rss'
--------Next cycle--------
COUNT:          5
TYPE:           4        (ATTRIBUTE)
LEVEL:          1
ELEMENT:        'rss'
ATTRIBUTE:               'version'
VALUE:          '2.0'
PATH:           '/rss'
--------Next cycle--------
COUNT:          6
TYPE:           1        (OPEN ELEMENT - Standard)
LEVEL:          2
ELEMENT:        'channel'
PATH:           '/rss/channel'
--------Next cycle--------
COUNT:          7
TYPE:           13       (COMMENT-ELEMENT)
LEVEL:          2
VALUE:          ' XML-Demo '
PATH:           '/rss/channel'
```

```
--------Next cycle--------
COUNT:          8
TYPE:           12          (CDATA)
LEVEL:          2
VALUE:          '<b>Current Conditions:</b><br />'
PATH:           '/rss/channel'
--------Next cycle--------
COUNT:          9
TYPE:           1           (OPEN ELEMENT - Standard)
LEVEL:          3
ELEMENT:        'title'
PATH:           '/rss/channel/title'
--------Next cycle--------
COUNT:          10
TYPE:           3           (TEXT)
LEVEL:          3
ELEMENT:        'title'
VALUE:          ' XML_Reader'
PATH:           '/rss/channel/title'
--------Next cycle--------
COUNT:          11
TYPE:           2           (CLOSE ELEMENT)
LEVEL:          2
ELEMENT:        'title'
PATH:           '/rss/channel'
--------Next cycle--------
COUNT:          12
TYPE:           1           (OPEN ELEMENT - Standard)
LEVEL:          3
ELEMENT:        'link'
PATH:           '/rss/channel/link'
--------Next cycle--------
COUNT:          13
TYPE:           3           (TEXT)
LEVEL:          3
ELEMENT:        'link'
VALUE:          'http://oscat.de'
PATH:           '/rss/channel/link'
--------Next cycle--------
COUNT:          14
TYPE:           2           (CLOSE ELEMENT)
LEVEL:          2
ELEMENT:        'link'
PATH:           '/rss/channel'
--------Next cycle--------
COUNT:          15
TYPE:           2           (CLOSE ELEMENT)
LEVEL:          1
```

ELEMENT:	'channel'	
PATH:	'/rss'	
--------Next cycle--------		
COUNT:	16	
TYPE:	2	(CLOSE ELEMENT)
LEVEL:	0	
ELEMENT:	'rss'	
PATH:	''	
--------Next cycle--------		
COUNT:	17	
TYPE:	99	(EXIT – END OF DATA)

9.26. WORLD_WEATHER

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (Transmit data)
	R_BUF: NETWORK_BUFFER (Receive data)
	WW: WORLD_WEATHER_DATA (Weather data)
INPUT	ACTIVATE: BOOL (positive edge starts the query)
	LATITUDE: REAL (latitude of the reference location)
	LONGITUDE : REAL (longitude of the reference location)
	KEY: STRING (30) (API-Key)
OUTPUT	BUSY: BOOL (Query is active)
	DONE: BOOL (Query completed without errors)
	ERROR_C: DWORD (Error code)
	ERROR_T: BYTE (error type)



The module loads the current weather data for the specified location of http://worldweather.com down, analyzes the data and stores the essential data processed in the WORLD_WEATHER_DATA data structure.

Following values are stored from the current day.

Observation time (UTC) Temperature (°C), Unique Weather code
Weather description text, wind speed in miles per hour, wind speed in kilometer per hour, wind direction in degree, 16-point wind direction compass, precipitation amount in millimeter, Humidity (%), Visibility (km) Atmospheric pressure in milibars
, Cloud cover (%)

From the current day and the next four days the following values are stored.

Date For which the weather is forecasted,
Day and night temperature in °C (Celsius) and °F (Fahrenheit)
Wind speed in mph (miles per hour) and kmph (kilometers per hour)
16-point compass wind direction, A unique weather condition code;
Weather description text , Precipitation Amount (millimetre)

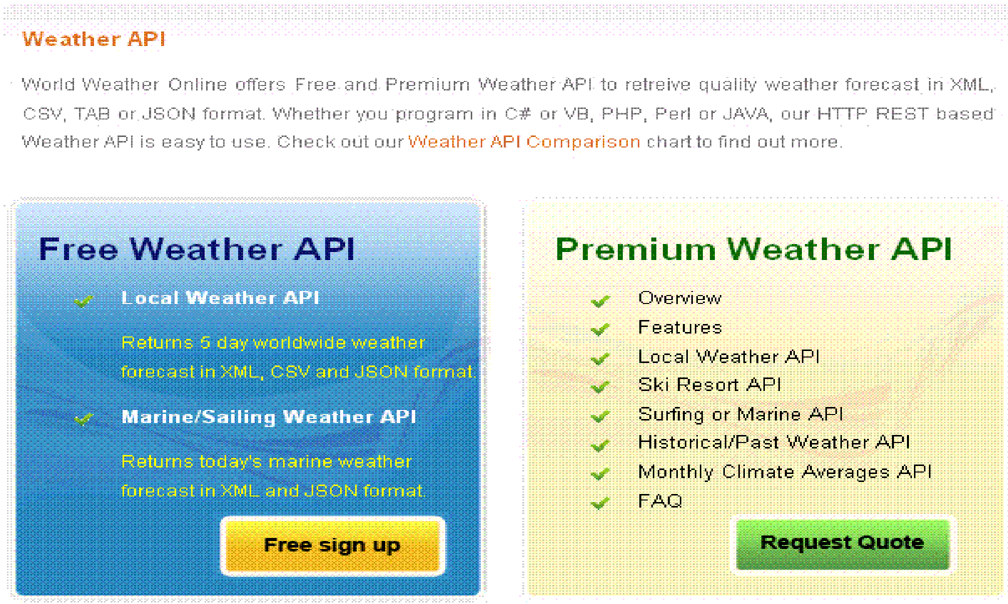
With a positive edge of ACTIVATE, the query started and process a DNS query with the following HTTP-GET. After successful receiving all data elements are processed and if necessary stored in the data structure in converted form. By the parameters of latitude and longitude the exact place (geographical position) of the weather is indicated. While the query is active, BUSY = TRUE is passed. After successful completion of the query DONE = TRUE is shown. If an error occurs during the query it is reported in ERROR_C in combination with ERROR_T.

ERROR_T:

Value	Properties
1	The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	The exact meaning of ERROR_C can be read at module HTTP_GET

Creating a new API KEY:

Use your Internet browser and call the page http://www.worldweatheronline.-com, call the "Free sign up" registration dialog, and fill out the required fields. After registering, an email is sent, in turn, has to be confirmed, and subsequently secon ad e-mail is sent with the personal API key. This API Key must be passed to the module-KEY API parameters.



Determine Latitude and longitude of a specific place:

Use your Internet browser to access http://www.mygeoposition.com/ page, and enter the name of the desired location and search the location using "calculate the spatial data" . Then the desired location is shown on the map, including the latitude and longitude needed in decimal notation. The determined position has to be passed to the block parameters, latitude and longitude.

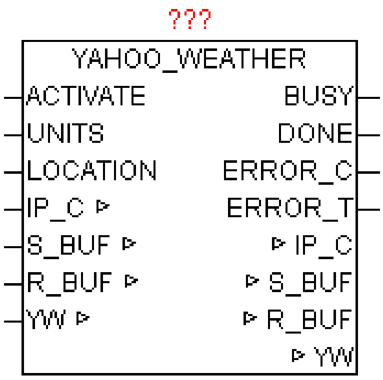
9.27. YAHOO_WEATHER

Type	Function module:
IN_OUT	IP_C: IP_C (parameterization)
	S_BUF: NETWORK_BUFFER (Transmit data)
	R_BUF: NETWORK_BUFFER (Receive data)
	YW: YAHOO_WEATHER (weather data)
INPUT	ACTIVATE: BOOL (positive edge starts the query)
	UNITS: BOOL (FALSE = Celsius, TRUE = Fahrenheit)
	LOCATION: STRING (20) (location specified by LOCATION-ID)
OUTPUT	BUSY: BOOL (Query is active)

DONE: BOOL (Query completed without errors)

ERROR_C: DWORD (Error code)

ERROR_T: BYTE (error type)



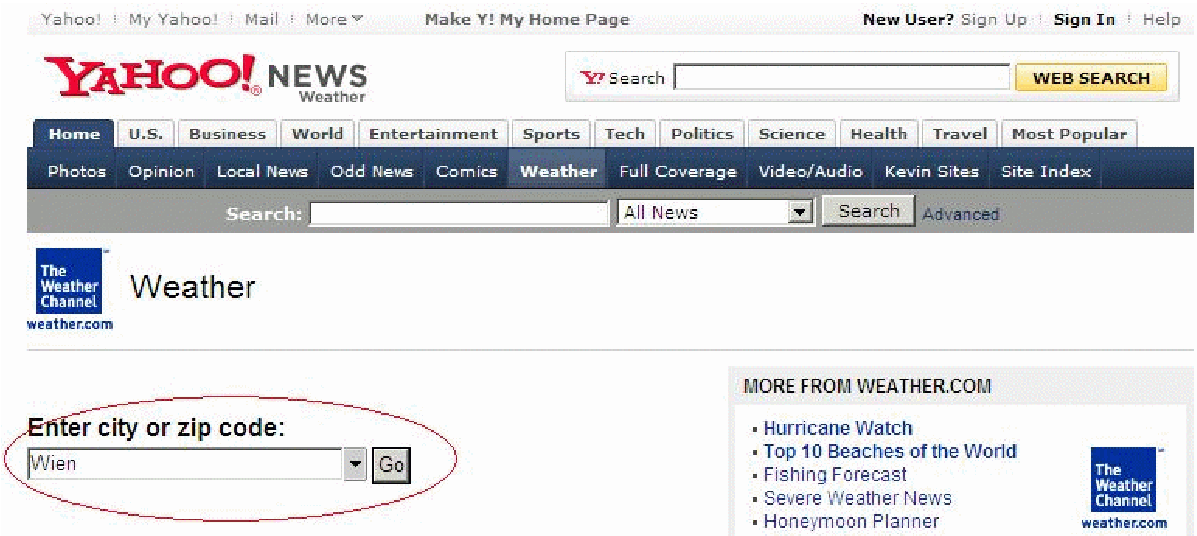
The module loads the current weather data for the specified location using an RSS feed (XML data structure) of http://weather.yahooapis.com down, analyzes the XML data and provides the essential data processed from the YAHOO_WEATHER data structure. With a positive edge of ACTIVATE, the query started and process a DNS query with the following HTTP-GET. After successful receipt of data by XML_READER all elements are processed and if necessary stored in the data structure in converted form. With UNITS may still be selected between Fahrenheit and Celsius as a unit. By specifying the precise LOCATION_ID the location of the weather is indicated. While the query is active, BUSY = TRUE is passed. After successful completion of the query DONE = TRUE is shown. If occur in the query, then this error is reported under ERROR_C in combination with ERROR_T.

ERROR_T:

Value	Properties
1	The exact meaning of ERROR_C can be read at module DNS_CLIENT
2	The exact meaning of ERROR_C can be read at module HTTP_GET

Find the Location ID of a specific place:

Use your Internet browser the page http://weather.yahoo.com/ and in the field: "Enter city or zip code" and enter the name of the desired location and search.



After being selected in the browser window displays the current weather information of the specified location. In the URL (web link) line is now the location ID can be seen.

Thus, the desired settlement "Wien (Vienna)" returns the Location ID "551801".
This code must be passed on the module as parameters.

Example of an RSS feed:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
<rss version="2.0" xmlns:yweather="http://weather.yahooapis.com/ns/rss/1.0"
 xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#">
<channel>
  <title>Yahoo! Weather - Sunnyvale, CA</title>
  <link>http://us.rd.yahoo.com/dailynews/rss/weather/Sunnyvale__CA/
  *http://weather.yahoo.com/forecast/94089_f.html</link>
  <description>Yahoo! Weather for Sunnyvale, CA</description>
  <language>en-us</language>
  <lastBuildDate>Tue, 29 Nov 2005 3:56 pm PST</lastBuildDate>
  <ttl>60</ttl>
  <yweather:location city="Sunnyvale" region="CA" country="US"></yweather:location>
  <yweather:units temperature="F" distance="mi" pressure="in" speed="mph"></yweather:units>
  <yweather:wind chill="57" direction="350" speed="7"></yweather:wind>
  <yweather:atmosphere humidity="93" visibility="1609" pressure="30.12"  rising="0"></yweather:atmosphere>
  <yweather:astronomy sunrise="7:02 am" sunset="4:51 pm"></yweather:astronomy>
```

```
  <image>
    <title>Yahoo! Weather</title>
    <width>142</width>
    <height>18</height>
    <link>http://weather.yahoo.com/</link>
    <url>http://us.i1.yimg.com/us.yimg.com/i/us/nws/th/main_142b.gif</url>
  </image>
  <item>
    <title>Conditions for Sunnyvale, CA at 3:56 pm PST</title>
    <geo:lat>37.39</geo:lat>
    <geo:long>-122.03</geo:long>
    <link>http://us.rd.yahoo.com/dailynews/rss/weather/
    <span style="font-size: 0px"> </span>Sunnyvale__CA/*
    <span style="font-size: 0px"> </span>http://weather.yahoo.com/<span style="font-size: 0px"> </span>forecast/94089_f.html
    </link>
    <pubDate>Tue, 29 Nov 2005 3:56 pm PST</pubDate>
    <yweather:condition text="Mostly Cloudy" code="26" temp="57" date="Tue, 29 Nov 2005 3:56
      pm PST"></yweather:condition>
    <description><![CDATA[
 <img src="http://us.i1.yimg.com/us.yimg.com/i/us/we/52/26.gif" /><br />
 <b>Current Conditions:</b><br />
 Mostly Cloudy, 57 F<p />
 <b>Forecast:</b><BR />
  Tue - Mostly Cloudy. High: 62 Low: 45<br />
  Wed - Mostly Cloudy. High: 60 Low: 52<br />
  Thu - Rain. High: 61 Low: 46<br />
<br />
<a href="http://us.rd.yahoo.com/dailynews/rss/weather/Sunnyvale__CA/*http://weather.yahoo.com/forecast/94089_f.html">Full
Forecast at Yahoo! Weather</a><BR/>
 (provided by The Weather Channel)<br/>]]>
    </description>
    <yweather:forecast day="Tue" date="29 Nov 2005" low="45" high="62" text="Mostly Cloudy"
      code="27"></yweather:forecast>
    <yweather:forecast day="Wed" date="30 Nov 2005" low="52" high="60" text="Mostly Cloudy"
      code="28"></yweather:forecast>
    <guid isPermaLink="false">94089_2005_11_29_15_56_PST</guid>
  </item>
</channel>
</rss>
```


The XML data the required elements are processed and stored in the YAHOO_WEATHER data structure.

10. File-System

10.1. CSV_PARSER_BUF

Type Function module

IN_OUT SEP : BYTE (devider)

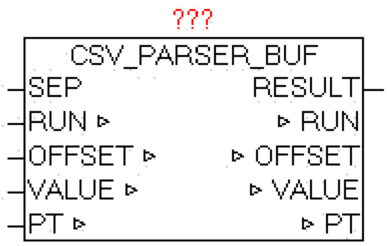
RUN: BYTE (command code for current action)

OFFSET: UDINT (current file offset of the query)

VALUE: STRING (STRING_LENGTH) (value of a key)

PT: NETWORK_BUFFER (read data buffer)

OUTPUT: RESULT: BYTE (result of query)



The module CSV_PARSER_BUF enables the analysis of the elements contained in the buffer. The number of data contained on PT.SIZE specified. The separator is specified in parameter "SEP". The search for elements that always begins, depending on the given "OFFSET", so it is very easy to look at certain points in order to not always have to search the entire buffer. At the beginning should be started with by default the OFFSET 0 (but need not).

At the beginning of the default should be started OFFSET 0 (but need not). Of course this is dependent on the content or the structure of the data.

Evaluate elements:

Will specify in SEP 0, lines are always evaluated completely and parameter "VALUE" is issued. If the elements in the buffer are structured as CSV (Excel), so at SEP the separator ',' or something else can be specified. RUN = 1 startes the evaluation. Since it is not foreseeable how long the search takes, a watchdog function is Integrated that stops the search for the current cycle, then RESULT = 5 and RUN remains unchanged. In the next cycle, the analysis proceeds automatically. As soon as the next element is detected, the element in VALUE is passed, and RESULT is 1. If the element is also the last in a line, then RESULT = 2 is the output. As soon as the end of the data has been reached at RESULT = 10 passed. Always if yet RUN = 0 is output, RESULT defines the result. If an item is longer than the maximum length (string_length) so the characters are cut off automatically. The parameter OFFSET is by the module automatically passed after each result, but can be defined individually before each evaluation.

Example 1

Analyze data by lines:

Zeile 1<CR,LF>
Line 2<CR,LF>

Default: offset 0, SEP = 0 and RUN = 1
VALUE = 'Line 1', RUN = 0, RESULT = 2
Default: RUN = 1
VALUE = 'line 2', RUN = 0, RESULT = 2
RUN set back to 1
VALUE = '', RUN = 0 , RESULT = 10

Example 2

Analyze data as individual elements:

10,20<CR,LF>
a,b<CR,LF>

Offset 0 , SEP = ',' und RUN = 1
VALUE = '10', RUN = 0 , RESULT = 1
RUN set back to 1
VALUE = '20', RUN = 0 , RESULT = 2
RUN set back to 1
VALUE = 'a', RUN = 0 , RESULT = 1
RUN set back to 1
VALUE = 'b', RUN = 0 , RESULT = 2
RUN set back to 1
VALUE = '', RUN = 0 , RESULT = 10

RUN: Feature List

RUN	Function
0	No function to perform - and last function is complete
1	Element to evaluate

RESULT: Result - Feedback

RESULT	Description
1	Element found
2	Element and the end of the line identified
5	Current query is still running - call module further cyclical!
10	Nothing found - reached the end of data

10.2. CSV_PARSER_FILE

Type Function module

IN_OUT SEP : BYTE (devider)

FILE NAME: STRING (file name)

FSD: FILE_SERVER_DATA (file interface)

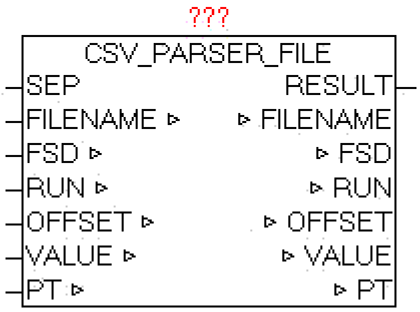
RUN: BYTE (command code for current action)

OFFSET: UDINT (current file offset of the query)

VALUE: STRING (STRING_LENGTH) (value of a key)

PT: NETWORK_BUFFER (read data buffer)

OUTPUT: RESULT: BYTE (result of query)



The module CSV_PARSER_FILE enables the analysis of the elements of an arbitrarily large file which is read into the read data buffer block by block for automatically processing. The separator is specified in parameter "SEP". The name of the file is passed in parameter "FILENAME". The search for elements that always begins, depending on the given "OFFSET", so it is very easy to look at certain points in order to not always have to search the entire buffer. At the beginning should be started with by default the OFFSET 0 (but need not).

When queried by elements of the file, there are various procedures. Of course this is dependent on the content or the structure of the data.

Evaluate elements:

Will specify in SEP 0, lines are always evaluated completely and parameter "VALUE" is issued. If the elements in the file are structured as CSV (Excel), so at SEP the separator ',' or something else can be specified. RUN = 1 startes the evaluation. Since it is not foreseeable how long the search takes, a watchdog function is Integrated that stops the search for the current cycle, then RESULT = 5 and RUN remains unchanged. In the next cycle, the analysis proceeds automatically. As soon as the next element is detected, the element in VALUE is passed, and RESULT is 1. If the element is also the last in a line, then RESULT = 2 is the output. As soon as the end of the data has been reached at RESULT = 10 passed. Always if yet RUN = 0 is output, RESULT defines the result. If an item is longer than the maximum length (string_length) so the characters are cut off automatically. The parameter OFFSET is by the module automatically passed after each result, but can be defined individually before each evaluation.

Example 1

evaluate Text file line by line:

Zeile 1<CR,LF>
Line 2<CR,LF>

Default: offset 0, SEP = 0 and RUN = 1
VALUE = 'Line 1', RUN = 0, RESULT = 2
Default: RUN = 1
VALUE = 'line 2', RUN = 0, RESULT = 2
RUN set back to 1
VALUE = '', RUN = 0 , RESULT = 10

Example 2

Analyze data as individual elements:

10,20<CR,LF>
a,b<CR,LF>

Offset 0 , SEP = ',' und RUN = 1
VALUE = '10', RUN = 0 , RESULT = 1
RUN set back to 1
VALUE = '20', RUN = 0 , RESULT = 2
RUN set back to 1
VALUE = 'a', RUN = 0 , RESULT = 1
RUN set back to 1
VALUE = 'b', RUN = 0 , RESULT = 2
RUN set back to 1
VALUE = '', RUN = 0 , RESULT = 10

If the file access is no longer needed, the user must close the file be either by use of AUTO_CLOSE or MODE 5 (close file) of the FILE_SERVER.

RUN: Feature List

RUN	Function
0	No function to perform - and last function is complete
1	Element to evaluate

RESULT: Result - Feedback

RESULT	Description
1	Element found
2	Element and the end of the line identified
5	Current query is still running - call module further cyclical!
10	Nothing found - reached the end of data

10.3. FILE_BLOCK

Type Function module

INPUT PT_SIZE: UINT (number of bytes in the buffer)

FILE NAME: STRING (file name)

POS: UDINT (current file reading position)

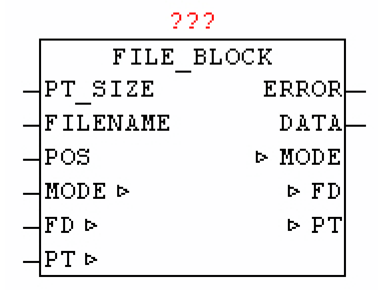
OUTPUT: ERROR: BYTE (error code - See module FILE_SERVER)

DATA: BYTE (BYTE of the requested file position)

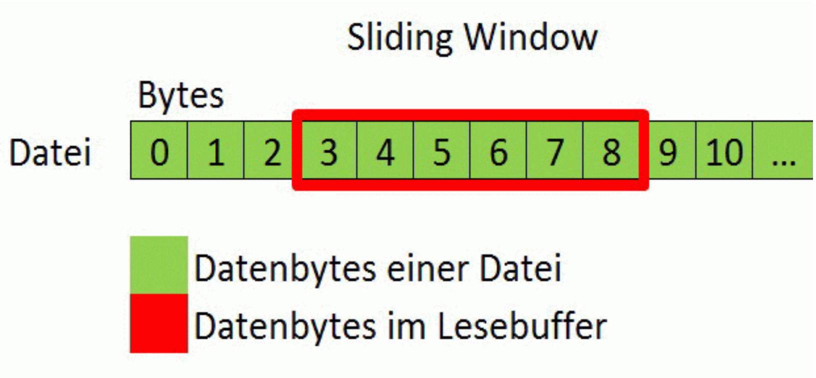
IN_OUT MODE: BYTE (Current mode)

FD: FILE_SERVER_DATA (File Interface)

PT: NETWORK_BUFFER (read data)



The module FILE_BLOCK provides access to files of any size by a data block that is always kept in a read buffer. If the requested byte of a file is not stored in last block of data, automatically a matching new data block is read and the desired byte is putted out. The greater the read buffer is the less frequently a block must be read again. Optimally it is a linear access to the bytes, so that as seldom as possible, a data block must be read anew.



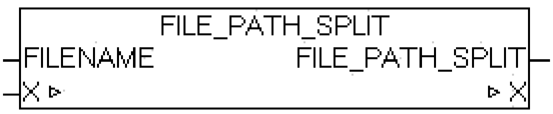
Procedure:

The Parameter FILENAME specifies the name of the file to be read, and with PT_SIZE the size of the read buffer is specified in bytes. The value for parameter POS is the exact data position within the file, which has to be read. The process is triggered by setting MODE to 1. Then the system automatically checks whether the desired data byte is already in the read buffer. If not, then a new matching block of data is copied into the read buffer, and the desired data byte is passed on the parameter DATA. As long as this operation is not finished yet, MODE remains at 1, and only after completion of the operation of module is reset to MODE = 0. If a specified data position is larger than the current length of the file or the file has length 0, so the output at ERROR is 255 (See ERROR codes from block FILE_SERVER).

If the file access is no longer needed, the user must close the file be either by use of AUTO_CLOSE or MODE 5 (close file) of the FILE_SERVER.

10.4. FILE_PATH_SPLIT

Type	Function: BOOL
INPUT	FILE NAME: STRING (string_length)
IN_OUT	X: FILE_PATH_DATA ' (Single path elements)



The module split a file path into its component elements. The drive, path and file name are extracted and stored in the data structure X. As directory separator "\" and "/" will be accepted. If the passed "File name" is not empty and elements can be evaluated, the module returns TRUE, otherwise FALSE.

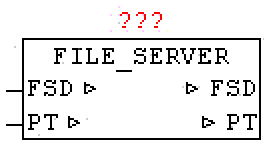
Example:

c: \folder1\dir2\oscat.txt

DRIVE DIRECTORY FILE NAME

10.5. FILE_SERVER

Type	Function module
IN_OUT	FSD: FILE_SERVER_DATA (file interface)
	PT: NETWORK_BUFFER (read / write data)



Available platforms and related dependencies

CoDeSys:

Does the library " SysLibFile.lib "

Runs on

WAGO 750-841

CoDeSys SP PLCWinNT V2.4

and compatible platforms

PCWORX:

No library required

Runs on all controllers with file system from firmware >= 3.5x

BECKHOFF:

Development Environment	Target Platform	PLC libraries to include
TwinCAT v2.8.0 or higher	PC or CX (x86)	TcSystem.Lib
TwinCAT v2.10.0 Build >= 1301 or higher	CX (ARM)	TcSystem.Lib

The module FILE_SERVER enables hardware and manufacturers a neutral access to the file system of PLC. Since at almost every hardware and software platform, the accessibility to the file system is sometimes very different, it is necessary to use a uniform and simplified functional interface, which is reduced to the necessary functions. The module is hardware-dependent and therefore it must be available for that platform are the appropriate implementation.

WIth FILE NAME the file is determined. Depending on the platform may be slightly different syntax (with or without the path). With MODE parameter the principle of access is given. At MODE 1,2 and 3 with "OFFSET" the position can be specified in the file. In the file system counting is always started with byte 0. The first step is always to check whether this file is already (still) open, and if not they will open and the current file size is observed and passed to the "FILE_SIZE". When specifying a time AUTO_CLOSE > 0ms, the file is automatically closed after each command and the expiration of the waiting time. Alternatively, using MODE = 5, the closing of the file is done manually. Each write command which change the size of the file automatically leads to a corrected "FILE_SIZE" entry, so it is always visible how large the file is right now. Once a file is open, this is reported on FILE_OPEN = TRUE.

Each write command at which the size of the file changes automatically leads to a corrected "FILE_SIZE" entry, so you can always how large the file is right now. In PT.SIZE is the actual amount of data automatically corrected or entered.

If the MODE 1,2 or 3 called with PT.SIZE = 0, the file is opened, the FILE_SIZE determined, but no read/write command is performed, and the file will remain open until manually closed or AUTO_CLOSE.

If data has to be written, the data has to be stored in PT.BUFFER and in PT.SIZE the bytes must exist. This data are written to the specified relative offset in the file. If a write mode is called with PT.SIZE = 0, then in turn the file is opened (if not already open, and made no write command, and these will remain open until a manual closing or AUTO_CLOSE is carried out.)

After every executed command that changes the position of the virtual read / write pointer, the current position in the data structure is written in the parameter "OFFSET". An automatic append function can be realized very easy. The parameter FILE_SIZE has to be written to the OFFSET parameter after opening the file. After that, all written bytes are appended to the end without changing the OFFSET parameter manually. The same principle can be applied of course when reading, the read pointer should be positioned first within the file (usually starting at offset 0).

If a command is executed and FILE NAME differs from the current FILE NAME, the old one, still open file, is closed automatically and the new one is opened then, and continued with the normal command. This can easily perform a flying change of the file without having to perform cumbersome and OPEN to CLOSE before.

When you delete a file with MODE 4 automatically a potentially outstanding file is closed before, and then deleted in sequence.

After a AUTO_CLOSE or manual closing by MODE 5 all data in FILE_SERVER_DATA is reseted.

The module FILE_SERVER should always be called periodically, at least as long as not all requests are completed safely.

Since some platforms perform a file-lock (eg CoDeSys) and do not always allow an asynchronous use, FILE_SERVER should run in a separate task so that the default application is not influenced in the time behavior. .

The FILE_SERVER provides the following commands in "MODE":

MODE	**Properties**
1	An existing file is opened for reading and reading data optional
2	An existing file is opened for write access and optional data is written
3	A file will be created for writing and data will be written optional
4	Delete file
5	Close file

ERROR: Error codes Beckhoff

Value	trigger	Description
0		No error
19	SYSTEMSERVICE_FOPEN	Unknown or invalid parameter
28	SYSTEMSERVICE_FOPEN	File not found. Invalid file name or file path
38	SYSTEMSERVICE_FOPEN	SYSTEMSERVICE_FOPEN

51	SYSTEMSERVICE_FCLOSE	unknown or invalid file handle.
62	SYSTEMSERVICE_FCLOSE	File was opened with the wrong method.
67	SYSTEMSERVICE_FREAD	unknown or invalid file handle.
74	SYSTEMSERVICE_FREAD	No memory for read buffer.
78	SYSTEMSERVICE_FREAD	File was opened with the wrong method.
83	SYSTEMSERVICE_FWRITE	unknown or invalid file handle
94	SYSTEMSERVICE_FWRITE	File was opened with the wrong method.
99	SYSTEMSERVICE_FSEEK	unknown or invalid file handle.
110	SYSTEMSERVICE_FSEEK	File was opened with the wrong method.
115	SYSTEMSERVICE_FTELL	unknown or invalid file handle.
126	SYSTEMSERVICE_FTELL	File was opened with the wrong method
140	SYSTEMSERVICE_FDELETE	File not found. Invalid file name or file path.
255	Application	Position is after the end of file

ERROR: Error codes PCWORX:

Value	trigger	Description
0		No error
2	File_open	The maximum number of files already open
4	File_open	The file is already open
5	File_open	The file is write-protected or access denied
6	File_open	File name not specified
11	File_close	Invalid file handle
30	File_close	File could not be closed
41	FILE_READ	Invalid file handle
50	FILE_READ	End of file reached
52	FILE_READ	The number of characters to read is larger than the data buffer
62	FILE_READ	Data could not be read
71	FILE_WRITE	Invalid file handle

81	FILE_WRITE	There is no memory available to write the data
82	FILE_WRITE	The count of characters to write is larger than the data buffer
93	FILE_WRITE	There were no written data
0	FILE_SEEK	Invalid file handle
113	FILE_SEEK	Invalid positioning mode or the specified position is before the start of the file
124	FILE_SEEK	The position could not be set
131	FILE_TELL	Invalid file handle
142	FILE_REMOVE	The maximum number of files already open
143	FILE_REMOVE	The file could not be found
145	FILE_REMOVE	The file is opened, readonly or access denied
146	FILE_REMOVE	File name not specified
161	FILE_REMOVE	File could not be deleted
255	Application	Position is after the end of file

ERROR: CoDeSys error codes:

Value	trigger	Description
0		No error
1	SysFileOpen	Error
2	SysFileClose	Error
3	SysFileRead	Error
4	SysFileWrite	Error
5	SysFileSetPos	Error
6	SysFileGetPos	Error
7	SysFileDelete	Error
8	SysFileGetSize	Error
255	Application	Position is after the end of file

10.6. INI-DATEIEN

An initialization file (INI file in short) is a text file, which Windows uses to store program settings (such as location of a program). Re-starting the program, the program settings can be imported to retake the state before the last closing.

Due to the very simple functional structuring and handling, this default standard ist used for program settings and for similar PLC with a file system..

An INI file can be divided into sections, which must be enclosed in square brackets. Information is read out as a key with an associated value.

When you create an ini file the following rules apply:

Each section must be unique.

Each key may appear only once per section.

Values are accessed by means of section and key.

A section may also contain no key

Comments start with a "#"

Comments must not be directly behind a key or a section.

Comments must always start on a new line

Is given no value for a key, an empty string is reported as the value.

Each section and each key or the following value must end with a newline. In this case, the type of newline character does not matter since all variants are accepted. Most common variant is <CR><LF> . All control characters (not printable characters) are interpreted as end of line.

Space is always considered as part of the elements and is evaluated in the same manner.

In principle any number of section and key can be used.

Basic structure:

#Comment <CR><LF>

[Section] <CR><LF>

#Comment <CR><LF>

Key = value <CR><LF>

Example:

```
[SYSTEM]
DEBUG_LEVEL=10
QUIT_TIME=5

#---------------------------
Station 1 Parameter -
#---------------------------
[Station_1]
NAME=ILC150 ETH
IP=192.168.15.100
M2=S2/M3/C1
#---------------------------
# Station 2 parameters -
#---------------------------
[Station_2]
NAME=ILC350PN
IP=192.168.15.108
M1=S1/M1
M2=S3/M2
```

10.7. INI_PARSER_BUF

Type Function module

OUTPUT: RESULT: BYTE (result of query)

IN_OUT STR: STRING(STRING_LENGTH) (searched item)

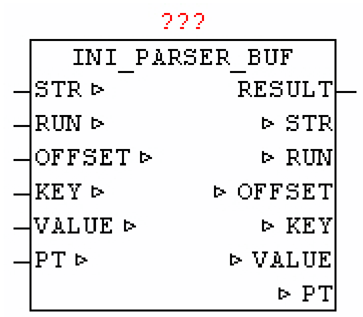
RUN: BYTE (command code for current action)

OFFSET: UDINT (current file offset of the query)

KEY: STRING(STRING_LENGTH) (found item)

VALUE: STRING (STRING_LENGTH) (value of a key)

PT: NETWORK_BUFFER (read data buffer)



The module INI_PARSER_BUF enables the analysis of elements of a INI file stored in a Byte-Array . Before queries can be processed the user must fill the byte array PT.BUFFER with the ini data, and the number of bytes has to be stored in PT.SIZE. The search for elements always begins on the given depended "OFFSET", and hence is very easy to look only at certain positions, or to repeat the search from a specific section to browse not always the entire byte array. At the initial search should start default to OFFSET 0 (but may not!). When querying sections and keys, there are various procedures. Either it is queried to a Section and evaluates all of the following keys by individual queries, or to use in very large initialization file the classic enumeration (listing), which means it will be report serially all the elements, and processed by the application.

Section Search:

To determine the OFFSET of a specific Section, STR must declare the name of the Section and the offset can be set to a position that is located before of the searched section. Should only the nearest available section be found, at STR an empty sting must be passed. The search query is started by RUN = 1. The search will take different time, depending on the structure and size of the INI data, it takes an indefinite number of cycles until a positive or negative result is achieved. Once the search is finished, the INI_PARSER_BUF sets the parameters of RUN to 0. RESULT passes the result of the search to output. Upon successful search the name of the section is shown at parameters KEY. And then the OFFSET parameter points to the end of the section line. Thus, immediately after that the key evaluation can be continued, without having to manually change the OFFSET.

Key Search:

Before a Key is evaluated, the OFFSET must have a correct value, this can be done by manual set of OFFSET or by a previously executed Section search. Before running the query at STR the name of the key must be are passed. If an empty string STR is handed over, the next key found is returned. RUN = 2 means the query can be started. Once the search is finished, the INI_PARSER_BUF sets the parameters of RUN to 0. With RESULT the search results will be issued. When in a query the key identified a new Section, this is reported by RESULT = 11. Upon successful search the output of the parameter KEY is the name of the found key , and VALUE is the key value. And then the OFFSET parameter points to the end of the key line. Thus, immediately after the next Key evaluation be continued, without having to manually change the OFFSET.

Enumeration - see next item:

For very large amount of data of an initialization file to be evaluated, with a enumeration (list) the user program can be build simple, and the evaluation be carried out more quickly because here no line must be used more than once. Before the start OFFSET must have a reasonable value, the default case to 0. With RUN = 3 the evaluation is started. Once a section or a key is found, it is also issued immediately. In a section KEY prints the name of the Section and RESULT = 1. With a found KEY, KEY has the key name and VALUE is the key value, and RESULT= 2.

If in a query, the end of the data array is reached, this will be reported by RESULT = 10.

RUN: Feature List

RUN	Function
0	No function to perform - and last function has finished
1	Specific section or evaluate next found section
2	evaluate specific Key or Key found next
3	evaluate next found element (section or key)

RESULT: Result - Feedback

RESULT	Description

1	Section found
2	Key found
5	Current query is still running - call module further cyclical!
10	Nothing found - reached the end of data
11	Key not found - reached the end of Section

10.8. INI_PARSER_FILE

Type Function module

OUTPUT: RESULT: BYTE (result of query)

IN_OUT FILE NAME: STRING (file name)

FSD: FILE_SERVER_DATA (file interface)

STR: STRING(STRING_LENGTH) (searched item)

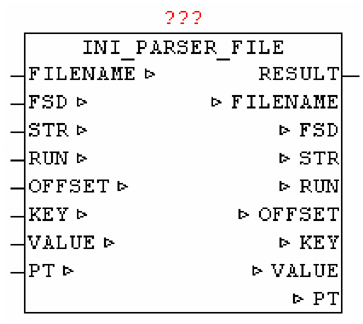
RUN: BYTE (command code for current action)

OFFSET: UDINT (current file offset of the query)

KEY: STRING(STRING_LENGTH) (found item)

VALUE: STRING (STRING_LENGTH) (value of a key)

PT: NETWORK_BUFFER (read data buffer)



The module INI_PARSER_FILE enables the analysis of the elements of an arbitrarily large INI file which is read into the read data buffer block by block for automatically processing. The name of the file is passed in parameter "FILENAME". The search for elements always begins on the given depended "OFFSET", and hence is very easy to look only at certain positions, or to repeat the search from a specific section to browse not always the entire byte array. At the initial search should start default to OFFSET 0 (but may not!). When querying sections and keys, there are various procedures. Either it is queried to a Section and evaluates all of the following keys by individual queries, or to use in very large initialization file the classic enumeration (listing), which means it will be report serially all the elements, and processed by the application.

Section Search:

To determine the OFFSET of a specific Section, STR must declare the name of the Section and the offset can be set to a position that is located before of the searched section. Should only the nearest available section be found, at STR an empty sting must be passed. The search query is started by RUN = 1. The search will take different time, depending on the structure and size of the INI data, it takes an indefinite number of cycles until a positive or negative result is achieved. Once the search is finished, the INI_PARSER_BUF sets the parameters of RUN to 0. RESULT passes the result of the search to output. Upon successful search the name of the section is shown at parameters KEY. And then the OFFSET parameter points to the end of the section line. Thus, immediately after that the key evaluation can be continued, without having to manually change the OFFSET.

Key Search:

Before a Key is evaluated, the OFFSET must have a correct value, this can be done by manual set of OFFSET or by a previously executed Section search. Before running the query at STR the name of the key must be are passed. If an empty string STR is handed over, the next key found is returned. RUN = 2 means the query can be started. Once the search is finished, it sets the parameters of RUN to 0. With RESULT the search results will be issued. When in a query the key identified a new Section, this is reported by RESULT = 11. Upon successful search the output of the parameter KEY is the name of the found key , and VALUE is the key value. And then the OFFSET parameter points to the end of the key line. Thus, immediately after the next Key evaluation be continued, without having to manually change the OFFSET.

Enumeration - see next item:

For very large amount of data of an initialization file to be evaluated, with a enumeration (list) the user program can be build simple, and the evaluation be carried out more quickly because here no line must be used more than once. Before the start OFFSET must have a reasonable value, the default case to 0. With RUN = 3 the evaluation is started. Once a section or a key is found, it is also issued immediately. In a section KEY prints the name of the Section and RESULT = 1. With a found KEY, KEY has the key name and VALUE is the key value, and RESULT= 2.

If in a query, the end of the data array is reached, this will be reported by RESULT = 10.

If the file access is no longer needed, the user must close the file be either by use of AUTO_CLOSE or MODE 5 (close file) of the FILE_SERVER.

RUN: Feature List

RUN	Function
0	No function to perform - and last function is complete
1	Evaluate specific section or evaluate next found section
2	evaluate specific Key or Key found next
3	evaluate next found element (section or key)

RESULT: Result - Feedback

RESULT	Description
1	Section found
2	Key found
5	Current query is still running - call module further cyclical!
10	Nothing found - reached the end of data
11	Key not found - reached the end of Section

11. Telnet-Vision

11.1. TELNET_VISION

The package TELNET_VISION is a framework comprising a plurality of function modules to enable simple means with a graphical interface based on the standard TELNET.

The GUI (Graphic User Interface) uses a screen of 80 characters wide and 24 lines down. At each coordinate (position) any displayable characters with selectable color attributes can be displayed.

The horizontal axis (from left to right) is called standard with X, and includes the positions 00-79. The vertical axis (from top to bottom) is called by default to Y and includes the positions 00-23. For pure coordinates specify the location with X and Y. If an area (rectangle) are indicated, the upper-left corner and lower right corner X1/Y1 with X2, Y2 defined.

The individual characters can be equipped with color attributes. A color attributes consist of a byte, where the left nibble (4 bits) the ink color (foreground color), and the right nibble (4 bits) the background color defines.

Example: BYTE #16 #74, (* foreground: white, and blue background *)

The following color attributes are defined:

Foreground color:

Nibble	**Color**	**Byte**	**Color**
0	Black	8	Flashing Black
1	Light Red	9	Flashing Light Red
2	Light Green	10	Flashing Light Green
3	Yellow	11	Flashing Yellow
4	Light Blue	12	Flashing Light Blue
5	Pink / Light Magenta	13	Flashing Pink / Light Magenta
6	Light Cyan	14	Flashing Light Cyan
7	White	15	Flashing White

Background color:

Nibble	Color
0	Black
1	Red
2	Green
3	Brown
4	Blue
5	Purple / Magenta
6	Cyan
7	Gray

For easy handling of the package the module TN_FRAMEWORK is responsible, it must be called cyclically in the application, as it manages the whole system and executes it. This communication with the telnet client is processed, at graphic changes the system always made an intelligent automatic update. The INPUT_CONTROL items are stored, and the keystrokes to the respective elements forwarded, and even an optional menu bar is available.

As this is a relatively complex interplay of many elements, in the library under / DEMO are two applications available, that perform with all the possibilities. It is to be advised at own projects to ocreate them based on these two templates to get as quickly as possible a working result, and to understand the interaction of the individual components and modules.

The program TN_VISION_DEMO_1 shows the following elements:

Graphical representation of lines, polygons, texts, and associated shadow, and color scheme of the layout

Representation of a Menu Bar

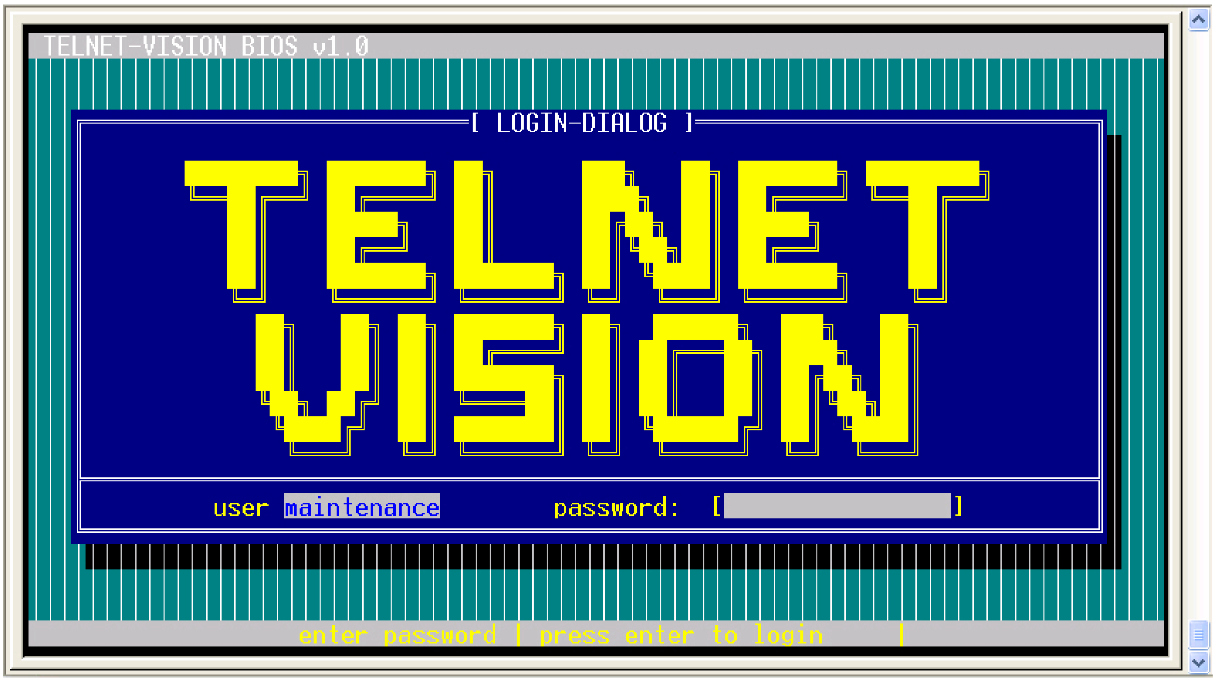
Elements: EDIT_LINE (normal and hidden input), SELECT_POPUP, SELECT_TEXT

TOOLTIP info line

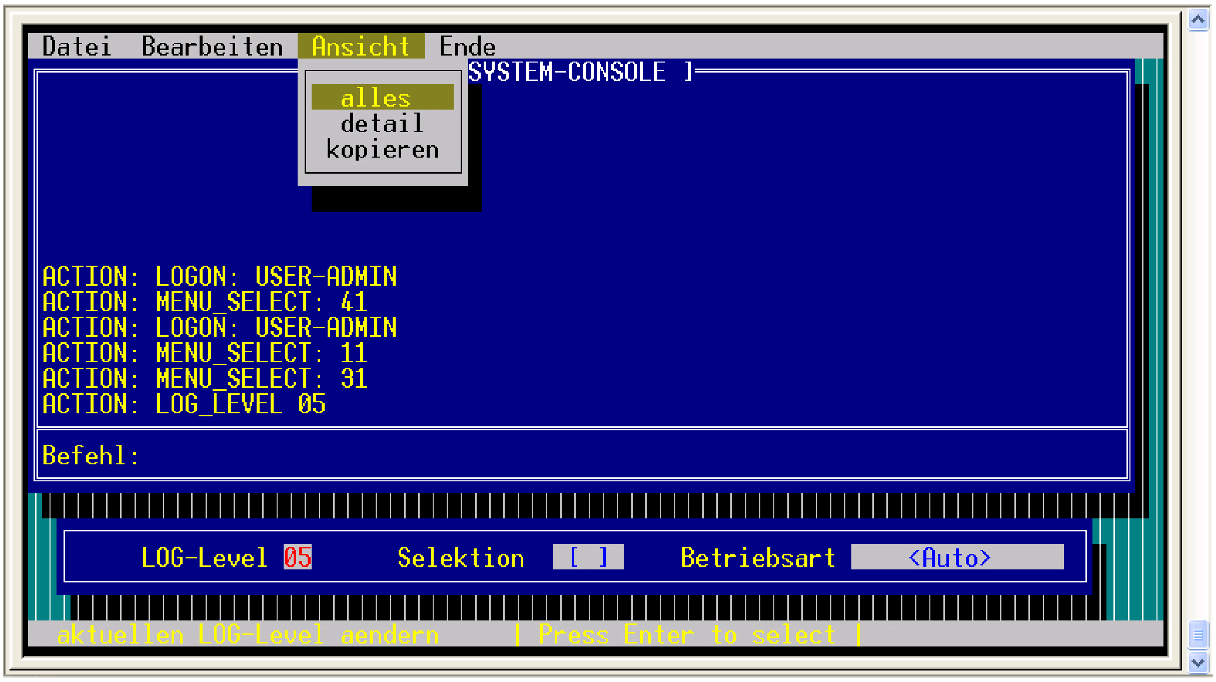
Illustration of a LOG_VIEWPORT with the message buffer, and navigation using keys.

On the home page a LOGIN function is realized. By entering the password 'oscat' you can switch to the next page. The main page can be changed using the cursor up / down button and with tab between the individual elements. The menu can be called with the Escape key. The individual menu items are only for demonstration purposes, and lead to a log message. Only the menu item "end/LOGOUT" leads back to the home page.

TN_VISION_DEMO_1 (screen page 1)



TN_VISION_DEMO_1 (screen 2)



The program TN_VISION_DEMO_2 shows the following elements:

Graphical representation of lines, polygons, texts and design the layout monchrome

Elements: EDIT_LINE (normal and hidden input, and using an input mask), SELECT_TEXT

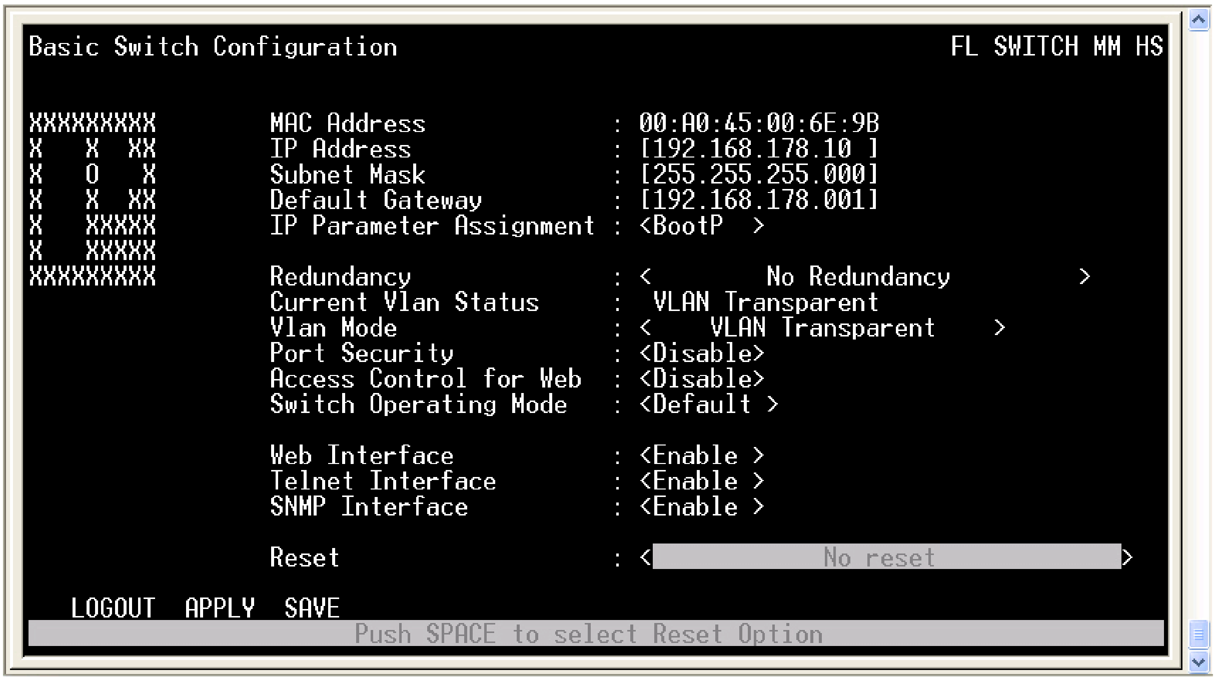
TOOLTIP info line

On the home page a LOGIN function is realized. By entering the password 'oscat' you can switch to the next page. The main page can be changed using the cursor up / down button and with tab between the individual elements. Only the item "LOGOUT" leads back to the home page.

The two sides have shown a replica of the Telnet page of a manageable switch from PHOENIX CONTACT, used to show that the TELNET VISION package can be used for for simple configuration pages.

TN_VISION_DEMO_2 (screen page 1)

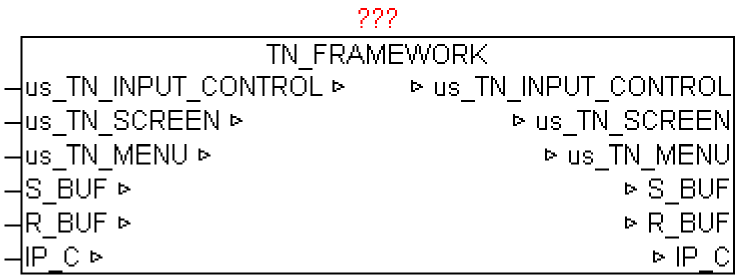
TN_VISION_DEMO_2 (screen 2)



11.2. TN_FRAMEWORK

Type Function module

IN_OUT Xus_TN_INPUT_CONTROL : Us_TN_INPUT_CONTROL
Xus_TN_SCREEN : Us_TN_SCREEN
Xus_TN_MENU: us_TN_MENU
S_BUF: NETWORK_BUFFER (transmit data)
R_BUF: NETWORK_BUFFER (receive data)
IP_C: IP_CONTROL (parameterization)



The module TN_FRAMEWORK is a frame structure, which provides a finished maturity model for TELNET-Vision .

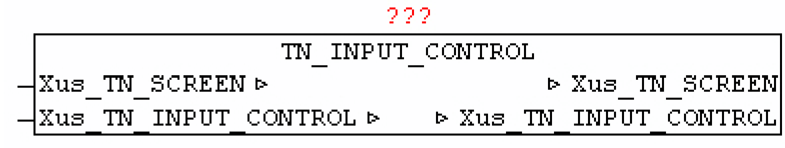
The following tasks and functions are treated.

Connection setup and breakdown with Telnet Client
Send and receive data
Data structures for graphics functions
INPUT_CONTROL elements
Intelligent automatic updating of the Telnet display
Menu bar display
Direct access to all data structures for user program

11.3. TN_INPUT_CONTROL

Type Function module

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN
Xus_TN_INPUT_CONTROL: us_TN_INPUT_CONTROL



The module TN_INPUT_CONTROL is used to manage the INPUT_CONTROL elements. If Xus_TN_INPUT_CONTROL.bo_Reset_Fokus = TRUE then the FOCUS is disabled on all elements and the first item gets to the focus. Using the cursor up / down buttons and tab, the individual elements can be selected or changed. The current element loses focus and then the next following item gets the input focus reallocated. At the focus change of the elements automatically a redraw of the respective elements is triggered. The image/flashing cursor is always positioned at each active element and is displayed. It always automatically displays and updates the ToolTip text, as this has been configured.

It supports the following elements.

TN_INPUT_EDIT_LINE

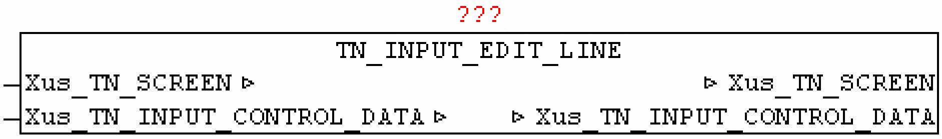
TN_INPUT_SELECT_TEXT

TN_INPUT_SELECT_POPUP

11.4. TN_INPUT_EDIT_LINE

Type Function module

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN
Xus_TN_INPUT_CONTROL: us_TN_INPUT_CONTROL



The module TN_INPUT_EDIT_LINE is used to manage a command line. This must be set *. in_TYPE = 1.

The item will be provided as *.in_X and *.in_Y. Every entry line can be provided with a title text. With *.in_Title_Y_Offset and *.in_Title_X_Offset the position relative to the element coordinates is expressed. The color can be determined with *.by_Title_Attr, and the text by *.st_Title_String. If a tool tip should appear at the element *. st_Input_ToolTip the text hast to be specified.

If the item has focus, using the keyboard cursor left / right the flashing cursor can be moved within the line. The backspace key can delete entered character. By pressing the Enter / Return key the input text is issued at *.st_Input_String and *.bo_Input_Entered ist set to TRUE. The input flag must be reset after receive by the user. Using *.bo_Input_Hidden = TRUE the hidden input is activated, thus, all input characters represented with a '*'.

Using *.st_Input_Mask determines at which position and how many characters can be entered. At each position which a space, character can be entered. During initialization *.st_Input_Mask must be copied once to *.st_Input_Data.

Is *.bo_Input_Only_Num = TRUE only numeric keys are accepted and adopted.

Example:

```
*.in_Type := INT#01;
*.in_Y := INT#16;
*.in_X := INT#09;
*.by_Attr_mF := BYTE#16#72; (* white, green *)
*.by_Attr_oF := BYTE#16#74; (* white, blue *)
*.in_Cursor_Pos := INT#0;
*.bo_Input_Only_Num := FALSE;
*.bo_Input_Hidden := FALSE;
*.st_Input_Mask := '                                                            ';
*.st_Input_Data :=  *.st_Input_Mask;
*.st_Input_ToolTip := 'inputline active       | SCROLL F1/F2/F3/F4 |';
*.in_Input_Option := INT#02;
*.in_Title_Y_Offset := INT#00;
*.in_Title_X_Offset := INT#00;
*.by_Title_Attr := BYTE#16#34;
*.st_Title_String := 'command: ';
```

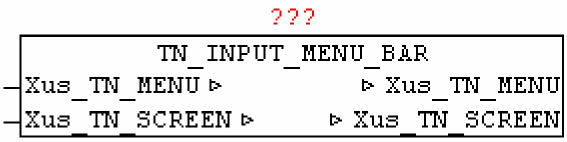
The following output:

11.5. TN_INPUT_MENU_BAR

Type Function module

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN

Xus_TN_MENU: us_TN_MENU



The module TN_INPUT_MENU_BAR is used to manage and view the Menu_Bar. The element is shown in *.in_X and *.in_Y. The menu items are stored as elements within verschachelte *.st_MENU_TEXT. Two different separators are used. A '$' separates the different menu lists, and each menu list is further divided by '#' into individual menu items. The first menu list is the actual menu bar, this implies the number of sub-menus, and the titles of the elements. Then all the sub-menu lists are follow and are separated by '%'. To devide individual sub-menu items from each other or providing them with a cut line, an '-' has to submitted as text menu-element.

By pressing the Escape key, the menu bar activated and the respective sub-menu is displayed using the module TN_INPUT_MENU_POPUP. Within the sub-menu can be navigated with up / down key. Within the sub-menu can be navigated with up / down cursor. If a sub-menu item is confirmed by pressing Enter / Return key, then in *.in_Menu_Selected the number of the selected menu-point is passed. The calculation of the menu item number is as following: Main menu index * 10 + Submenu-index. The entry in *.in_Menu_Selected needs set again to 0 after acceptance by users.

Thus, a maximum of 9 main menu items and each 9-Submenu items are executable. Means of escape key at any time the menu can be hided again.

Active Menu automatically backs up the background before it is drawn, and restores the background after ending.

As long as a menu is display, the user program may not make graphical changes. This can be checked by TN_SCREEN.bo_Menue_Bar_Dialog = TRUE.

Example:

*.in_X := INT#00;

*.in_Y := INT#00;

.by_Attr_mF := BYTE#16#33; (yellow + brown *)

.by_Attr_oF := BYTE#16#0F; (black + grey *)

*.st_MENU_TEXT := 'File#Edit#View#End';

.st_MENU_TEXT := CONCAT(.st_MENU_TEXT,

'%oeffnen#-#speichern#beenden%loeschen#-#einfuegen#-#kopieren');

.st_MENU_TEXT := CONCAT(.st_MENU_TEXT,

'%alles#detail#kopieren%Logout');

*.bo_Create := TRUE;

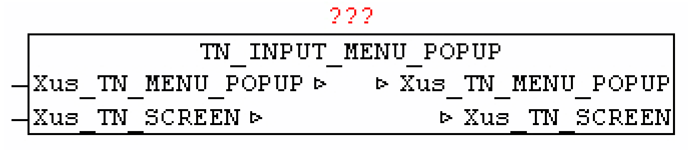
The following output:

11.6. TN_INPUT_MENU_POPUP

Type Function module

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN

Xus_TN_MENU: us_TN_MENU



The module TN_INPUT_MENU_POPUP is used to manage and display the Menu_Bar Submenu and for the representation of TN_INPUT_SELECT_POPUP elements. The element is shown in *.in_X and *.in_Y. The menu items are stored as elements within *.st_Menu_Text. The individual element are devided from each other using '#'. To devide individual sub-menu items from each other or providing them with a cut line, an '-' has to submitted as text menu-element.

Within the sub-menu can be navigated with up / down key. If a sub-menu item is confirmed by pressing Enter / Return key, then in *.in_Menu_Selected the number of the selected menu-point is passed.

An active Menu automatically backs up the background before it is drawn, and restores the background after ending.

As long as a menu is display, the user program may not make graphical changes. This can be checked by TN_SCREEN.bo_Menue_Bar_Dialog = TRUE or TN_SCREEN.bo_Modal_Dialog = TRUE.

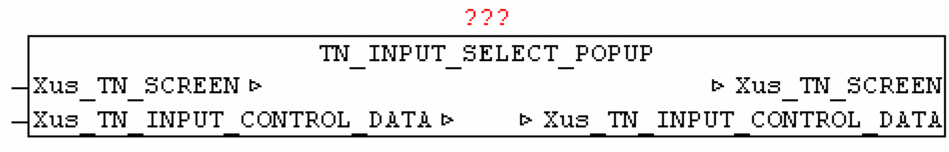
The module is primarily from TN_INPUT_MENU_BAR and TN_INPUT_SELECT_POPUP used internally, and need not be executed directly by the user.

11.7. TN_INPUT_SELECT_POPUP

Type Function module

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN

Xus_TN_INPUT_CONTROL: us_TN_INPUT_CONTROL



The module TN_INPUT_SELECT_POPUP is used to manage a selection of texts, by displaying a pop-up dialogue. This must be set *.IN_TYPE = 3.

The item will be provided as *.in_X and *.in_Y. Every entry line can be provided with a title text. With *.in_Title_Y_Offset and *.in_Title_X_Offset the position relative to the element coordinates is expressed. The color can be determined with *.by_Title_Attr, and the text by *.st_Title_String. If a tool tip should appear at the element *. st_Input_ToolTip the text hast to be specified.

The selection of texts will be handed over in *.st_Input_Data. The text element should be separated from each other by the character '#'.

If the focus is on an element, using the Enter / Return key selection dialog can be activated.

With the cursor up/down can be changed between the individual elements. If the beginning or the end of the list will be reachted, it continues at the opposite side.

The text-element is connected by means *.st_Input_Mask, meaning that the output text length are affected later.

By pressing the Enter / Return key is the text of the selected element is passed to *. st_Input_String and *. bo_Input_Entered = TRUE. The input flag must be reset after receive by the user.

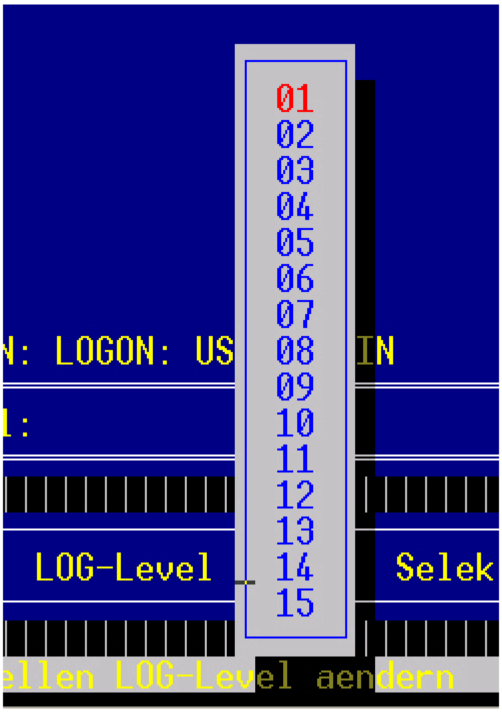
An active selection (selection dialog) can always be canceled with the Escape key.

Example:

```
*.in_Type := 03;
*.in_Y := 20;
*.in_X := 18;
*.by_Attr_mF := 16#17;
*.by_Attr_oF := 16#47;
*.st_Input_ToolTip :=
            'Change the current log level     | Press enter to select | ';
*.in_Input_Option := 00;
*.in_Title_Y_Offset := 00;
*.in_Title_X_Offset := 00;
*.by_Title_Attr := 16#34;
```

```
*.st_Title_String := ' LOG-Level ';
*.st_Input_Mask := '  ';
*.st_Input_Data :=
          '01#02#03#04#05#06#07#08#09#10#11#12#13#14#15';
```

The following output:

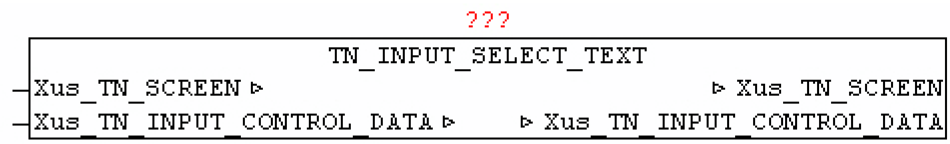


11.8. TN_INPUT_SELECT_TEXT

Type Function module

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN

Xus_TN_INPUT_CONTROL: us_TN_INPUT_CONTROL



The module TN_INPUT_SELECT_TEXT is used to manage a selection of texts. This must be set *.IN_TYPE = 2.

The item will be provided as *.in_X and *.in_Y. Every entry line can be provided with a title text. With *.in_Title_Y_Offset and *.in_Title_X_Offset the position relative to the element coordinates is expressed. The color can be determined with *.by_Title_Attr, and the text by *.st_Title_String. If a tool tip should appear at the element *. st_Input_ToolTip the text hast to be specified.

The selection of texts will be handed over in *.st_Input_Data. The text element should be separated from each other by the character '#'.

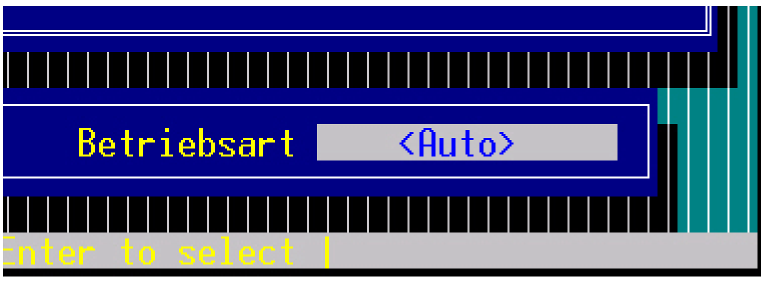
If the Element has the focus, by using the spacebar (space) can be changed between the individual texts. The text-element is connected by means *.st_Input_Mask, meaning that the output text length are affected later.

By pressing the Enter / Return key the input text is issued at *.st_Input_String and *.bo_Input_Entered ist set to TRUE. The input flag must be reset after receive by the user.

Example:

*.in_Type := 2;
*.in_Y := 20;
*.in_X := 58;
*.by_Attr_mF := 16#17;
*.by_Attr_oF := 16#47;
*.st_Input_ToolTip := ' selection text active | press space to select |';
*.in_Input_Option := 02;
*.in_Title_Y_Offset := 00;
*.in_Title_X_Offset := 00;
*.by_Title_Attr := 16#34;
*.st_Title_String := ' operation mode';
*.st_Input_Mask := '   ';
*.st_Input_Data := '<Auto>#<Hand>#<Stop>#<Restart>';

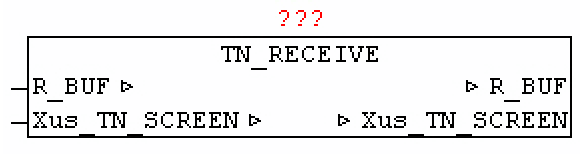
The following output:



11.9. TN_RECEIVE

Type Function module

IN_OUT Xus_TN_SCREEN: us_TN_SCREEN
R_BUF: NETWORK_BUFFER (Telnet receive buffer)



The module TN_RECEIVE receives input data from the Telnet client, and evaluates the key codes.

If the key code in the range 32-126 it shall be stored as ASCII code under Xus_TN_SCREEN, by_Input_ASCII_Code. In addition, Xus_TN_SCREEN.bo_Input_ASCII_IsNum = TRUE if this corresponds to a number between 0 and 9.

If the key code is of the following extended code then this is filed under Xus_TN_SCREEN,by_Input_Exten_Code.

Exten_code	**Button name**
65	Cursor up
66	Cursor down
67	Cursor RIGHT
68	Cursor left
72	Pos1
75	End
80	F1
81	F2
82	F3
83	F4
8	Backspace
9	Tabulator
13	Return (Enter)
27	Escape

11.10. TN_SEND_ROWS

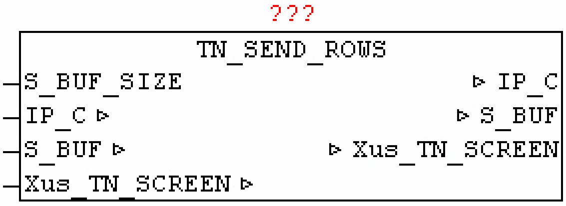
Type Function module

INPUT S_BUF_SIZE: UINT (number of bytes in S_BUF.BUFFER)

IN_OUT IP_C: IP_CONTROL (Connection data)

S_BUF: NETWORK_BUFFER (transmit data)

Xus_TN_SCREEN: us_TN_SCREEN



The module TN_SEND_ROWS is used to automatically update the graphical changes to the Telnet screen, by send the modified lines to the Telnet client.

If you change the Telnet screen a color or a character in a line, this line is always automatically selected for update. The module checks if marked at Xus_TN_SCREEN.bya_Line_Update [0..23] one or more lines, and generates an ANSI-code byte-stream which is sent to the Telnet client. Furthermore, when Xus_TN_SCREEN.bo_Clear_Screen = TRUE a clear screen is triggered. Upon detection of a new Telnet client connection automatically all the rows are marked for update, so that the whole screen content is rendered. If the required amount of data greater than S_BUF.BUFFER the data is automatically output in blocks.

11.11. TN_SC_ADD_SHADOW

Type Function module

INPUT: Iin_Y1: INT (Y1 coordinate of the area)

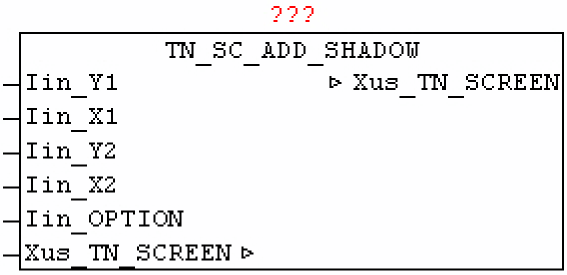
Iin_X1: INT: (X1 coordinate of the area)

Iin_Y2: INT (Y2 coordinate of the area)

Iin_X2: INT: (X2-coordinate of the area)

Iin_OPTION: INT: (kind of the shadow)

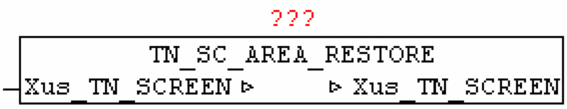
IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_ADD_SHADOW allows you to add optical shadow to rectangular glyphs. By specifying a rectangular area by means of the parameters X1, Y1 and X2, Y2, a basic framework is defined, at which at the right and bottom color darkened lines are drawn (shadow). The shadow coordinates X1, Y1 and X2, Y2 are always given +1 for proper primitive. OPTION means you can choose between two shadow variations. If OPITION = 0 then the shadow is reached by pure color adjustment (darkening of the character) . If an OPTION > 0, in the area of the shadow all the characters replaced by black filled characters.

11.12. TN_SC_AREA_RESTORE

Type	Function module
IN_OUT	Xus_TN_SCREEN : Us_TN_SCREEN

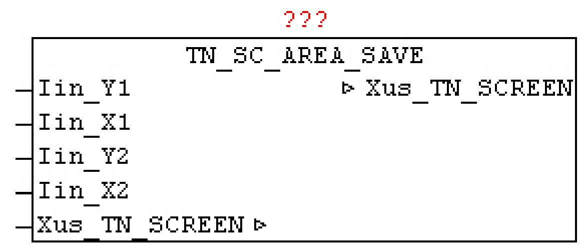


The module TN_SC_AREA_RESTORE enables recovery of previously saved screen area. The screen data in Xus_TN_SCREEN.bya_BACKUP [x] is restored using the stored coordinates. This is done mainly done after the call from the module MENU-BAR amd MENU-POPUP, to restore the modified screen.

11.13. TN_SC_AREA_SAVE

Type	Function module

INPUT: Iin_Y1: INT (Y1 coordinate of the area)
Iin_X1: INT: (X1 coordinate of the area)
Iin_Y2: INT (Y2 coordinate of the area)
Iin_X2: INT: (X2-coordinate of the area)
IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN

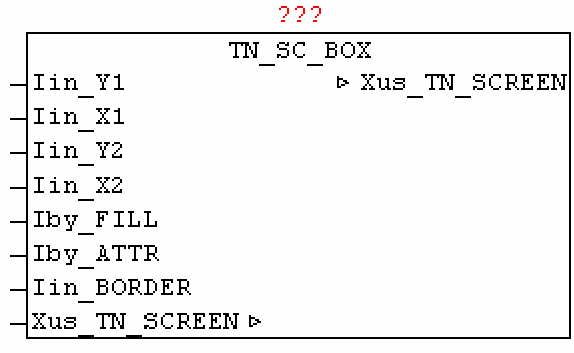


The module TN_SC_AREA_SAVE allows you to save of rectangular areas of the screen before it is modified by other drawing operations. This is mainly done before the call from the module BAR-MENU and MENU-POPUP , because these are the elements as an overlay graphic. Means X1, Y1 and X2, Y2 are given the coordinates of the secured area of the screen. The data are saved in the data area Xus_TN_SCREEN.bya_BACKUP [x]. Here the coordinates and the actual characters and color information is stored. The buffer can hold up half the area of the screen.

11.14. TN_SC_BOX

Type Function module

INPUT: Iin_Y1: INT (Y1 coordinate of the area)
Iin_X1: INT: (X1 coordinate of the area)
Iin_Y2: INT (Y2 coordinate of the area)
Iin_X2: INT: (X2-coordinate of the area)
Iby_FILL: BYTE: (fill in the character of the area)
Iby_ATTR: BYTE: (color code to fill the area)
Iby_BORDER: BYTE: (type of frame)
IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_BOX is used to draw a rectangular area, that is filled with the specified character in Iby_FILL. With parameter Iby_ATTR fill color can be specified. The fill area is drawn with a border that is given by Iin_BORDER.

Border types:

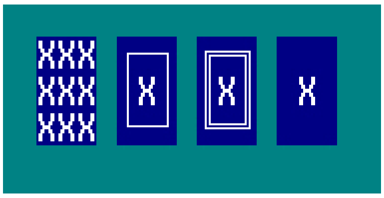
0 = no border

1 = frame with a single line

2 = frame double line

3 = frame with spaces

Example: Box with leaders 'X' and white color to blue



Representation with Iin_BORDER value 0,1,2 and 3 (from left to right)

11.15. TN_SC_FILL

Type Function module

INPUT: Iin_Y1: INT (Y1 coordinate of the area)

Iin_X1: INT: (X1 coordinate of the area)

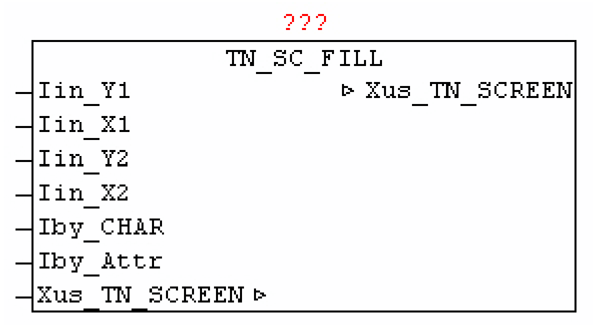
Iin_Y2: INT (Y2 coordinate of the area)

Iin_X2: INT: (X2-coordinate of the area)

Iby_CHAR: BYTE: (character to fill in the the area)

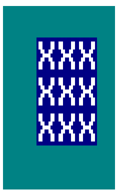
Iby_ATTR: BYTE: (color code to fill the area)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_FILL is used to draw a rectangular area, that is filled with the specified character in Iby_FILL.

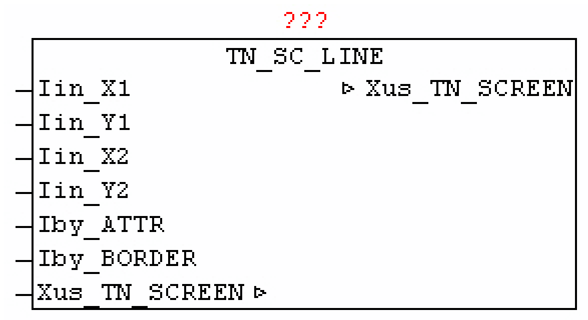
Example: Box with leaders 'X' and white color to blue



11.16. TN_SC_LINE

Type Function module

INPUT: Iin_Y1: INT (Y1 coordinate of the line)
Iin_X1: INT: (X1 coordinate of the line)
Iin_Y2: INT (Y2 coordinate of the line)
Iin_X2: INT: (X2-coordinate of the line)
Iby_ATTR: BYTE: (color code of the line)
Iby_BORDER: BYTE: (type of line)
IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_LINE is used to draw horizontal and vertical lines. By means of the X1/Y1 and X2/Y2 coordinates defines the beginning and the end of the line. The line type is passed by Iin_BORDER and the color code with Iby_ATTR. If when drawing a line and another line of this type cut, automatically the appropriate crossing sign is used.

Border types:

1 = line with single line

2 = line with double line

> 2 = line is drawn with the specified character in Iin_BORDER

Example:

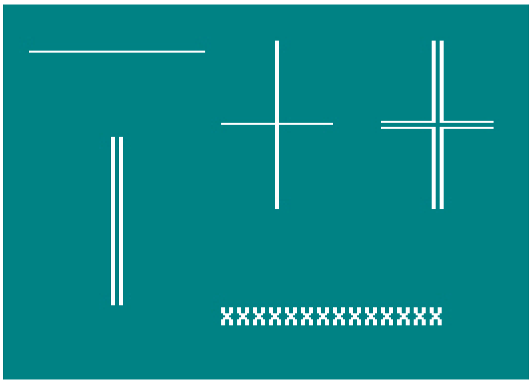
Horizontal line: type single-line

Vertical line: Type Double-Line

Horizontal and vertical lines crossed: Single-Line Type

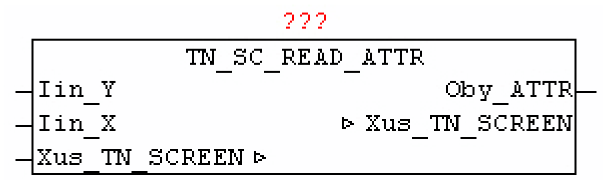
Horizontal and vertical lines crossed: Type Double-Line

Horizontal line: type character (X)



11.17. TN_SC_READ_ATTR

Type	Function module
INPUT	Iin_Y: INT: (Y coordinate)
	Iin_X: INT: (X coordinate)
OUTPUT	Oby_ATTR: BYTE: (color information at position X / Y)
IN_OUT	Xus_TN_SCREEN : Us_TN_SCREEN



The block TN_SC_READ_ATTR is used to read the current color of the character at the specified location X / Y.

11.18. TN_SC_READ_CHAR

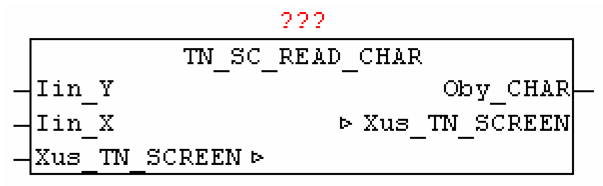
Type Function module

INPUT Iin_Y: INT: (Y coordinate)

Iin_X: INT: (X coordinate)

OUTPUT Oby_CHAR: BYTE: (character at position X / Y)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_READ_CHAR is used to read the current character at the specified location X / Y.

11.19. TN_SC_SHADOW_ATTR

Type Function: BYTE

INPUT Iby_ATTR: BYTE: (Color Information)

The block TN_SC_SHADOW_ATTR converts a light color to a dark color.

11.20. TN_SC_VIEWPORT

Type Function module

INPUT Iin_Y: INT: (Y coordinate)

Iin_X: INT: (X coordinate)

Iin_Width: INT: (width of the window - the number of characters)

Idw_ATTR_1: DWORD: (color 1,2,3 and 4)

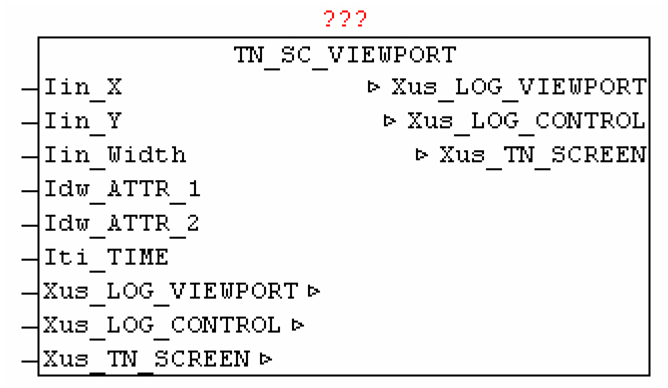
Idw_ATTR_2: DWORD: (color 5,6,7 and 8)

Iti_TIME: TIME: (update time)

IN_OUT Xus_LOG_VIEWPORT: LOG_VIEWPORT

Xus_LOG_CONTROL: LOG_CONTROL

Xus_TN_SCREEN: us_TN_SCREEN



The module TN_SC_VIEWPORT is used to display messages from the data structure LOG_CONTROL within a rectangular area on the screen. The desired messages are processed before using with Block LOG_VIEWPORT, and if necessary, with Xus_LOG_VIEWPORT.UPDATE an update is triggered. Means Iin_X and Iin_Y defines the upper-left corner of the window, and with Iin_Width the width if of the viewing window is defined. The number of rows to be displayed is determined by Xus_LOG_VIEWPORT.COUNT. The color information is stored in Xus_LOG_CONTROL.MSG_OPTION [x] per message. It is converted to the configured color codes from Idw_ATTR_1 and Idw_ATTR2 automatically, so the colors in the presentation can always be adjusted individually. The messages are always automatically reduced to the width of the window or cut off.

11.21. TN_SC_WRITE

Type Function module

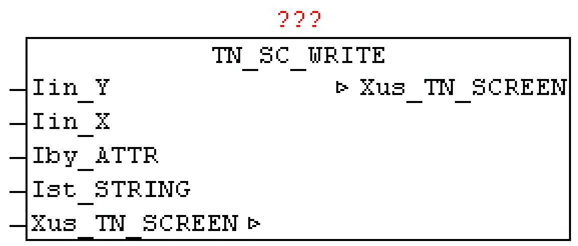
INPUT Iin_Y: INT (Y coordinate)

[fzy] Iin_X: INT: (X coordinate)

Iby_ATTR: BYTE: (color code - font color)

Ist_STRING: STRING (text)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_WRITE passes the text Ist_STRING at the coordinates Iin_Y, Iin_Y and the color of Iby_ATTR.

Is specified color code = 0, then the string is displayed without change the existing old color information at he respective character positions.

11.22. TN_SC_WRITE_ATTR

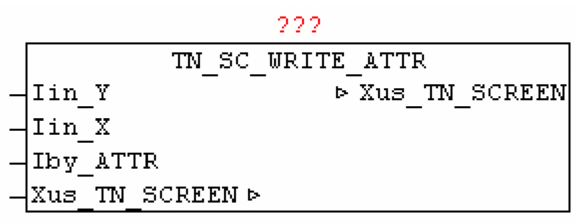
Type Function module

INPUT Iin_Y: INT (Y coordinate)

[fzy] Iin_X: INT: (X coordinate)

Iby_ATTR: BYTE: (color code)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_WRITE_ATTR changes at the given coordinates Iin_Y, Iin_Y the colorcode to change without changing the existing character at that position.

11.23. TN_SC_WRITE_C

Type Function module

INPUT Iin_Y: INT (Y coordinate)

[fzy] Iin_X: INT: (X coordinate)

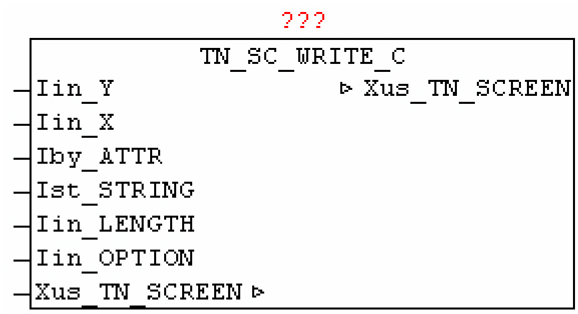
Iby_ATTR: BYTE: (color code)

Ist_STRING: STRING: (text)

Iin_LENGTH: INT: (text will be adjusted to this length)

Iin_OPTION: INT: (option-length adaptation of the text)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_WRITE_C is at the given coordinates Iin_Y, Iin_Y Ist_STRING the text with the color of Iby_ATTR. The text is adapted before output on the length Iin_LENGTH, and by Iin_OPTION, the text position is determined.

Iin_OPTION

0 = right fill with spaces eg 'TEST '

1 = left fill with blanks eg ' TEST '

2 = center and fill with blanks eg ' TEST '

11.24. TN_SC_WRITE_CHAR

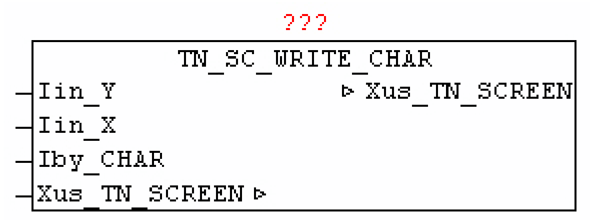
Type Function module

INPUT Iin_Y: INT: (Y coordinate)

Iin_X: INT: (X coordinate)

OUTPUT Iby_CHAR: BYTE: (sign)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_WRITE_CHAR passes the character Iby_CHAR at the given coordinates Iin_Y, Iin_X, and does not change the color information at the specified position.

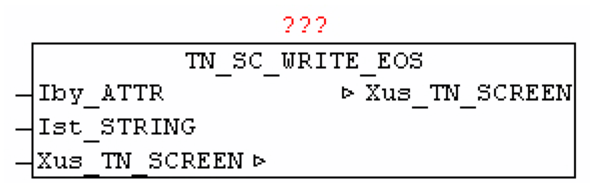
11.25. TN_SC_WRITE_EOS

Type Function module

INPUT Iby_ATTR: BYTE: (color code - font color)

Ist_STRING: STRING (text)

IN_OUT Xus_TN_SCREEN : Us_TN_SCREEN



The module TN_SC_WRITE_EOS passes at the end position of the last, with TN_SC_WRITE or TN_SC_WRITE_EOS issued text, the text Ist_STRING with the color of Iby_ATTR.

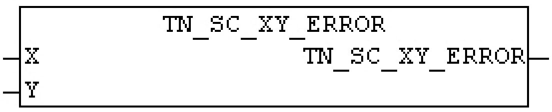
This allows to continuous passes texts without the need to always pass the new coordinates.

11.26. TN_SC_XY_ERROR

Type Function: BOOL

INPUT: X: INT: (X coordinate)

Y: INT: (Y coordinate)



The module TN_SC_XY_ERROR checks whether the specified coordinate is within the screen area. If the check fails, as result is passes TRUE.

11.27. TN_SC_XY2_ERROR

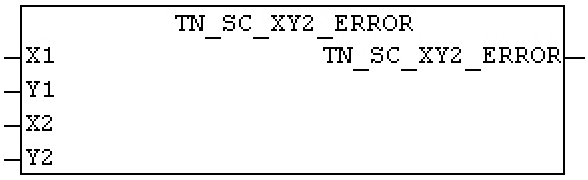
Type Function: BOOL

INPUT: X1: INT: (X1 coordinate of the area)

Y1: INT: (Y1 coordinate of the area)

X2: INT: (X2-coordinate of the area)

Y2: INT: (Y2 coordinate of the area)



The module TN_SC_XY2_ERROR checks whether the specified coordinates are within the screen area. The area may not cross off the screen. If the check fails, as result is passes TRUE.

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